

# QED on $S^3$ : Spectral Geometry of Perturbative Transcendentals

GeoVac Project

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## Abstract

We compute perturbative QED spectral sums on the three-sphere  $S^3$  using the Camporesi–Higuchi Dirac spectrum and the Hodge-1 (photon) spectrum with  $\text{SO}(4)$  vertex selection rules. Eight structural results emerge. (1) **T<sub>9</sub> theorem**: the spectral zeta function of the squared Dirac operator  $D^2$  is a two-term polynomial in  $\pi^2$  with rational coefficients at every integer  $s$ ; no odd-zeta content appears (Theorem 1). (2) **Even/odd parity discriminant**: operator order (2nd vs 1st) is the primary transcendental discriminant—even  $s$  produces  $\pi^{\text{even}}$ , odd  $s$  produces  $\zeta(3)$ ,  $\zeta(5)$ , etc. (Theorem 2). (3)  **$\chi_{-4}$  identity**:  $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$  at every integer  $s \geq 2$ , a concrete spectral-zeta to Dirichlet- $L$  bridge (Theorem 3). (4) **Self-energy structural zero**: the one-loop electron self-energy on  $S^3$  vanishes identically at the ground state  $n_{\text{ext}} = 0$  as a consequence of vertex parity (Theorem 4). (5) **Anomalous magnetic moment**: the one-loop vertex correction with a  $q = 1$  probe photon yields  $F_2/[\alpha/(2\pi)] = 1.084$  at the lowest external mode ( $n_{\text{ext}} = 1$ ), approximately 8.4% above the flat-space Schwinger value, matching the Parker–Toms leading curvature correction within 0.4%. The next-order coefficient  $c_3 = -5.946(3) \times 10^{-7}$  is nonzero at  $200.9\sigma$ , so the curvature expansion does not terminate at second order (Sec. 7.7). (6) **Depth- $k$  tower**: loop order bounds the motivic depth of the level-2 (alternating-MZV) content; it does not automatically mint new irreducibles. The two-loop  $S_{\text{min}}$  decomposes exactly into Hoffman double  $t$ -values (classical periods of  $\mathcal{MT}(\mathbb{Z}[1/2])$ ; the prior “PSLQ-irreducible” claim was a basis-coverage artifact); the three-loop  $S^{(3)} = 31.5726\dots$  (rigorous bracket  $[31.57063, 31.57300]$ ) has realized depth  $\leq 2$  on every identified component: the catalogued depth-2 generators  $t(5, 1)$ ,  $t(5, 3)$ ,  $t(7, 1)$  enter at coefficients  $\pm \frac{1}{2}$ , and nothing new is minted (Proposition 4, Sec. 10.4). (7)  **$\zeta(3)$  complementarity**: the degeneracy-weighted fine-structure shell sum  $\mathcal{S}(n) = n(5n-4)/2$  and the Dirac mode degeneracy  $g_n^{\text{Dirac}} = 2n(n+1)$  satisfy  $\mathcal{S}(n) + g_n^{\text{Dirac}} = 9n^2/2$  exactly, so the Dirichlet series identity  $D_{\mathcal{S}}(s) + D_g(s) = \frac{9}{2}\zeta(s-2)$  holds for all  $s$ : the  $\zeta(s-1)$  terms—all odd-zeta content—cancel identically (Theorem 7). (8) **Fine-structure Euler sum cancellation and product survival rule**: the degeneracy-weighted Dirichlet sums  $D_p$  at order  $(Z\alpha)^{2p}$  in the exact Dirac hydrogen spectrum, computed through  $D_6$  ( $(Z\alpha)^{12}$ , weight 11), satisfy a product survival rule:  $\zeta(2) \times \zeta(\text{odd})$  cancels algebraically at every order, while exactly  $\max(0, \lfloor (2p-5)/2 \rfloor)$  products survive (Observations 1, 2). The cancellation mechanism is the  $n^2$  Fock degeneracy—the same weighting that produces  $F = \zeta(2) = D_{n^2}(4)$  in Paper 2. However, PSLQ confirms that Paper 2’s  $K = \pi(B + F - \Delta)$  is NOT a  $\mathbb{Q}$ -linear combination of  $D_2, \dots, D_6$ : the spectral-action ingredients of  $K$  inhabit the  $\pi$ -polynomial ring of Theorem 1, while the Sommerfeld sums carry irreducible odd-zeta from the Euler–Zagier MZV algebra. A complementary *graph-native* formulation (Sec. 15) computes all one-loop diagrams as finite matrix traces on the GeoVac Fock graph, requiring no infinite sums, regularization, or renormalization. All graph-native QED quantities are  $\pi$ -free (algebraic); the projection exchange constant relating graph and continuum VP is *rational* at every finite truncation, confirming Paper 18’s calibration-tier classification. The continuum self-energy structural zero (Theorem 4) is broken on the finite graph: the pendant-edge theorem (Proposition 5) proves  $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\text{max}} - 1)/n_{\text{max}} \rightarrow 2$  exactly, never healing in the graph limit. The root cause is that the graph photon is a scalar 1-cochain, lacking the spin-1 vector structure whose  $\text{SO}(4)$  selection rules enforce the continuum zero. The graph-native anomalous moment  $F_2(t)$  is an

even rational function of the hopping coupling  $t$  over  $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$ , with  $F_2(\kappa)$  decreasing as a power law  $\sim 3.5 n_{\max}^{-0.57}$  ( $F_2 = 2.353, 1.873, 1.589, 1.396, 1.253$  at  $n_{\max} = 2, \dots, 6$ ). A systematic selection rule census shows that only 1 of 8 continuum QED selection rules survives on the scalar Fock graph (Gaunt/CG sparsity). Building QED on the native Dirac graph with  $(n, \kappa, m_j)$  node labels recovers 4 of 8 rules, establishing a three-tier partition: 1 always survives, 3 are spinor-recoverable (intrinsic to spinor quantum numbers), and 4 require vector photon structure (calibration tier). A five-track vector QED sprint (VQ-1–5) confirms this partition from a complementary direction: neither Pauli spin matrices at the vertex, nor direction-resolved edge Hodge decomposition, nor their combination recovers any of the 4 vector-required rules. However, the transverse photon propagator  $G_T = (d_1 d_1^\top)^+$  built from Paper 30’s plaquette boundary operator *does* recover the ground-state structural zero ( $\Sigma_T(\text{GS}) = 0$  exactly), refining the partition to 1+3+1+3: the plaquette-recoverable tier is topological (pendant-edge exclusion from closed walks), requiring no photon quantum numbers. Adding explicit vector photon quantum numbers  $(q, m_q)$  with Clebsch–Gordan vertex coupling recovers 7 of 8 continuum selection rules for scalar electrons and **8 of 8 for Dirac electrons** (Sec. 15.12). The calibration cost is exactly one transcendental per loop:  $\pi$ , entering as the  $S^2$  Weyl exchange constant  $1/(4\pi)$  from Paper 18’s taxonomy. For scalar electrons, the sole failure is Furry’s theorem; for Dirac electrons, the diagonal vertex coupling  $V(a, a, q, m_q) = 0$  vanishes identically by the *Dirac spinor phase constraint*: the  $\alpha$ -matrix is off-diagonal in large/small component space, and for  $a = b$  the cross terms combine, via Hermiticity and the  $i$ -factor on the small component, into an imaginary part that  $m_l$  conservation forces to vanish. This is a kinematic identity of the single-particle vertex, not the field-theoretic charge conjugation mechanism. For Dirac electrons, the 8 rules partition as 1–7–8: graph topology  $\rightarrow$  angular momentum conservation  $\rightarrow$  Dirac spinor phase constraint (zero additional cost). The VP projection constant  $C$  does *not* serve as a universal graph-to-continuum projector:  $C \times F_2$  diverges from  $\alpha/(2\pi)$ , confirming that different diagram topologies require independent projection factors. Comprehensive transcendental tables for  $D(s)$ ,  $D_{\text{even}}(s)$ ,  $D_{\text{odd}}(s)$ , and the one-loop self-energy are presented, all backed by the test suite.

## 1 Introduction

The GeoVac framework [1, 2] computes quantum-mechanical observables from the spectral data of graph Laplacians on compact manifolds. On the three-sphere  $S^3$ , the Dirac operator has a discrete spectrum with known eigenvalues and degeneracies [12], and the framework produces structurally sparse qubit Hamiltonians [2] with  $O(Q^{2.5})$  Pauli scaling.

Paper 18 [3] introduced an exchange-constant taxonomy that classifies where transcendental numbers enter quantum-mechanical computations: intrinsic (from angular structure), calibration (from the  $S^3$  projection), and embedding (from the electron–electron cusp). The present paper extends this taxonomy to perturbative QED by computing spectral sums that probe the interaction structure of quantum electrodynamics on  $S^3$ .

All calculations in the paper use four modules in the `geovac` codebase: `qed_vacuum_polarization` (one-loop heat kernel), `qed_vertex` (two-loop sunset with vertex parity), `qed_self_energy` (one-loop self-energy and vertex correction), and `qed_three_loop` (three-loop chain topology with  $O(N^3)$  factorization). Every stated numerical value is backed by a test in `tests/test_qed*.py` (the four named modules’ suites collect 150 tests; the full glob collects 645 as of 2026-07-04 — all passing). Two scope notes (2026-07-04): the Sommerfeld product-survival closed forms  $D_4$ – $D_6$  are pinned at value level in `tests/test_paper28_gap_closure_pins.py` (their Euler-sum derivation pins are a named follow-up), and the flat-space consistency checks are frozen in `tests/test_qed_flat_limit.py`.

The central finding is that the  $S^3$  spectral geometry produces the *same* transcendental constants that appear in flat-space perturbative QED— $\zeta(3)$ ,  $\zeta(5)$ , Catalan’s  $G$ , Dirichlet  $\beta(4)$ —through a purely geometric mechanism (operator order plus vertex topology), together with multi-loop

constants that climb the depth filtration of the classical level-2 (alternating) MZV ring: the two-loop  $S_{\min}$  is identified in closed form in  $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$  (Sec. 9).

## 2 Spectral Data on $S^3$

### 2.1 Dirac spectrum

On the unit three-sphere, the Dirac operator has eigenvalues and degeneracies [12]

$$|\lambda_n| = n + \frac{3}{2}, \quad g_n = 2(n+1)(n+2), \quad n = 0, 1, 2, \dots \quad (1)$$

The Dirac Dirichlet series is  $D(s) = \sum_{n=0}^{\infty} g_n / |\lambda_n|^s$ . Via Hurwitz zeta,

$$D(s) = 2\zeta(s-2, \frac{3}{2}) - \frac{1}{2}\zeta(s, \frac{3}{2}). \quad (2)$$

Using  $\zeta(s, 3/2) = (2^s - 1)\zeta_R(s) - 2^s$ , this becomes

$$D(s) = 2(2^{s-2} - 1)\zeta_R(s-2) - \frac{1}{2}(2^s - 1)\zeta_R(s). \quad (3)$$

We define the even- $n$  and odd- $n$  sub-sums:

$$D_{\text{even}}(s) = \sum_{\substack{n=0 \\ n \text{ even}}}^{\infty} \frac{g_n}{|\lambda_n|^s}, \quad D_{\text{odd}}(s) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{g_n}{|\lambda_n|^s}, \quad (4)$$

so that  $D(s) = D_{\text{even}}(s) + D_{\text{odd}}(s)$ .

### 2.2 Hodge-1 (photon) spectrum

The transverse vector Laplacian (Hodge-1 operator) on  $S^3$  has eigenvalues and degeneracies

$$\mu_q = q(q+2), \quad d_q^T = q(q+2), \quad q = 1, 2, 3, \dots \quad (5)$$

(Labeling-convention note, 2026-07-04: this is the framework's  $q$ -labeling as implemented and documented in `geovac/hodge1_s3.py`; the literature's coclosed/transverse family — Ikeda–Taniguchi 1978, Camporesi 1990 — is indexed so that the eigenvalue reads  $(k+1)^2$  with multiplicity  $2k(k+2)$ . Reconciling the two labelings at the operator level is the named `hodge1/P28` follow-up, deferred as its own item; the module's anchor test documents the divergence.) The Ricci shift +2 relative to the scalar Laplacian eigenvalues  $\ell(\ell+2)$  follows from the Bochner–Weitzenböck identity  $\Delta_1 = \nabla^* \nabla + \text{Ric}$  on  $S^3$  (Ricci scalar  $R = 6$  on unit  $S^3$ , giving  $\text{Ric} = 2g$ ). There are no zero modes ( $\beta_1 = 0$  for  $S^3$ ).

### 2.3 Vertex selection rule

The QED vertex on  $S^3$  couples spinor harmonics at levels  $n_1, n_2$  to a vector harmonic at level  $q$  with the  $\text{SO}(4)$  selection rule:

$$|n_1 - n_2| \leq q \leq n_1 + n_2, \quad n_1 + n_2 + q \text{ is odd}. \quad (6)$$

The parity constraint arises from the  $\gamma^\mu$  coupling, which flips the chirality representation of  $\text{SO}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$ .

The  $\text{SO}(4)$  Clebsch–Gordan channel count  $W(n_1, n_2, q) \in \{0, 1, 2\}$  weights each allowed triple by the number of independent coupling channels (the two components of the transverse vector harmonic under  $\text{SU}(2)_L \times \text{SU}(2)_R$ ).

### 3 Theorem 1: Squared Dirac Spectral Zeta (T9)

**Theorem 1** (T9:  $D^2$  spectral zeta is  $\pi^{\text{even}}$ ). *The spectral zeta function of the squared Dirac operator on unit  $S^3$  satisfies*

$$\zeta_{D^2}(s) = 2^{2s-1} [\lambda(2s-2) - \lambda(2s)], \quad (7)$$

where  $\lambda(2k) = (1 - 2^{-2k}) \zeta_R(2k) = \text{rational} \times \pi^{2k}$ . At every integer  $s \geq 1$ ,  $\zeta_{D^2}(s)$  is a two-term polynomial in  $\pi^2$  with rational coefficients. No odd-zeta content appears at any  $s$ .

*Proof.* The squared Dirac eigenvalues are  $\lambda_n^2 = (n + 3/2)^2 = (2n + 3)^2/4$ , so  $\zeta_{D^2}(s) = \sum_{n \geq 0} g_n \cdot (4/(2n + 3)^2)^s = 4^s \sum_{n \geq 0} 2(n + 1)(n + 2)/(2n + 3)^{2s}$ . Substituting  $m = 2n + 3$  (which runs over odd integers  $\geq 3$ ), the sum becomes  $\sum_{m \geq 3, m \text{ odd}} (m^2 - 1)/2 \cdot 4^s/m^{2s}$ . The  $(m^2 - 1)/2$  factor splits as  $(m^{2-2s} - m^{-2s})/2$ , giving two Dirichlet-type sums over odd integers. Each odd-integer sum is  $\lambda(2k) = (1 - 2^{-2k}) \zeta_R(2k)$  with the first three odd values ( $m = 1$ ) added or subtracted to correct the lower limit. By the Euler–Bernoulli formula,  $\zeta_R(2k) = (-1)^{k+1} (2\pi)^{2k} B_{2k}/(2 \cdot (2k)!)$ , so  $\lambda(2k) = \text{rational} \times \pi^{2k}$ . Odd-zeta values  $\zeta_R(2k + 1)$  never appear because the squared spectrum converts every odd power to an even power.  $\square$

**Verification:** Symbolic identity confirmed in `sympy` for  $s = 1, \dots, 10$ . Numerical agreement to  $10^{-60}$  at  $s = 2, \dots, 20$ .

Table 1: Squared Dirac spectral zeta  $\zeta_{D^2}(s)$ .

| $s$ | $\zeta_{D^2}(s)$           |
|-----|----------------------------|
| 1   | $-\pi^2/4$                 |
| 2   | $\pi^2 - \pi^4/12$         |
| 3   | $\pi^4/3 - \pi^6/30$       |
| 4   | $2\pi^6/15 - 17\pi^8/1260$ |

**Remark 1** ( $\sqrt{\pi}$  slot at half-integer  $s$  vs integer  $s$ : clarification (Sprint Mixed-Tate Test, June 2026)). *The two-term ring  $\sqrt{\pi} \cdot \mathbb{Q} \oplus \pi^2 \cdot \mathbb{Q}$  that arises in heat-kernel-side analyses of  $\zeta_{D^2}$  is populated only at half-integer  $s$  ( $s = 1/2, 3/2, \dots$ ), corresponding to the leading heat-trace poles where the spinor-fiber-dimension factor  $\sqrt{\pi}$  enters before the standard  $(4\pi)^{-d/2}$  volume normalization. At integer  $s \geq 1$  the  $\sqrt{\pi}$  slot is exactly empty (Table 1 above), and the values are pure  $\pi^{2k} \cdot \mathbb{Q}$ . After the standard Connes–Chamseddine volume normalization (multiplication by  $(4\pi)^{d/2}$  with  $d = 3$ ), the heat-trace coefficients also cleanly enter  $\mathbb{Q}[\pi^2]$ : Paper 51 [8] Cor. 2.1 gives  $a_0^{D^2} = 4\pi^2$ ,  $a_1^{D^2} = -2\pi^2$ ,  $a_k = 0$  for  $k \geq 2$ . The GeoVac  $S^3$  M2 sub-mechanism is therefore mixed-Tate over  $\mathbb{Q}$  in the  $F$ – $M$  sense (Paper 32 [7] § VIII Corollary), in the strictly smaller pure-Tate sub-ring  $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$ .*

**Remark 2** (No functional equation for  $\zeta_{D^2}$ : the quadratic multiplicity is the obstruction (Sprint GB-2, 2026-05-29)). *Sprint RH-O (Apr. 2026) reported, by a 13,080-template numerical search, that the Dirac spectral zeta  $D(s)$  admits no spectral-triple functional equation (best residual 0.594 vs. classical anchors at  $10^{-51}$ ). The closed form (7) upgrades this to an analytic statement and pinpoints the mechanism. Since  $\zeta_{D^2}(s) = D(2s)$ , both  $\zeta_{D^2}(s)$  and its reflection  $\zeta_{D^2}(c-s)$  lie in the two-dimensional space  $\text{span}\{\zeta_R(2s), \zeta_R(2s-2)\}$  over the field of elementary (gamma/exponential/trigonometric) functions; a functional equation  $\zeta_{D^2}(s) = \chi(s) \zeta_{D^2}(c-s)$  with  $\chi$  elementary exists iff the two coefficient vectors are parallel, i.e.  $\det = AD - BC = 0$ .*

*The Riemann reflection  $z \mapsto 1 - z$  matches the argument set  $\{2s-2, 2s\}$  to its reflected partner only at the natural axis  $c = 3/2$  (equivalently  $c = 1/2$ ), the Dirac ground-state value  $|\lambda_0| = 3/2$  —*

so the axis is not the problem. The obstruction is the multiplier: the two terms acquire the gamma ratio  $\Gamma(2s)/\Gamma(2s-2) = (2s-1)(2s-2)$ , which forces  $\det \neq 0$  (verified  $|\det| = 2.8, 8.4, 60.3, 8.2$  at  $s = 1+i, \frac{3}{2}+3i, 2+5i, \frac{1}{2}+10i$ ; the candidate-multiplier ratio  $\chi_1/\chi_2$  tracks  $(2s-1)(2s-2)$ ). The exponent gap of 2 between the two  $\lambda$ -terms is produced precisely by the quadratic Weyl multiplicity  $g_n = 2(n+1)(n+2)$ : a constant-multiplicity (single-power) spectrum lies in a one-dimensional span, is trivially parallel, and does inherit a functional equation. The reflection algebra is cross-checked against direct evaluation of (7) to  $10^{-41}$ .

Consequence (unification of two RH walls). The “wrong Weyl law” wall (RH-N:  $\gamma_n \sim \sqrt{n}$ , harmonic-oscillator density) and the “no functional equation” wall (RH-O) are the same structural fact: both are consequences of the  $S^3$  quadratic degeneracy. The very multiplicity  $n^2$  that places the framework’s even- $s$  values on the Bernoulli ladder (Paper 32’s Bernoulli-ladder remark:  $F = \zeta(2) = \pi^2 B_2$ , etc.) is the multiplicity that denies  $\zeta_{D^2}$  a critical line. This is consistent with — and sharpens — WH6: the framework’s zeta carries  $\zeta_R$ ’s special values but not its zero-controlling functional equation. Source: `debug/sprint_gb2_zeta_dsq_functional_eq.py`, `debug/sprint_gb2_zeta_dsq_memo.md`.

## 4 Theorem 2: Even/Odd Parity Discriminant

**Theorem 2** (Parity discriminant). *At even integer  $s$ , the Dirac Dirichlet series  $D(s)$  is purely  $\pi^{\text{even}}$ . At odd integer  $s$ ,  $D(s)$  involves  $\zeta_R(s-2)$  and  $\zeta_R(s)$  (odd-zeta values). Operator order (2nd vs 1st) is the primary transcendental discriminant: second-order operators produce  $\pi^{\text{even}}$  exclusively; first-order operators produce odd-zeta values at odd  $s$ .*

*Proof.* From Eq. (3),  $D(s) = 2(2^{s-2} - 1)\zeta_R(s-2) - \frac{1}{2}(2^s - 1)\zeta_R(s)$ . At even  $s$ , both  $\zeta_R(s-2)$  and  $\zeta_R(s)$  are Bernoulli-type:  $\zeta_R(2k) = (-1)^{k+1}(2\pi)^{2k}B_{2k}/(2 \cdot (2k)!) = \text{rational} \times \pi^{2k}$ . At odd  $s$ , both  $\zeta_R(s-2)$  and  $\zeta_R(s)$  are genuine odd-zeta values ( $\zeta(3), \zeta(5), \dots$ ) that are  $\mathbb{Q}$ -linearly independent of all  $\pi^{2k}$  by Apéry’s theorem and its extensions [27, 28].  $\square$

Table 2: Dirac Dirichlet series  $D(s)$ : PSLQ identifications at 80 dps.

| $s$ | $D(s)$                                | Class               |
|-----|---------------------------------------|---------------------|
| 4   | $\pi^2 - \pi^4/12$                    | $\pi^{\text{even}}$ |
| 5   | $14\zeta(3) - \frac{31}{2}\zeta(5)$   | odd-zeta            |
| 6   | $\pi^4/3 - \pi^6/30$                  | $\pi^{\text{even}}$ |
| 7   | $62\zeta(5) - \frac{127}{2}\zeta(7)$  | odd-zeta            |
| 8   | $2\pi^6/15 - 17\pi^8/1260$            | $\pi^{\text{even}}$ |
| 9   | $254\zeta(7) - \frac{511}{2}\zeta(9)$ | odd-zeta            |
| 10  | $17\pi^8/315 - 31\pi^{10}/4725$       | $\pi^{\text{even}}$ |

The appearance of  $\zeta(3)$  at  $s = 5$  is the  $S^3$  analog of the  $\zeta(3)$  that enters flat-space two-loop QED through the Schwinger–Rosner vertex correction [13].

## 5 Theorem 3: The $\chi_{-4}$ Identity

**Theorem 3** ( $\chi_{-4}$  identity). *For every integer  $s \geq 2$ ,*

$$D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2)), \quad (8)$$

where  $\beta(s) = L(s, \chi_{-4})$  is the Dirichlet beta function.

*Proof.* Express  $D_{\text{even}}(s)$  and  $D_{\text{odd}}(s)$  via Hurwitz zeta at quarter-integer shifts. With  $m = n + 3/2$  and full degeneracy  $g_n = 2(n+1)(n+2) = 2(m^2 - 1/4)$ , the even- $n$  eigenvalues are  $m = 2(2k+3/4)$  and the odd- $n$  ones  $m = 2(2k+5/4)$ ,  $k \geq 0$ , so  $D_{\text{even}}(s) = 2^{-(s-3)}\zeta(s-2, 3/4) - 2^{-(s+1)}\zeta(s, 3/4)$  and  $D_{\text{odd}}(s) = 2^{-(s-3)}\zeta(s-2, 5/4) - 2^{-(s+1)}\zeta(s, 5/4)$ . Two elementary Hurwitz identities:  $\zeta(s, 5/4) = \zeta(s, 1/4) - 4^s$  (shift recurrence) and  $\zeta(s, 3/4) - \zeta(s, 1/4) = -4^s\beta(s)$  (the Dirichlet beta definition). Together,  $\zeta(s, 3/4) - \zeta(s, 5/4) = 4^s(1 - \beta(s))$ ; substituting into the difference,  $D_{\text{even}} - D_{\text{odd}} = 2^{s-1}(1 - \beta(s-2)) - 2^{s-1}(1 - \beta(s)) = 2^{s-1}(\beta(s) - \beta(s-2))$ , which is (8). The Hurwitz decompositions and the closed form are verified bit-exactly against direct spectral sums at  $s \in \{4, 5, 6, 7\}$  to 40 digits.  $\square$

**Verification:** Eq. (8) verified symbolically (**sympy** residual zero) at  $s \in \{2, 3, \dots, 10, 15\}$  and by PSLQ at 100-digit precision at every integer  $s$  in the tested range. At  $s = 4$ ,  $D_{\text{even}}(4) - D_{\text{odd}}(4) = 8(\beta(4) - G)$  (since  $G = \beta(2)$ ), consistent with the explicit decompositions (9)–(10) below.

### 5.1 Vertex-parity decomposition at $s = 4$

The two-loop sunset diagram couples spinor pairs  $(n_1, n_2)$  subject to the vertex parity constraint. The even- $n$  and odd- $n$  sub-sums of  $D(4)$  decompose as:

$$D_{\text{even}}(4) = \frac{\pi^2}{2} - \frac{\pi^4}{24} - 4G + 4\beta(4), \quad (9)$$

$$D_{\text{odd}}(4) = \frac{\pi^2}{2} - \frac{\pi^4}{24} + 4G - 4\beta(4), \quad (10)$$

$$D(4) = \pi^2 - \frac{\pi^4}{12} \quad (\text{Catalan and } \beta(4) \text{ cancel}). \quad (11)$$

Here  $G = \beta(2) = \sum_{n=0}^{\infty} (-1)^n / (2n+1)^2 \approx 0.9159656$  is Catalan's constant and  $\beta(4) \approx 0.9889446$  is the Dirichlet beta function at  $s = 4$ . The mechanism: vertex parity acts as the Dirichlet character  $\chi_{-4}$  on the mode sum, and the quarter-integer Hurwitz shifts expose  $L(s, \chi_{-4})$  in the even/odd difference. Verified by PSLQ at 80-digit precision with basis  $\{1, \pi^2, \pi^4, G, \beta(4)\}$ .

**Corollary 1.** *The  $\chi_{-4}$  identity is a structural realization of the vertex-parity-as- $\chi_{-4}$  mechanism, but does not constrain the zeros of  $\beta(s)$ : a zero of  $\beta$  at  $s = \rho$  propagates to zeros of  $D_{\text{diff}}$  at both  $\rho$  and  $\rho + 2$ , consistent with but not enforcing  $\Re\rho = 1/2$ .*

**Remark 3** (Motivic home of Theorem 3). *Each of  $D_{\text{even}}(s)$ ,  $D_{\text{odd}}(s)$ , and their difference is a period of mixed Tate motives at level 4 over  $\mathcal{O}_4[1/4] = \mathbb{Z}[i, 1/2]$ , by direct application of Deligne's classification at  $N = 4$  and the explicit basis of Glanois (Sprint M3 Cyclotomic Mixed-Tate, June 2026, memo `debug/sprint_m3_cyclotomic_mixed_tate_memo.md`; Paper 32 § VIII Corollary `cor:m3_cyclotomic_mixed_tate`; Paper 18 § III.7). The vertex-parity-as- $\chi_{-4}$  correspondence IS the Deligne–Glanois Galois descent from level 4 to level 2 on the framework's natural QED observables: the un-restricted  $D(s) = D_{\text{even}} + D_{\text{odd}}$  lives in the smaller level- $\leq 2$  ring (Theorem 1 and Theorem 2); the  $\zeta(s, 1/4)$  pieces in  $D_{\text{even}}, D_{\text{odd}}$  individually cancel in the sum but survive in the difference, exposing the level-4 Dirichlet  $L$ -value content. Catalan  $G = \beta(2)$  and  $\beta(4)$  are periods in  $\mathcal{MT}(\mathbb{Z}[i, 1/2])$  at depth 1 but conjecturally not in  $\mathcal{MT}(\mathbb{Z}[1/2])$  at any depth — the level-4 lift is structural to the vertex-parity mechanism.*

### 5.2 Three-axis taxonomy

The transcendental content of any QED spectral sum on  $S^3$  is classified by three independent axes:

1. **Operator order.** Second-order  $(D^2) \rightarrow \pi^{\text{even}}$  (Theorem 1); first-order  $(|D|) \rightarrow \text{odd-zeta}$  (Theorem 2).
2. **Bundle type.** Scalar vs spinor modulates coefficients but not the transcendental class (Paper 18 Sec IV).
3. **Diagram topology.** Vertex parity introduces Dirichlet  $L$ -function characters  $(G, \beta(4))$  that are absent from unrestricted sums (Theorem 3).

## 6 Theorem 4: Self-Energy Structural Zero

The one-loop electron self-energy on  $S^3$  (Feynman gauge) is:

$$\Sigma(n_{\text{ext}}) = \sum_{n_{\text{int}}, q} \frac{W(n_{\text{ext}}, n_{\text{int}}, q) g_{n_{\text{int}}} d_q^T}{|\lambda_{n_{\text{int}}}|^{2s_e} \mu_q^{s_\gamma}}, \quad (12)$$

where the sum is over internal electron mode  $n_{\text{int}}$  and photon mode  $q$  subject to the vertex selection rule (6),  $s_e = 2$  (so  $|\lambda|^4$  in the denominator),  $s_\gamma = 1$ , and  $W$  is the  $\text{SO}(4)$  channel count.

**Theorem 4** (Self-energy structural zero).  $\Sigma(0) = 0$  *exactly*. *The one-loop self-energy vanishes identically at the Dirac ground state  $n_{\text{ext}} = 0$ .*

*Proof.* At  $n_{\text{ext}} = 0$ , the vertex selection rule requires  $|0 - n_{\text{int}}| \leq q \leq 0 + n_{\text{int}}$  and  $0 + n_{\text{int}} + q$  odd. The first condition forces  $q = n_{\text{int}}$ . The second then requires  $2n_{\text{int}}$  to be odd, which is impossible for any integer  $n_{\text{int}}$ . Therefore  $W(0, n_{\text{int}}, q) = 0$  for all  $(n_{\text{int}}, q)$ , and the sum is empty.  $\square$

This is the  $S^3$  analog of the fact that the ground state of a compact system cannot emit into a vacuum mode: the vertex parity constraint is a spectral selection rule, not a dynamical cancellation.

### 6.1 Self-energy for excited states

For  $n_{\text{ext}} \geq 1$ , the self-energy is nonzero and convergent. We compute  $\Sigma(n_{\text{ext}})$  at physical propagator powers ( $s_e = 2$ ,  $s_\gamma = 1$ ) for  $n_{\text{ext}} = 0, 1, \dots, 5$  at cutoff  $n_{\text{max}} = 30$ .

Table 3: One-loop self-energy  $\Sigma(n_{\text{ext}})$  on  $S^3$ ,  $n_{\text{max}} = 30$ .

| $n_{\text{ext}}$ | $\Sigma(n_{\text{ext}})$ |
|------------------|--------------------------|
| 0                | 0 (exact, Theorem 4)     |
| 1                | 1.2854                   |
| 2                | 1.7862                   |
| 3                | 2.0843                   |
| 4                | 2.2884                   |
| 5                | 2.4398                   |

The self-energy increases monotonically with  $n_{\text{ext}}$ , consistent with stronger coupling at higher momentum shells. At  $n_{\text{ext}} = 1$ , PSLQ at 50 dps identifies  $\Sigma(1)$  as containing  $\pi^{\text{even}}$  and Catalan  $G$  components (vertex parity active on the asymmetric coupling to the  $n_{\text{ext}} = 1$  shell).

## 6.2 Vertex correction

The one-loop vertex correction involves two vertex insertions:

$$\Lambda(n_{\text{ext}}) = \sum_{n_{\text{int}}, q_1, q_2} \frac{W(n_{\text{ext}}, n_{\text{int}}, q_1) W(n_{\text{int}}, n_{\text{ext}}, q_2) g_{n_{\text{int}}} d_{q_1}^T d_{q_2}^T}{|\lambda_{n_{\text{int}}}|^{2s_e} \mu_{q_1}^{s_\gamma} \mu_{q_2}^{s_\gamma}}. \quad (13)$$

This is  $O(N^3)$  in the cutoff. Both  $\Lambda(n_{\text{ext}})$  and  $\Sigma(n_{\text{ext}})$  are backed by 35 tests in `tests/test_qed_self_energy.py`. **Note:** The shell-summed vertex correction  $\Lambda(n_{\text{ext}})$  defined above uses the CG-weight channel count  $W$  and is  $m_j$ -independent (it gives  $B = 0$  by the Wigner–Eckart theorem). The anomalous magnetic moment extraction requires the  $m_j$ -resolved three-point function developed in the next section.

## 7 Anomalous Magnetic Moment on $S^3$

### 7.1 Why the two-point vertex correction gives $B = 0$

The two-point vertex correction  $\Lambda(n_{\text{ext}})$  defined in Eq. (13) sums over all internal modes and photon polarizations at *fixed* external quantum numbers  $(n_{\text{ext}}, j_{\text{ext}}, m_j)$ . On  $S^3$ , the vertex amplitude at each junction involves a product of four  $\text{SU}(2)_L \times \text{SU}(2)_R$  Clebsch–Gordan coefficients. When the photon polarizations  $(m_{g,L}, m_{g,R})$  are summed over at both vertices, the  $\text{SO}(4)$  Wigner–Eckart theorem forces the result to be independent of  $m_j$ :

$$\Lambda(n_{\text{ext}}, j, m_j = +\tfrac{1}{2}) = \Lambda(n_{\text{ext}}, j, m_j = -\tfrac{1}{2}). \quad (14)$$

Therefore the spin-dependent piece  $B_{2\text{pt}} = \Lambda(+\tfrac{1}{2}) - \Lambda(-\tfrac{1}{2}) = 0$  identically. The two-point vertex correction is a scalar under  $\text{SO}(4)$  rotations and cannot distinguish spin orientations. To extract the anomalous magnetic moment ( $F_2$  form factor), an external probe must break the rotational symmetry.

### 7.2 Three-point vertex correction with probe photon

The anomalous magnetic moment is extracted from a three-point function in which an external probe photon of mode  $q_{\text{probe}} = 1$  (the lowest transverse vector harmonic, the  $S^3$  analog of a uniform magnetic field) is inserted between the two loop-photon vertices. The physical topology is:

$$\langle \text{ext}, m_j | \Gamma_{\text{probe}}^{1\text{-loop}} | \text{ext}, m_j \rangle = \sum_{\substack{n_{\text{int}}, j_{\text{int}} \\ m_{j,\text{int}}, m'_{j,\text{int}} \\ q_{\text{loop}}}} \frac{V_1 \cdot V_{\text{probe}} \cdot V_3}{|\lambda_{n_{\text{int}}}|^4 \mu_{q_{\text{loop}}}}, \quad (15)$$

where:

- $V_1$ : amplitude for the external electron ( $m_j$ ) to emit a loop photon and transition to the internal state  $(m_{j,\text{int}})$ ;
- $V_{\text{probe}}$ : amplitude for the probe photon at  $q = 1$  to scatter the internal electron from  $m_{j,\text{int}}$  to  $m'_{j,\text{int}}$ , summed over probe polarizations;
- $V_3$ : amplitude for the internal state  $(m'_{j,\text{int}})$  to reabsorb the loop photon and return to the external state  $(m_j)$ .



Each vertex amplitude is a product of four Clebsch–Gordan coefficients coupling the  $SU(2)_L \times SU(2)_R$  representations of source, target, and photon:

$$V = \sum_{m_{L,1}} C_{j_{s,L} j_{s,R}}^{j_s}(m_{L,1}, m_j - m_{L,1}; m_j) \cdot C_{j_{t,L} j_{t,R}}^{j_t}(m_{L,2}, m_{R,2}; m'_j) \cdot C_{j_{s,L} j_{g,L}}^{j_{t,L}}(m_{L,1}, m_{g,L}; m_{L,2}) \cdot C_{j_{s,R} j_{g,R}}^{j_{t,R}}(m_{R,1}, m_{g,R}; m_{R,2}) \quad (16)$$

where  $m_{L,2} = m_{L,1} + m_{g,L}$  and  $m_{R,2} = m_{R,1} + m_{g,R}$ , and the sum runs over all  $m_{L,1} \in \{-j_{s,L}, \dots, +j_{s,L}\}$ .

The probe photon at  $q = 1$  has channels  $(j_{g,L}, j_{g,R}) = (1, 0)$  or  $(0, 1)$ , corresponding to the two components of the transverse vector harmonic. The probe couples internal states  $m_{j,\text{int}} \rightarrow m'_{j,\text{int}}$ , enabling spin-dependent amplitudes that the polarization-summed two-point function cannot access.

The spin-dependent piece of the vertex correction is:

$$B = \Gamma_{\text{probe}}^{1-\text{loop}}(m_j = +\frac{1}{2}) - \Gamma_{\text{probe}}^{1-\text{loop}}(m_j = -\frac{1}{2}). \quad (17)$$

### 7.3 Tree-level normalization

The tree-level (zeroth-order) coupling of the external electron to the  $q = 1$  probe photon provides the normalization for  $F_2$  extraction. At the external state  $n_{\text{ext}} = 1$ ,  $j = 1/2$  (Camporesi–Higuchi level  $n_{\text{CH}} = 1$ , with  $j_L = 1$ ,  $j_R = 1/2$ ):

$$V_{\text{tree}}(m_j = +\frac{1}{2}) = -V_{\text{tree}}(m_j = -\frac{1}{2}) = 0.2790, \quad (18)$$

$$V_{\text{magnetic}} = V_{\text{tree}}(+\frac{1}{2}) - V_{\text{tree}}(-\frac{1}{2}) = 0.5579. \quad (19)$$

The charge piece  $[V_{\text{tree}}(+\frac{1}{2}) + V_{\text{tree}}(-\frac{1}{2})]/2 = 0$  vanishes—the  $q = 1$  probe photon couples purely to the magnetic moment at tree level, as expected for the  $S^3$  analog of a uniform magnetic field.

Analytically,  $V_{\text{magnetic}} = 2(\sqrt{2}/3 - \sqrt{3}/9)$ , a pure algebraic number from the Clebsch–Gordan coefficients.

The Pauli form factor is then:

$$F_2 = \frac{B}{V_{\text{magnetic}}}, \quad (20)$$

and the anomalous magnetic moment on  $S^3$  is  $a_e^{S^3} = F_2$ .

### 7.4 Convergence and results

The three-point vertex correction converges rapidly. Table 4 shows the per-level contributions for the external state at  $n_{\text{ext}} = 1$ .

Three structural features are visible:

1. **Rapid convergence.** The  $n_{\text{int}} = 1$  level contributes 99% of the final value. Each subsequent allowed level pair contributes roughly  $100\times$  less than the previous one.
2. **Vertex parity filtering.** The  $n_{\text{int}} = 0$  contribution vanishes by Theorem 4 (the loop-photon vertex at  $n_{\text{ext}} = 1$ ,  $n_{\text{int}} = 0$  requires  $q_{\text{loop}} = 1$  and  $1 + 0 + 1 = 2$  is even, forbidden). The dominant contribution comes from  $n_{\text{int}} = 1$  (the nearest allowed level to the external state). Higher even- $n_{\text{int}}$  levels contribute but are suppressed relative to odd- $n_{\text{int}}$  neighbors.
3. **Curvature correction.** The ratio  $F_2/[\alpha/(2\pi)]$  converges to 1.084, not to 1.

Table 4: Per-level contributions to the anomalous magnetic moment on  $S^3$ . External state  $n_{\text{ext}} = 1$ ,  $j = 1/2$ . The  $n_{\text{int}} = 0$  contribution vanishes by the self-energy structural zero (Theorem 4). Contributions drop by approximately  $100\times$  per successive allowed level pair.

| $n_{\text{int}}$ | $\Delta B$            | $B_{\text{cumul}}$     | $F_2/[\alpha/(2\pi)]$ | Status                          |
|------------------|-----------------------|------------------------|-----------------------|---------------------------------|
| 0                | 0                     | 0                      | —                     | $\Sigma(0) = 0$ structural zero |
| 1                | $6.96 \times 10^{-4}$ | $6.96 \times 10^{-4}$  | 1.074                 | 99% of final                    |
| 2                | $3.76 \times 10^{-6}$ | $7.00 \times 10^{-4}$  | 1.080                 |                                 |
| 3                | $2.85 \times 10^{-6}$ | $7.02 \times 10^{-4}$  | 1.084                 |                                 |
| 4                | $1.34 \times 10^{-7}$ | $7.025 \times 10^{-4}$ | 1.084                 |                                 |
| 5                | $1.25 \times 10^{-7}$ | $7.027 \times 10^{-4}$ | 1.084                 |                                 |
| 6                | $1.25 \times 10^{-8}$ | $7.027 \times 10^{-4}$ | 1.084                 |                                 |
| 7                | $1.36 \times 10^{-8}$ | $7.027 \times 10^{-4}$ | 1.084                 |                                 |
| 8                | $2.00 \times 10^{-9}$ | $7.027 \times 10^{-4}$ | 1.084                 |                                 |

## 7.5 Headline result

At  $n_{\text{max}} = 12$  (converged to 5 significant figures; the value is already stable at  $n_{\text{max}} = 8$ ):

$$\frac{F_2}{\alpha/(2\pi)} = 1.084, \quad (21)$$

with  $B = 7.027 \times 10^{-4}$  and  $V_{\text{magnetic}} = 0.5579$ .

The anomalous magnetic moment on unit  $S^3$  exceeds the flat-space Schwinger value by approximately 8.4%. This excess is a curvature correction: on a three-sphere of radius  $R$ , the electron propagator samples a compact geometry where the shortest-wavelength virtual photon modes are modified relative to flat space. In the flat-space limit  $R \rightarrow \infty$ , the curvature correction vanishes and  $F_2/[\alpha/(2\pi)] \rightarrow 1$ , recovering the Schwinger result [10] exactly.

## 7.6 Mode dependence

The anomalous magnetic moment depends strongly on the external electron mode. Table 5 shows  $F_2/[\alpha/(2\pi)]$  computed at  $n_{\text{ext}} = 1, \dots, 5$ .

Table 5: Mode-dependent anomalous magnetic moment on unit  $S^3$ . Each row reports the tree-level magnetic coupling  $V_{\text{mag}}$ , the one-loop correction  $B$ , and the normalized form factor ratio. The  $n_{\text{ext}} = 1$  result uses  $n_{\text{max}} = 12$  (converged to 5 digits);  $n_{\text{ext}} = 2-5$  use  $n_{\text{max}} = 5$  (converged to 3 digits via  $n^{-7}$  power-law tail). Exact algebraic forms for  $V_{\text{mag}}$  and the dominant  $B(n_{\text{int}} = 1)$  are available at each  $n_{\text{ext}}$ ; all expressions are sums of square roots with rational coefficients (no transcendentals).

| $n_{\text{ext}}$ | $\lambda_{\text{ext}}$ | $V_{\text{mag}}$ | $B$                    | $F_2/[\alpha/(2\pi)]$ | vs. Schwinger |
|------------------|------------------------|------------------|------------------------|-----------------------|---------------|
| 1                | 5/2                    | 0.5579           | $7.027 \times 10^{-4}$ | 1.084                 | +8.4%         |
| 2                | 7/2                    | 0.3893           | $2.772 \times 10^{-4}$ | 0.613                 | −39%          |
| 3                | 9/2                    | 0.3001           | $1.296 \times 10^{-4}$ | 0.370                 | −63%          |
| 4                | 11/2                   | 0.2445           | $6.940 \times 10^{-5}$ | 0.244                 | −76%          |
| 5                | 13/2                   | 0.2064           | $4.139 \times 10^{-5}$ | 0.173                 | −83%          |

The physical mechanism is that the loop correction  $B$  drops faster with  $n_{\text{ext}}$  than the tree-level coupling  $V_{\text{mag}}$ . Across the five modes,  $B$  spans a factor of  $17\times$  while  $V_{\text{mag}}$  spans only  $2.7\times$ . The tree-level amplitude follows  $V_{\text{mag}} \rightarrow (4/3)/|\lambda|$  asymptotically (Richardson-extrapolated to 4-digit agreement with  $4/3$ ), while the loop correction decays as a steeper power of  $|\lambda|$ . Higher-mode external electrons are “heavier” relative to the curvature scale ( $\lambda = n + 3/2$  sets the effective mass), and the loop integral is progressively suppressed.

**Comparison with Parker–Toms.** The leading curvature correction to the anomalous magnetic moment in a general curved spacetime is [11]:

$$\frac{F_2}{\alpha/(2\pi)} = 1 + \frac{R_{\text{scalar}}}{12|\lambda|^2} + O(|\lambda|^{-4}), \quad (22)$$

where  $R_{\text{scalar}} = 6$  on unit  $S^3$  and  $|\lambda|$  is the Dirac eigenvalue of the external state. At  $n_{\text{ext}} = 1$  ( $|\lambda| = 5/2$ ), this predicts  $1 + 6/(12 \cdot 25/4) = 1 + 2/25 = 1.080$ , which matches the computed 1.084 within 0.4%. The expansion parameter is  $\xi = R_{\text{scalar}}/(12|\lambda|^2) \approx 0.08$  at  $n_{\text{ext}} = 1$ , small enough for the leading term to dominate.

However, the perturbative expansion in  $1/|\lambda|^2$  diverges badly for the full mode-dependent curve: a two-term fit gives  $c_1 \approx -20$ , far from the Parker–Toms  $c_1 = 1/2$ . The discrepancy arises because the expansion (22) applies to a *single external mode at fixed curvature*, treating  $R/\lambda^2$  as the small parameter. When we vary  $n_{\text{ext}}$  at fixed  $R = 1$ , we change not only  $|\lambda|$  but also the angular structure of *every* CG coefficient in the vertex amplitude and the internal sum. The resulting mode dependence is non-perturbative in  $1/|\lambda|^2$ .

The five computed values are well described by a power law:

$$\frac{F_2}{\alpha/(2\pi)} = (5.91 \pm 0.36) |\lambda|^{-1.84 \pm 0.06}. \quad (23)$$

The fitted exponent is  $2.8\sigma$  from  $-2$ ; the forced  $p = -2$  fit has twice the RMS residual. Analysis of the dominant internal-mode contribution ( $n_{\text{int}} = 1$ ) shows its local log-log slope converging toward  $-2$  ( $-2.57 \rightarrow -2.26 \rightarrow -2.16 \rightarrow -2.11$  across  $n_{\text{ext}} = 1, \dots, 5$ ), but higher internal modes contribute an increasing fraction at larger  $n_{\text{ext}}$  (99% at  $n_{\text{ext}} = 1$  vs. 69% at  $n_{\text{ext}} = 3$ ), pulling the effective exponent above  $-2$ . Whether the true asymptotic exponent is exactly  $-2$  (with finite-curvature corrections) or a distinct value remains an open question.

The power law implies  $F_2 \rightarrow 0$  as  $|\lambda| \rightarrow \infty$  on unit  $S^3$ . This is not inconsistent with the flat-space Schwinger limit: the flat-space limit is  $R \rightarrow \infty$  at fixed physical mass (requiring  $n \sim mR \rightarrow \infty$  simultaneously), not  $|\lambda| \rightarrow \infty$  on a fixed-size sphere. Taking  $|\lambda|$  large at fixed  $R = 1$  corresponds to a heavy particle on a small sphere, where form-factor suppression is expected.

**Verification:** The computation uses exact `sympy` Clebsch–Gordan coefficients at every vertex, with no numerical approximation in the angular algebra. The propagator denominators are exact rationals ( $|\lambda_n|^4$  and  $\mu_q$  from Eqs. (1) and (5)). Computation scripts: `debug/alpha_g_minus_2.vertex.py`, `debug/alpha_g_minus_2.curvature_fit.py`.

## 7.7 Curvature expansion coefficients

The Parker–Toms expansion (22) at  $n_{\text{ext}} = 1$  provides a high-precision extraction of the curvature coefficients. Writing

$$\frac{F_2}{\alpha/(2\pi)} = 1 + c_1 x + c_2 x^2 + c_3 x^3 + \dots, \quad x \equiv \frac{1}{|\lambda|^2} = \frac{4}{25} \quad (24)$$

at  $n_{\text{ext}} = 1$  ( $|\lambda| = 5/2$ ), the leading coefficient  $c_1 = R_{\text{scalar}}/12 = 1/2$  follows from the Parker–Toms theorem [11] for a Dirac field on unit  $S^3$ .

The converged  $n_{\text{ext}} = 1$  spectral sum ( $n_{\text{int}} = 0 \dots 25$  with power-law tail correction, 13 significant figures) gives a corrected ratio of 1.0844529226552, hence  $\delta \equiv F_2/[\alpha/(2\pi)] - 1 = 0.0844529226552$ . Subtracting the leading term  $c_1 x = 0.08$  and dividing by  $x^2$ :

$$c_2^{\text{apparent}} = \frac{\delta - c_1 x}{x^2} = 0.173942291220598 \pm 10^{-12}. \quad (25)$$

By the T9 theorem (Theorem 1), one-loop quantities on  $S^3$  are polynomials in  $\pi^2$  with rational coefficients; hence  $c_2 = a + b\pi^2$  for rationals  $a, b$ .

PSLQ identification against the Paper 2 spectral invariants  $B = 42$ ,  $F = \pi^2/6$ ,  $\Delta = 1/40$  [9] yields

$$c_2 = \frac{2 - B\Delta - F\Delta - F/B}{5} = \frac{19}{100} - \frac{41\pi^2}{25200}, \quad (26)$$

numerically 0.173942310300. The match to (25) is 8 digits (residual  $1.9 \times 10^{-8}$ ), compared with 5.5 digits for the next-best candidate  $(6\pi^2 - 58)/7$  and 3 digits for the purely rational  $25/144$ .

**Structural decomposition.** The three Paper 2 invariants enter (26) through three bilinear combinations:

$$B\Delta = \frac{42}{40} = \frac{21}{20}, \quad F\Delta = \frac{\pi^2}{240}, \quad \frac{F}{B} = \frac{\pi^2}{252}. \quad (27)$$

The numerator  $2 - B\Delta - F\Delta - F/B = \frac{19}{20} - \frac{41\pi^2}{5040}$  is rational plus a single  $\pi^2$  term, consistent with the T9 constraint. The denominator 5 is a simple integer.

**High-precision  $c_3$  extraction.** Extending the  $n_{\text{ext}} = 1$  spectral sum to  $n_{\text{int}} = 50$  with interleaved even/odd power-law tail correction (Hurwitz zeta analytic tail;  $p_{\text{even}} \approx 7.62$ ,  $p_{\text{odd}} \approx 7.71$ ) gives a converged ratio of 1.0844529207082, hence  $\delta = 0.0844529207082$ . The residual after subtracting  $c_1 x + c_2 x^2$  is  $-2.44 \times 10^{-9}$ , yielding

$$c_3 = \frac{\delta - c_1 x - c_2 x^2}{x^3} = -5.946(3) \times 10^{-7}, \quad (28)$$

nonzero at  $200.9\sigma$  (uncertainty from the tail-correction power-law fit). The convergence is slow: the estimate drifts by  $\sim 10^{-8}$  per five internal levels at  $n_{\text{int}} = 50$ , so the true asymptote lies near  $-5.95$  to  $-5.97 \times 10^{-7}$ . The ratio  $|c_3 x^3|/|c_2 x^2| \approx 5.4 \times 10^{-6}$  confirms the expansion is rapidly converging but *does not* terminate at second order. PSLQ identification against the T9 basis  $\{1, \pi^2, \pi^4, B\Delta, F\Delta, F/B, c_2, c_2^2, c_1 c_2, \Delta, \Delta^2, B\Delta^2, F\Delta^2, F^2/B\}$  at  $n_{\text{int}} = 50$  finds no relation with integer coefficients bounded by  $10^4$ ;  $c_3$  is not expressible in these invariants at current precision.

**Multi- $n_{\text{ext}}$  obstruction.** The curvature coefficients in (24) are defined for the  $j = 1/2$  sector of the external Dirac mode. At  $n_{\text{ext}} \geq 2$ , the Dirac level carries multiple total angular momentum sectors ( $j = 1/2, 3/2, \dots, n_{\text{ext}} + 1/2$ ) with distinct form factors. The  $j = 1/2$  component is a  $2/[(n_{\text{ext}} + 1)(n_{\text{ext}} + 2)]$  fraction of the total degeneracy, shrinking as  $\sim 1/n^2$ ; extracting universal curvature coefficients from a full trace requires Wigner–Eckart decomposition into multipole form factors, which is not yet implemented. Consequently, the high-precision extraction is currently restricted to  $n_{\text{ext}} = 1$ , where  $j = 1/2$  is the unique sector.

**Paper 2 invariants in  $c_2$ : a numerical coincidence, not a derivation.** Since  $\alpha$  cancels in the ratio  $F_2/[\alpha/(2\pi)]$ ,  $c_2$  is a pure geometric invariant of the Dirac spectral sum on  $S^3$ , and the T9 theorem (Theorem 1) constrains it to the form  $a + b\pi^2$  with rationals  $a, b$ . The PSLQ identification

(Eq. 26) writes  $a$  and  $b$  as a small-integer combination of  $B$ ,  $F$ ,  $\Delta$ , but this should be read with caution.

First, the form  $(c_0 - c_1 B \Delta - c_2 F \Delta - c_3 F/B)/c_4$  has more free small-integer coefficients than the two real numbers  $(a, b)$  it explains. Finding *some* small-integer combination matching  $a + b\pi^2$  to high precision is expected by chance at this search size; the cross-invariant beats the next-best PSLQ candidate  $(6\pi^2 - 58)/7$  by roughly 2.4 digits, which is suggestive but not conclusive.

Second, and decisively: the *same* form does not predict  $c_3$ . Eq. (28) measures  $c_3 = -5.946(3) \times 10^{-7}$  at  $200\sigma$ , and the extended T9 basis  $\{1, \pi^2, \pi^4, B\Delta, F\Delta, F/B, c_2, c_2^2, c_1 c_2, \Delta, \Delta^2, B\Delta^2, F\Delta^2, F^2/B\}$  admits no PSLQ relation at integer coefficients bounded by  $10^4$ . A genuine structural identity should extend. This single-term failure is consistent with  $c_2$ 's identification being a numerical coincidence between two natural Tier-A spectral invariants of the same manifold, both built from  $|\lambda_n|$  and  $g_n$ , rather than evidence of shared structural origin.

A genuine bridge between the Paper 2 invariants and the curvature expansion would require deriving  $c_2$  (and ideally  $c_3$ ) from the Schwinger–DeWitt heat-kernel expansion of the spinor on  $S^3$ , with  $B$ ,  $F$ ,  $\Delta$  appearing as identifiable terms in the expansion rather than via post-hoc PSLQ matching. This derivation is not done in this work and is flagged as an open question.

Computation scripts: `debug/g2_c2_verification.py`, `debug/g2_c2_diagnostic.py`.

## 8 One-Loop Vacuum Structure

The one-loop QED effective action involves traces of  $D^2$ . The Seeley–DeWitt heat kernel expansion on unit  $S^3$  gives

$$a_0 = a_1 = \sqrt{\pi}, \quad a_2 = \sqrt{\pi}/8 \quad (29)$$

(exact in `sympy`, computed from the Dirac spectral data). The vacuum polarization coefficient and QED beta function follow:

$$\Pi = \frac{1}{48\pi^2}, \quad \beta(\alpha) = \frac{2\alpha^2}{3\pi}. \quad (30)$$

Both reproduce the standard flat-space one-loop QED results from  $S^3$  spectral data alone, confirming the flat-space limit of the spectral framework at one loop.

### 8.1 SD two-term exactness for the Dirac heat kernel

The Dirac heat kernel on unit  $S^3$ ,  $K(t) = \text{Tr } e^{-tD^2} = \sum_{n=0}^{\infty} 2(n+1)(n+2) e^{-(n+3/2)^2 t}$ , has a Seeley–DeWitt expansion that terminates at exactly two terms:

$$K(t) = \frac{\sqrt{\pi}}{2} t^{-3/2} - \frac{\sqrt{\pi}}{4} t^{-1/2} + O(e^{-\pi^2/t}). \quad (31)$$

All higher SD coefficients  $a_k = 0$  identically for  $k \geq 2$ . The proof uses the Jacobi theta modular identity  $\theta_2(0, e^{-t}) = \sqrt{\pi/t} \theta_4(0, e^{-\pi^2/t})$ , which maps the small- $t$  expansion to a large- $\pi^2/t$  regime where  $\theta_4 \rightarrow 1 + O(e^{-\pi^2/t})$ . Numerical verification at 80 dps (50,000 terms) confirms  $a_2 = -3.88 \times 10^{-74}$  (numerical noise, not a physical coefficient), and the residual does not scale as  $t^2$ .

Setting  $\text{Tr } e^{-D^2/\Lambda^2} = K/\pi$  gives the depressed cubic

$$2\Lambda^3 - \Lambda = \frac{4(K/\pi)}{\sqrt{\pi}}, \quad (32)$$

with Cardano closed form  $\Lambda = \sqrt[3]{A + \sqrt{A^2 - 1/216}} + \sqrt[3]{A - \sqrt{A^2 - 1/216}}$ , where  $A = 2(B + F - \Delta)/\sqrt{\pi} \approx 24.611$ . This yields  $\Lambda_{\infty} = 3.7102454679060528505$  (50 dps). At this cutoff, a sharp

spectral truncation includes exactly  $40 = \Delta^{-1}$  modes ( $n = 0, 1, 2$ ), since  $|\lambda_2| = 3.5 < 3.71 < 4.5 = |\lambda_3|$ .

The SD ratio  $a_0(D^2)/a_0(\Delta_{\text{LB}}) = 2 = \text{rank}(\text{spinor bundle})$  clarifies the supertrace sprint F1 result (ratio = 4): the discrepancy is the Dirac vs. Weyl counting convention ( $4 = 2 \times 2$ ).

*Structural conclusion:* the Connes–Chamseddine framework provides an exact 2-term functional form for the spectral action but does *not* provide a mechanism to fix  $\Lambda$ . The value  $\Lambda = 3.710\dots$  is determined by inverting the spectral action at  $K/\pi$ , not selected autonomously.

## 8.2 Parametric spectral action on $S_R^3$ and $\zeta_{\text{unit}}(-k) = 0$

The two-term exactness theorem (31) has an equivalent formulation on the spectral-zeta side which makes the spectral action on the family  $\{S_R^3 : R > 0\}$  *exactly* two-term for any reasonable cutoff function  $f$ .

**The  $\zeta_{\text{unit}}(-k) = 0$  identity.** The spectral zeta of  $D^2$  on unit  $S^3$  admits a Hurwitz decomposition:

$$\zeta_{\text{unit}}(s) = \sum_n 2(n+1)(n+2) \left(n + \frac{3}{2}\right)^{-2s} = 2\zeta_H(2s-2, \frac{3}{2}) - \frac{1}{2}\zeta_H(2s, \frac{3}{2}). \quad (33)$$

At non-positive integers  $s = -k$  ( $k = 0, 1, 2, \dots$ ), using  $\zeta_H(-n, a) = -B_{n+1}(a)/(n+1)$ :

$$\zeta_{\text{unit}}(-k) = -\frac{2B_{2k+3}(3/2)}{2k+3} + \frac{B_{2k+1}(3/2)}{2(2k+1)}. \quad (34)$$

**Theorem 5** ( $\zeta_{\text{unit}}(-k) = 0$  for all  $k \geq 0$ ). *The right-hand side of (34) vanishes identically for every non-negative integer  $k$ . Equivalently,*

$$4(2k+1)B_{2k+3}(3/2) = (2k+3)B_{2k+1}(3/2) \quad \forall k \geq 0. \quad (35)$$

*Proof sketch.* Via the Mellin transform  $\text{Tr } e^{-tD^2} = (1/2\pi i) \int \Gamma(s) \zeta_{\text{unit}}(s) t^{-s} ds$ , shifting the contour through the right half-plane picks up contributions from poles of  $\Gamma(s)\zeta_{\text{unit}}(s)$ . The two-term exactness theorem (31) states that the asymptotic expansion at small  $t$  has only the leading  $t^{-3/2}$  and  $t^{-1/2}$  terms, with no Taylor terms  $t^k$  ( $k \geq 0$ ). The Taylor coefficients would otherwise be sourced by poles of  $\Gamma(s)$  at  $s = -k$  with residue  $(-1)^k/k!$ , giving contributions proportional to  $\zeta_{\text{unit}}(-k)$ . Their absence forces  $\zeta_{\text{unit}}(-k) = 0$ . Verified symbolically in **sympy** for  $k = 0, 1, 2, 3, 4, 5$ .  $\square$

**Exact two-term spectral action on  $S_R^3$ .** For any cutoff  $f$  with Mellin transform  $\phi(s) = \int_0^\infty f(x) x^{s-1} dx$  that is meromorphic with poles only at non-positive integers (e.g. Gaussian  $e^{-x}$ , polynomial  $e^{-x^2}$ , sharp  $\Theta(1-x)$ ), the spectral action on  $S_R^3$  is

$$S(R, \Lambda) := \text{Tr } f(D^2/\Lambda^2) = \phi(3/2) (\Lambda R)^3 - \frac{1}{4} \phi(1/2) (\Lambda R) + O(\text{exp small}) \quad (36)$$

modulo exponentially small corrections in  $(\Lambda R)^2$ . The two-term form is *exact*: there are no power-law corrections at any order. Corrections from poles of  $\phi(s)$  at non-positive integers vanish identically because each one is proportional to  $\zeta_{\text{unit}}(-k) = 0$ .

This is a clean structural consequence of two-term exactness: on the family of round  $S_R^3$  at fixed cutoff function, the spectral action is EXACTLY of Einstein–Hilbert plus cosmological-constant form ( $\phi(3/2)(\Lambda R)^3$  is the cosmological-constant slot via  $\text{Vol}(S_R^3) \propto R^3$ ;  $-\frac{1}{4}\phi(1/2)\Lambda R$  is the Einstein–Hilbert slot via  $R_{\text{scalar}} \cdot V \propto R$ ).

**Formal extremum and the CC spectral-action reading.** Three natural cutoffs give:

| Cutoff                | $\phi(3/2)$              | $\phi(1/2)$              | $u_{\text{crit}} = \sqrt{\phi(1/2)/(12\phi(3/2))}$ |
|-----------------------|--------------------------|--------------------------|----------------------------------------------------|
| Gaussian $e^{-x}$     | $\sqrt{\pi}/2$           | $\sqrt{\pi}$             | $1/\sqrt{6} \approx 0.408$                         |
| Polynomial $e^{-x^2}$ | $\frac{1}{2}\Gamma(3/4)$ | $\frac{1}{2}\Gamma(1/4)$ | $\approx 0.497$                                    |
| Sharp $\Theta(1-x)$   | $2/3$                    | $2$                      | $1/2$                                              |

The asymptotic spectral action  $S_{\text{asympt}}(u) = Au^3 - Bu$  with  $A = \phi(3/2) > 0$ ,  $B = \frac{1}{4}\phi(1/2) > 0$  has a unique minimum at  $u_{\text{crit}} = \sqrt{B/(3A)} = O(1)$ . Under the Connes–Chamseddine spectral action principle (Chamseddine–Connes 1997),  $S_{\text{asympt}}$  IS the effective classical action for the geometry; this gives a formal de-Sitter-like preferred radius  $R_{\text{crit}} \sim 1/\Lambda$ , the GeoVac analog of CC’s spectral-action-selects-metric mechanism.

**Honest scope.** The literal QM action  $S_{\text{QM}}(u) = \sum_n g_n f(\lambda_n^2/\Lambda^2)$  is monotonically increasing in  $u$  for all three cutoffs and has no literal extremum: at  $u \rightarrow 0$  all modes are Boltzmann-suppressed by the cutoff and  $S_{\text{QM}} \rightarrow 0$ , while at  $u \rightarrow \infty$  many modes pass the cutoff and  $S_{\text{QM}} \rightarrow Au^3$ . The asymptotic  $S_{\text{asympt}} = Au^3 - Bu$  agrees with  $S_{\text{QM}}$  to machine precision for  $u \gtrsim 2$  (Gaussian); the formal extremum at  $u_{\text{crit}} \sim 0.4\text{--}0.5$  sits in the IR regime where the asymptotic does NOT faithfully represent the QM exact sum.

Under the CC spectral-action reading (asymptotic = classical action), GeoVac on  $S_R^3$  reproduces CC-style Einstein–Hilbert + cosmological constant with a stronger structural result: *the asymptotic terminates at two terms with no higher-curvature corrections at any order*, by Theorem 5. CC’s continuum spectral action on a generic 4-manifold has higher-curvature corrections at every order; the GeoVac substrate eliminates them structurally. Sprint G1 driver: `debug/g1_spectral_action_S3_radius.py` and memo `debug/g1_spectral_action_S3_radius_memo.md`.

### 8.3 Extension to $S_R^3 \times S_\beta^1$ : factorization and 4D CC structure

The two-term exactness theorem (31) and the  $\zeta_{\text{unit}}(-k) = 0$  identity (34) PROPAGATE from  $S^3$  to the thermal product  $S_R^3 \times S_\beta^1$  (Euclidean compactified time with antiperiodic boundary conditions for fermions). On the product, the spectral action acquires the exact 4D Connes–Chamseddine form: Einstein–Hilbert + cosmological constant with no higher-curvature corrections at any order.

**Factorization.** The Dirac operator on  $S_R^3 \times S_\beta^1$  is  $D = \gamma^0 \otimes \partial_\tau + I \otimes D_{S^3}$  with  $\{\gamma^0, D_{S^3}\} = 0$ , so  $D^2 = D_{S^3}^2 \otimes I + I \otimes (-\partial_\tau^2)$ . The heat kernel factorizes:

$$\text{Tr } e^{-tD^2} \Big|_{S_R^3 \times S_\beta^1} = K_{S^3}(t, R) \cdot K_{S^1}(t, \beta). \quad (37)$$

$K_{S^3}(t, R) = K_{\text{unit}}(t/R^2)$  has the two-term form (31).  $K_{S^1}(t, \beta) = \sum_k e^{-t\omega_k^2}$  with Matsubara frequencies  $\omega_k = 2\pi(k + 1/2)/\beta$  admits Poisson resummation:

$$K_{S^1}(t, \beta) = \frac{\beta}{2\sqrt{\pi t}} \theta_4\left(0, e^{-\beta^2/(4t)}\right) = \frac{\beta}{2\sqrt{\pi t}} + O\left(e^{-\beta^2/(4t)}\right). \quad (38)$$

**Two-term exactness on the product.** Both factors have clean asymptotic expansions modulo exp-small. Multiplying:

$$\text{Tr } e^{-tD^2} \Big|_{S_R^3 \times S_\beta^1} = \frac{\beta R^3}{4} t^{-2} - \frac{\beta R}{8} t^{-1} + O(\text{exp small in } \min(R^2/t, \beta^2/t)). \quad (39)$$

EXACTLY two power-law terms, no Taylor terms in  $t$ , no higher curvature corrections. The cross-product of "exp small" with "power-law" remains exp small.

**Corollary 2** ( $\zeta_{S^3 \times S^1}(-k) = 0$ ). *For the product space  $S_R^3 \times S_\beta^1$  with antiperiodic boundary conditions for fermions,  $\zeta_{S^3 \times S^1}(-k) = 0$  for all non-negative integers  $k$ .*

*Proof.*  $\zeta_{S^3 \times S^1}(-k)$  corresponds to the Taylor coefficient of  $\text{Tr } e^{-tD^2}$  at  $t = 0$  via Mellin inversion (cf. proof of Theorem 5). By (39),  $\text{Tr } e^{-tD^2}$  has no Taylor terms in  $t$ , only power-law  $t^{-2}, t^{-1}$  and exp-small corrections.  $\square$

**4D Connes–Chamseddine form.** For Gaussian cutoff  $f(x) = e^{-x}$ , the spectral action factorizes and matches the asymptotic to bit-exact precision at large  $\Lambda$ :

$$S(R, \beta, \Lambda) = \text{Tr } e^{-D^2/\Lambda^2} = \frac{\beta R^3}{4} \Lambda^4 - \frac{\beta R}{8} \Lambda^2 + O(\text{exp small}). \quad (40)$$

This is the standard 4D CC form:  $\Lambda^4$  cosmological constant +  $\Lambda^2$  Einstein–Hilbert.

- $\Lambda^4$  coefficient:  $\beta R^3/4 = \text{Vol}(S_R^3 \times S_\beta^1)/(8\pi^2)$  where  $\text{Vol} = 2\pi^2 R^3 \beta$ .
- $\Lambda^2$  coefficient:  $-\beta R/8 = -\frac{1}{96\pi^2} \int R_{\text{scalar}} \sqrt{g} d^4x$  where  $\int R_{\text{scalar}} \sqrt{g} d^4x = 12\pi^2 R \beta$  (only  $S^3$  contributes;  $S^1$  is flat).
- Higher  $\Lambda$  powers: exactly zero by Corollary 2.

**Formal extremum and de Sitter vacuum.** The asymptotic  $S(R, \beta, \Lambda) = (\beta R^3/4)\Lambda^4 - (\beta R/8)\Lambda^2$  has  $\partial S/\partial R = 0$  at  $R_{\text{crit}}\Lambda = 1/\sqrt{6}$  (identical to the one-parameter G1 extremum). At  $R = R_{\text{crit}}$ ,  $\partial S/\partial \beta < 0$ , so the joint minimum is at  $\beta \rightarrow \infty$ . This corresponds to the decompactified  $S^1 = \mathbb{R}_\tau$  limit, i.e., the **\*\*zero-temperature de Sitter vacuum\*\*** with radius  $R_{\text{crit}} \sim 1/\Lambda$ . Same standard de Sitter prediction (and same cosmological-constant-problem gap) as CC’s continuum spectral action.

**Stefan–Boltzmann lives in the exp-small.** The thermal free energy  $F_{\text{SB}}(T) = -7\pi^2/180 \cdot VT^4$  for free fermions does NOT appear in the UV asymptotic (40). It lives in the exp-small corrections from higher Matsubara modes ( $m \geq 1$  in Poisson resum). The CC spectral-action expansion (UV / asymptotic) and Stefan–Boltzmann thermal physics (IR / convergent) are structurally distinct contributions, as already known in CC’s continuum literature.

Sprint G2 driver: `debug/g2_spectral_action_S3_x_S1.py` and memo `debug/g2_spectral_action_S3_x_S1_memo`

## 8.4 Spinor-bundle specificity: scalar and TT-tensor heat traces

The two-term exactness theorem (31), which underwrites both the parametric spectral action on  $S_R^3$  (§8.2) and the 4D thermal extension (§8.3), is *specific to the spinor bundle on  $S^3$* . The scalar Laplacian and the Lichnerowicz Laplacian on transverse-traceless symmetric rank-2 tensors (the graviton sector) have heat traces with infinitely many Seeley–DeWitt coefficients, the standard CC continuum expansion.

This sharpens the structural reading of two-term exactness: it reflects the half-integer-shifted Dirac spectrum, NOT the geometry of  $S^3$  per se. We derive closed-form heat traces for the three operators of interest.



**Three closed forms. Dirac** (Theorem (31), half-integer spectrum  $(n + 3/2)^2$  with multiplicity  $2(n + 1)(n + 2)$ , Jacobi  $\theta_2$  inversion):

$$K_{D^2}(t)|_{S^3} = \frac{\sqrt{\pi}}{2} t^{-3/2} - \frac{\sqrt{\pi}}{4} t^{-1/2} + O(e^{-\pi^2/t}). \quad (41)$$

**Scalar Laplacian** (integer spectrum  $n(n + 2)$  with multiplicity  $(n + 1)^2$ ,  $n \geq 0$ , Jacobi  $\theta_3$  inversion + exponential prefactor):

$$K_{\Delta}(t)|_{S^3} = \frac{\sqrt{\pi}}{4} \frac{e^t}{t^{3/2}} + O(e^{-\pi^2/t}). \quad (42)$$

**TT-tensor Lichnerowicz** (integer spectrum  $n(n + 2) - 2$  for  $n \geq 2$  with multiplicity  $2(n - 1)(n + 3)$ , Jacobi  $\theta_3$  inversion + discrete corrections from missing  $n = 0, 1$  levels):

$$K_{\Delta_L^{TT}}(t)|_{S^3} = e^{3t} \left[ \frac{\sqrt{\pi}}{2} t^{-3/2} - 4\sqrt{\pi} t^{-1/2} + 4 \right] + 6e^{2t} + O(e^{-\pi^2/t}). \quad (43)$$

All three closed forms are verified bit-exactly (relative error  $< 10^{-50}$  at  $t \leq 0.1$ ) numerically in `debug/g3_graviton_lichnerowicz_S3.py`.

**Seeley–DeWitt coefficient comparison.** Expanding into standard form  $K(t) = (4\pi t)^{-3/2} \sum_k a_k t^k$ :

**Theorem 6** (Scalar SD closed form on  $S^3$ ). *For the scalar Laplacian on unit  $S^3$ , all Seeley–DeWitt coefficients are nonzero and given by*

$$a_k^{\Delta} = \frac{2\pi^2}{k!} \quad \forall k = 0, 1, 2, \dots \quad (44)$$

*Proof.* From (42), expanding  $e^t = \sum_k t^k/k!$  gives  $K_{\Delta}(t) = \sum_k (\sqrt{\pi}/(4k!)) t^{k-3/2}$ . Matching to  $K_{\Delta} = (4\pi t)^{-3/2} \sum_k a_k t^k$  yields  $a_k = (4\pi)^{3/2} \sqrt{\pi}/(4k!) = 2\pi^2/k!$ .  $\square$

For the TT-tensor, the heat trace (43) contains BOTH half-integer-power terms (from the smooth Weyl asymptotic with  $e^{3t}$  prefactor) AND integer-power terms (from the constant 4 and the discrete correction  $6e^{2t}$ ). The standard SD form captures only the half-integer-power content; the integer-power content is "non-SD," reflecting the truncation of the spectrum to  $n \geq 2$ .

Half-integer-power coefficients from  $e^{3t}[(\sqrt{\pi}/2) t^{-3/2} - 4\sqrt{\pi} t^{-1/2}]$ :

$$\begin{aligned} a_0^{TT} &= 4\pi^2, \\ a_1^{TT} &= -20\pi^2 = (\sqrt{\pi}/2)(3) - 4\sqrt{\pi} \text{ scaled by } (4\pi)^{3/2}, \\ a_2^{TT} &= -78\pi^2, \text{ etc.} \end{aligned}$$

All nonzero, with  $a_k^{TT} \propto \pi^2 \cdot p_k(3)$  for explicit polynomials  $p_k$  from the  $e^{3t}$  expansion.

**Structural distinction.**

| Operator                     | Spectrum shift      | Two-term exact? | Higher SD coefficients      |
|------------------------------|---------------------|-----------------|-----------------------------|
| $D^2$ Dirac                  | half-integer        | YES (Paper 28)  | $a_k = 0$ for $k \geq 2$    |
| $\Delta$ scalar              | integer             | NO              | $a_k = 2\pi^2/k!$ (Thm 6)   |
| $\Delta_L^{TT}$ Lichnerowicz | integer (truncated) | NO              | all nonzero + non-SD pieces |

**Implication.** The clean CC Einstein–Hilbert + cosmological constant structure on  $S^3$  that emerges from the spectral action (36) (Sprint G1) and (40) (Sprint G2) is a feature of the *spinor bundle*, not of  $S^3$  as a manifold. Gravitons (spin-2 TT modes) on  $S^3$  have a heat trace with infinite higher-curvature SD coefficients plus non-SD discrete corrections from the spectrum’s  $n \geq 2$  cutoff. The graviton sector on the GeoVac substrate, were it constructed via CC-style metric variation, would inherit the standard CC continuum expansion with corrections at every order, NOT the clean two-term form of the Dirac sector.

This is consistent with the framework’s structural-skeleton scope: clean structure lives in the Dirac sector sourced by the Fock projection (Paper 23 rigidity), with gravity-side coupling constants remaining external calibration data.

Sprint G3 driver: `debug/g3_graviton_lichnerowicz_S3.py` and memo `debug/g3_graviton_lichnerowicz_S3`

## 8.5 Graviton diagnostic: physical $(1, 1)$ modes on the truncated substrate

The structural-test phase of G1–G3 establishes that the spectral action on the GeoVac substrate has EXACT Einstein–Hilbert + cosmological constant form on the Dirac sector but inherits standard CC continuum behavior on the spin-2 sector. The natural follow-up question: does the discrete substrate host graviton-like perturbations of  $D$  at the operator level, or are gravitons structurally blocked?

**Setup.** Perturb  $D = D_0 + \epsilon V$  where  $V \in \text{Herm}(\mathcal{H}_{n_{\max}})$ . Expand the spectral action:  $S[D_0 + \epsilon V] = S[D_0] + \epsilon S^{(1)}[V] + \frac{1}{2}\epsilon^2 S^{(2)}[V, V] + O(\epsilon^3)$ . The quadratic form  $S^{(2)}$  is diagonal in the  $D_0$  eigenbasis.

**Gauge–physical classification.** Modes  $V = i[X, D_0]$  for Hermitian  $X$  are tangent to the gauge orbit  $D_0 \rightarrow UD_0U^*$ , and the spectral action is invariant under unitary conjugation. Therefore  $S^{(2)}$  eigenvalues on gauge modes measure the curvature of the gauge orbit along the linear direction, not physical kinetic content. For the truncated CH Dirac, the gauge subspace equals the cross-sector off-block Hermitian matrices, and within-sector modes are physical.

**Spin-2 content.** The  $SU(2)_L \times SU(2)_R$  irreps at level  $n$  are  $((n+1)/2, n/2)$  for +chirality and  $(n/2, (n+1)/2)$  for –chirality. Within-sector tensor products  $\rho \otimes \rho$  contain the graviton irrep  $(j_L, j_R) = (1, 1)$  for  $n \geq 1$ .

**Proposition 1** (Physical  $(1, 1)$  modes on the substrate). *At every  $n_{\max} \geq 1$ , the second variation  $S^{(2)}$  has PHYSICAL (within-sector) eigenmodes carrying  $(j_L, j_R) = (1, 1)$  irrep content with POSITIVE eigenvalue*

$$A_\lambda = a_\lambda \left( \frac{4\lambda^2}{\Lambda^4} - \frac{2}{\Lambda^2} \right), \quad a_\lambda = e^{-\lambda^2/\Lambda^2}, \quad (45)$$

for each within-sector block at CH eigenvalue  $|\lambda|$ . Numerically verified by 5-point finite-difference stencil to rel diff  $\sim 10^{-9}$  at  $\Lambda^2 = 6$ ,  $n_{\max} \in \{1, 2, 3\}$ :

| $ \lambda $ | blocks containing $(1, 1)$         | $A_\lambda$ |
|-------------|------------------------------------|-------------|
| 5/2         | $S_3 \otimes S_3, S_4 \otimes S_4$ | +0.127      |
| 7/2         | $S_5 \otimes S_5, S_6 \otimes S_6$ | +0.133      |
| 9/2         | $S_7 \otimes S_7, S_8 \otimes S_8$ | +0.066      |

*Eigenvalues are stable across  $n_{\max}$  (no sign changes), and the  $|\lambda|$ -dependence reflects the standard Gaussian-regulated kinetic behavior of the CC spectral action.*

**Honest scope.** Proposition 1 establishes the NECESSARY condition for graviton dynamics on the discrete substrate: physical  $(1, 1)$ -irrep eigenmodes exist and propagate with positive kinetic eigenvalue. SUFFICIENT conditions — Fierz–Pauli kinetic structure separating TT graviton (2 polarizations) from longitudinal/scalar modes (7 of the 9 within  $(1, 1)$ ), gauge invariance of the TT projector, propagator structure, and continuum graviton emergence via Paper 38-style state-space GH convergence — are not verified here; the J-blindness leg has since been completed (Paper 51, §Fierz–Pauli + the J-blindness theorem), and the remaining sufficiency legs constitute the extended G6 program. The  $S^3$  Fock projection rigidity (Paper 23) fixes the background metric; graviton candidates here are operator-level perturbations of  $D$ , not direct metric perturbations.

*Structural reading.* Sprint G3 (§8.4) established that the spin-2 sector inherits standard CC continuum expansion. Sprint G6-Diag-Full establishes that this expansion’s graviton modes are accessible on the discrete substrate at the necessary-condition level. The clean structure of G1/G2 (Einstein–Hilbert + cosmological constant exactly on Dirac sector) does NOT extend to gravitons; gravitons inherit CC continuum behavior with infinite higher-curvature corrections. Substrate-level existence of  $(1, 1)$  modes confirms gravitons are not structurally blocked.

**DOF count is a continuum-limit statement (Sprint GD-1).** The reduction of the 9-dimensional  $(1, 1)$  irrep to the 2 transverse-traceless physical polarizations does *not* occur at finite truncation. The continuum count is standard: under the diagonal  $SU(2)$ ,  $(1, 1) = \text{spin-2} \oplus \text{spin-1} \oplus \text{spin-0} = 5 \oplus 3 \oplus 1$ , and diffeomorphism gauge (4) plus constraints (3) leave 2 TT. On the discrete substrate, however, the inner-automorphism (diffeomorphism) gauge tangent  $i[X, D_0]$  has entries  $(\lambda_b - \lambda_a)X_{ab}$ , which vanish *within* a sector ( $\lambda_a = \lambda_b$ ); the inner gauge is therefore *entirely cross-sector* (verified zero within-sector gauge entries at  $n_{\max} = 1, 2, 3$ ). Within a within-sector  $(1, 1)$  block the gauge does not act, so all 9 modes are non-inner-gauge at finite truncation. The continuum diffeomorphism gauge acts within the tensor field at each point; the discrete inner gauge connects different  $\lambda$ -sectors. These reconcile only as  $n_{\max} \rightarrow \infty$ , when the cross-sector directions assemble into continuum diffeomorphisms that remove 7 of the 9 within-sector modes. Thus the graviton DOF reduction is a *continuum-limit phenomenon* (Paper 38-style state-space GH), not an internal-to-the-block projection — which is the structural reason the Fierz–Pauli decomposition is the extended G6 task and not a sprint-scale one. Source: `debug/sprint_gd1_graviton_dof_memo.md`.

**Which irrep carries the TT graviton (Sprint GD-4).** A rep-theory diagnostic sharpens *which* sector the necessary-condition test should examine. The helicity of an  $SO(4) = SU(2)_L \times SU(2)_R$  irrep  $(j_L, j_R)$  is  $|j_L - j_R|$ ; the TT spin-2 graviton is helicity  $\pm 2$ , hence lives in  $|j_L - j_R| = 2$  reps (the TT tensor harmonics on  $S^3$  are  $(k+2, k) \oplus (k, k+2)$ , lowest  $(2, 0) \oplus (0, 2)$ ). The  $(1, 1)$  irrep that Proposition 1 tested has  $|j_L - j_R| = 0$  — *helicity zero*, the scalar/trace-class sector. The within-sector blocks  $\text{sector}_i \otimes \text{sector}_i$  do contain the helicity-2 reps  $(2, 0) \oplus (0, 2)$  (verified present at  $n = 1, 2, 3, 4$ ), but these were not examined. *Caveat:* in Connes–Chamseddine the physical graviton enters through the  $\delta D \leftrightarrow h_{\mu\nu}$  dictionary, which can shift irrep assignments — the *local* polarization space is  $(1, 1)$  while the *global* TT harmonic modes are  $|\Delta j| = 2$ . So GD-4 does not refute the necessary-condition result; it identifies a concrete gap: the extended G6 must either examine the  $|\Delta j| = 2$   $\delta D$ -modes directly or establish the dictionary showing the  $(1, 1)$  bilinear carries the physical graviton. Together with the GD-1 finding, the extended G6 has two sharply identified must-dos: (i) test the helicity-2 sector and fix the dictionary; (ii) realize the  $9 \rightarrow 2$  gauge reduction in the continuum (state-space GH) limit. Source: `debug/sprint_gd4_graviton_irrep_memo.md`.

**Helicity-2 modes are positive too (Sprint GD-5).** The necessary-condition half of GD-4’s concern is resolved directly. The coefficient  $A_\lambda = a(\lambda)(4\lambda^2/\Lambda^4 - 2/\Lambda^2)$  is a function of the sector eigenvalue  $\lambda$  alone (the spectral action sees only the degenerate-sector eigenvalue), hence *irrep-blind* within a block: every irrep —  $(1, 1), (2, 0), (0, 2), \dots$  — inherits the same  $A_\lambda$ . Therefore the helicity-2 (TT-graviton-class) reps carry the *same positive*  $A_\lambda$  as the  $(1, 1)$  (+0.127, +0.133, +0.066 at  $n = 1, 2, 3, \Lambda^2 = 6$ ). So Proposition 1’s positivity is *not* confined to the scalar/trace sector; it extends to the actual TT-graviton irrep class. *Honest scope:* the irrep-blindness that makes this immediate is exactly the G6 approximation — the *sufficient* condition (a TT kinetic structure *distinct* from the trace, Fierz–Pauli) requires lifting  $A_\lambda$  to an irrep-resolved second variation, the extended-G6 task where the blindness must break. GD-5 closes the necessary condition for the TT sector; it does not close the TT-vs-trace distinction. Source: `debug/sprint_gd5_helicity2_memo.md`.

Sprint G6-Diag-Full driver: `debug/g6_diag_full.py`, companion first-pass `debug/g6_diag_quadratic_form.py`, scoping memo `debug/g6_scoping_memo.md`, and full memo `debug/g6_diag_full_memo.md`.

## 8.6 Cigar spectral action and Bekenstein–Hawking entropy

The gravity arc beyond the structural-test phase (§8.2–§8.5) includes the gravity-side targets of Bekenstein–Hawking entropy ( $S = A/(4G)$  from the cigar) and decompactified de Sitter vacuum (the  $\beta \rightarrow \infty$  limit of §8.3). Sprint TD Track 4 (CLAUDE.md §2) already reproduced  $T_H = 1/(8\pi M)$  from regularity of the cigar’s  $\tau$ -circle. Sprint G4 first-pass establishes the entropy via the standard Gibbons–Hawking–York thermodynamic derivation and identifies the load-bearing structural mechanism.

**Cigar geometry.** Euclidean Schwarzschild with metric  $ds^2 = (1 - 2M/r) d\tau^2 + (1 - 2M/r)^{-1} dr^2 + r^2 d\Omega_2^2$  outside the horizon  $r > 2M$ . Regularity at  $r = 2M$  forces  $\tau$ -period  $\beta = 8\pi M$ . Horizon area  $A = 4\pi r_h^2 = 16\pi M^2$ .

**Bekenstein–Hawking from spectral action.** After Gibbons–Hawking subtraction of asymptotic flat-space reference, the Euclidean on-shell action of Schwarzschild is

$$I_E^{\text{Schw}} = \frac{A}{4G}, \quad (46)$$

generated by the horizon boundary heat-kernel coefficient at the smooth conical tip. Bulk  $R = 0$  (Ricci-flat) and the asymptotic boundary cancels against the flat-space subtraction. The thermodynamic entropy is

$$S_{\text{thermo}} = -I_E + \beta \frac{\partial I_E}{\partial \beta} = \frac{A}{4G} = 4\pi M^2/G, \quad (47)$$

the Bekenstein–Hawking result. Symbolically verified in `debug/g4_cigar_BH_entropy.py` (sympy, residual exact zero).

**Load-bearing mechanism.** The structural piece producing  $A/(4G)$  is the boundary heat-kernel coefficient at the smooth conical horizon, generating the GHY  $(1/8\pi G) \oint K \sqrt{h} d^3x = A/(4G)$ . This is the standard CC continuum derivation.

**GeoVac discrete substrate.** The GeoVac substrate (CH Dirac on truncated  $S^3$  at finite  $n_{\text{max}}$ ) does NOT natively describe the cigar: the cigar’s spatial section is  $\mathbb{R} \times S^2$  (non-compact radial,  $S^2$  instead of  $S^3$ ), and the horizon is a smooth conical tip in the  $(\tau, r)$  plane. Constructing the discrete

spectral triple analog on the cigar requires substantial new infrastructure: spatial  $S^2$  truncation, non-compact radial regularization, horizon tip preservation under discretization, and verification that the boundary heat-kernel coefficient generates the correct GHY term. This is the full G4 program.

Sprint G4 first-pass: `debug/g4_cigar_BH_entropy.py` and `debug/g4_cigar_BH_entropy_memo.md`.

## 8.7 Decompactified $S^3 \times \mathbb{R}_\tau$ and de Sitter vacuum

Sprint G2 (§8.3) found that the joint extremum of the spectral action on  $S_R^3 \times S_\beta^1$  sits at  $R_{\text{crit}}\Lambda = 1/\sqrt{6}$  (in  $R$ ) and  $\beta \rightarrow \infty$  (in  $\beta$ ), corresponding to the zero-temperature de Sitter vacuum. Sprint G5 makes this  $\beta \rightarrow \infty$  limit explicit by decompactifying the temporal direction.

**Heat-kernel decompactification.** On  $S^3 \times \mathbb{R}_\tau$ , the heat kernel factorizes per unit length of  $\mathbb{R}_\tau$ :

$$K_{\text{full}}(t, R) = K_{S^3}(t, R) \cdot K_{\mathbb{R}}(t), \quad K_{\mathbb{R}}(t) = \int \frac{dk}{2\pi} e^{-k^2 t} = \frac{1}{2\sqrt{\pi t}}. \quad (48)$$

This matches  $K_{S^1}(t, \beta)/\beta$  from §8.3 in the  $\beta \rightarrow \infty$  limit. Combined with the two-term exactness of  $K_{S^3}$ :

$$K_{\text{full}}(t, R) = \frac{R^3}{4} t^{-2} - \frac{R}{8} t^{-1} + O(\text{exp small}). \quad (49)$$

Two-term exactness propagates to the decompactified case.

**Spectral action density.** For Gaussian cutoff  $f(x) = e^{-x}$ :

$$s(R, \Lambda) := K_{\text{full}}(1/\Lambda^2, R) = \frac{R^3}{4} \Lambda^4 - \frac{R}{8} \Lambda^2. \quad (50)$$

**De Sitter vacuum.**

$$\frac{\partial s}{\partial R} = 0 \implies R_{\text{crit}}\Lambda = \frac{1}{\sqrt{6}}, \quad (51)$$

identical to G1 (§8.2). Minimum action density:

$$s_{\text{min}} = s(R_{\text{crit}}, \Lambda) = -\frac{\Lambda}{12\sqrt{6}}. \quad (52)$$

Symbolically verified (sympy,  $-\sqrt{6}\Lambda/72 = -\Lambda/(12\sqrt{6})$ , residual exact zero).

**Physical reading.** The framework's preferred zero-temperature de Sitter vacuum has Planck-scale radius  $R_{\text{crit}} \sim 1/\Lambda$  and vacuum action density  $-\Lambda/(12\sqrt{6})$ . This is the standard CC spectral-action-principle prediction, with the well-known cosmological-constant-scale gap. The decompactification is clean: G2's  $\beta \rightarrow \infty$  minimum is manifest in G5 as the  $\mathbb{R}_\tau$  vacuum.

Sprint G5: `debug/g5_decompactified_S3_x_R.py` and `debug/g5_decompactified_memo.md`.

## 8.8 Newton constant, cosmological constant, and background extremality

Sprint G7 consolidates the gravity arc's connection to standard Connes–Chamseddine gravity by deriving the framework's gravitational parameters in standard QFT language and verifying a structural background-extremality consistency check.

**Background extremality.** From the G6-Diag-Full physical  $(1, 1)$ -irrep eigenvalues  $A_\lambda = a_\lambda(4\lambda^2/\Lambda^4 - 2/\Lambda^2)$ , the Gaussian-weighted integral over the spectrum vanishes exactly:

$$\int_0^\infty (4u^2 - 2) e^{-u^2} du = 4 \cdot \frac{\sqrt{\pi}}{4} - 2 \cdot \frac{\sqrt{\pi}}{2} = 0. \quad (53)$$

This confirms the standard background-extremality condition: the spectral action is stationary at the CH Dirac background, with sub-leading perturbations giving the actual graviton kinetic structure. Sympy-verified.

**Newton constant and cosmological constant from G2.** Matching G2's spectral action coefficients  $(\beta R^3/4)\Lambda^4 - (\beta R/8)\Lambda^2$  (40) against the standard Einstein–Hilbert + cosmological constant action  $S_{EH} = -(1/16\pi G_{\text{eff}}) \int (R_{\text{scalar}} - 2\Lambda_{cc}) \sqrt{g} d^4x$  using  $\text{Vol}(S_R^3 \times S_\beta^1) = 2\pi^2 R^3 \beta$  and  $\int R_{\text{scalar}} \sqrt{g} = 12\pi^2 R \beta$ :

**Proposition 2** (Newton constant + cosmological constant). *The Gaussian-cutoff spectral action on  $S_R^3 \times S_\beta^1$  identifies the framework's effective gravitational parameters as*

$$G_{\text{eff}} = \frac{6\pi}{\Lambda^2}, \quad \Lambda_{cc} = 6\Lambda^2. \quad (54)$$

**Planck-scale identification.** Setting  $\Lambda \rightarrow M_{\text{Planck}}$ :  $G_{\text{eff}} = 6\pi/M_{\text{Planck}}^2 = 6\pi G_{\text{Newton}}$  (Planck-scale Newton's constant);  $\Lambda_{cc} = 6M_{\text{Planck}}^2 = 6/G_{\text{eff}}$  (Planck-scale cosmological constant).

**Cosmological constant scale gap.** The predicted ratio  $\Lambda_{cc}/M_{\text{Planck}}^2 = 6$  vs. the observed cosmological dark energy  $\Lambda_{cc}/M_{\text{Planck}}^2 \sim 10^{-120}$  gives a  $\sim 120$  orders of magnitude gap. This is the standard CC cosmological-constant-scale problem, inherited from continuum Connes–Chamseddine and not GeoVac-specific. The framework's discrete substrate does not modify this leading-order prediction. Standard interpretation: cosmological constant scale is Class 1 calibration data (see `memory/external_input_three_class_partition.md`).

**Master Mellin engine classification.** Gravity content is M2 (Seeley–DeWitt heat-kernel coefficients  $\sqrt{\pi} \cdot \mathbb{Q} \oplus \pi^2 \cdot \mathbb{Q}$ ; see Paper 18 §III.7). G2's coefficients  $1/(8\pi^2)$  and  $-1/(96\pi^2)$  are M2 ring members. On the Dirac substrate, M2 terminates at two terms by (31); on scalar/tensor sectors, M2 is the full infinite series (§8.4, eq (44)). The gravity arc's content is thus the M2 sub-mechanism of the master Mellin engine, with substrate-specific termination on the Dirac bundle.

Sprint G7 driver: `debug/g7_extremality_newton.py`, memo `debug/g7_extremality_newton_memo.md`.

## 8.9 $S^2$ Dirac and the cigar's spatial spinor structure

Sprint G4-1 (first sub-sprint of the full G4 program) tests whether the framework's two-term exactness on  $S^3$  Dirac (§8.3, Theorem 5) propagates to the  $S^2$  Dirac that appears in the cigar's spatial section  $\mathbb{R}_+ \times S_r^2$ . Answer: NO.

**$S^2$  Dirac spectrum.** Camporesi-Higuchi for  $S^d$  Dirac:  $|\lambda_n^{S^d}| = n + d/2$ . For  $d = 2$ :

$$|\lambda_n^{S^2}| = n + 1, \quad (\text{INTEGER shift}) \quad (55)$$

$$g_n^{\text{Dirac}}|_{S^2} = 4(n + 1). \quad (56)$$

In contrast,  $S^3$  Dirac has  $|\lambda_n^{S^3}| = n + 3/2$  (HALF-INTEGER shift) with  $g_n^{\text{Dirac}}|_{S^3} = 2(n + 1)(n + 2)$ .

**Heat trace asymptotic.** The  $S^2$  Dirac heat trace is  $K(t) = 4 \sum_{u \geq 1} u e^{-u^2 t}$ . The leading asymptotic

$$K_{S^2}^{\text{Dirac}}(t) \sim \frac{2}{t} + O(1) \quad (57)$$

matches the standard CC form  $(4\pi t)^{-1} \cdot \dim_S \cdot \text{Vol}(S^2)$  with  $\dim_S = 2$ ,  $V = 4\pi$ . Numerically verified to rel diff  $10^{-5}$  at  $t = 10^{-4}$  (`debug/g4_1.S2.dirac.py`).

**First subleading.** The Seeley-DeWitt coefficient  $a_1 = \dim_S \cdot (R_{\text{scalar}}/6 - E_{\text{Lich}}) \cdot V$  on unit  $S^2$  ( $R = 2$ ,  $E = R/4 = 1/2$ ):

$$a_1^{S^2, \text{Dirac}} = 2(1/3 - 1/2) \cdot 4\pi = -\frac{4\pi}{3} \neq 0. \quad (58)$$

The numerical extraction  $\lim_{t \rightarrow 0} [K(t) - 2/t] = a_1/(4\pi) = -1/3$  matches at rel diff  $\sim 10^{-3}$  across  $t = 10^{-1}$  to  $10^{-3}$ .

**Structural distinction.** The two-term exactness theorem on  $S^3$  Dirac comes from the half-integer spectrum shift, allowing Jacobi  $\theta_2$  inversion to give exactly two power-law terms.  $S^2$  Dirac (integer shift) uses Jacobi  $\theta_3$  inversion + exponential prefactor, giving the standard CC infinite Seeley-DeWitt series with nonzero  $a_k$  at every order.

This is consistent with Sprint G3 (§8.4): the scalar Laplacian on  $S^3$  (integer spectrum from  $n = 0$ ) also has full SD series with  $a_k = 2\pi^2/k!$  (Theorem 6). The pattern crystallizes as:

| Operator              | Spectrum shift         | Higher SD coefficients                |
|-----------------------|------------------------|---------------------------------------|
| $S^3$ Dirac           | half-integer $n + 3/2$ | $a_k = 0$ for $k \geq 2$              |
| $S^3$ scalar $\Delta$ | integer $n(n+2)$       | $a_k = 2\pi^2/k!$ (G3)                |
| $S^2$ Dirac           | integer $n + 1$        | $a_1 = -4\pi/3 \neq 0$ , $a_k \neq 0$ |

**Implication for G4 full.** The cigar’s spatial section  $\mathbb{R}_+ \times S_r^2$  has the  $S^2$  component inheriting standard CC continuum behavior with higher-curvature corrections at all orders. The discrete-substrate BH entropy derivation cannot exploit a clean two-term form on the spatial  $S^2$ ; it must follow standard CC heat-kernel asymptotic structure. The load-bearing piece for  $A/(4G)$  remains the horizon boundary  $a_2$  coefficient (§8.6).

The framework’s clean two-term exactness is *specific to the  $S^3$  substrate sourced by the Fock projection*. It does not extend to other dimensions or the gravity arc’s cigar geometry.

Sprint G4-1 driver: `debug/g4_1.S2.dirac.py` and memo `debug/g4_1.S2.dirac.memo.md`.

## 8.10 Conical defect / replica method for BH entropy

Sprint G4-2 (second sub-sprint of G4 full) builds on §8.9 to derive the Bekenstein–Hawking entropy via the standard conical-defect / replica method on the cigar’s near-horizon geometry  $D_\alpha^2 \times S_{r_h}^2$ .

**Setup.** The cigar’s near-horizon geometry is  $ds^2 = d\rho^2 + \rho^2 d\varphi^2 + r_h^2 d\Omega_2^2$  with  $\varphi \in [0, 2\pi)$  (smooth tip) or  $\varphi \in [0, 2\pi\alpha)$  (conical singularity for  $\alpha \neq 1$ ). The Bekenstein–Hawking entropy is extracted via the replica formula

$$S_{BH} = - \left. \frac{dI_E(\alpha)}{d\alpha} \right|_{\alpha=1}. \quad (59)$$

**Conical heat trace contribution.** The Sommerfeld/Cheeger formula for a 2D cone with apex angle  $2\pi\alpha$ :

$$K_{\text{cone}}(t) = K_{\text{smooth bulk}}(t) + \frac{1}{12} \left( \frac{1}{\alpha} - \alpha \right) + O(t). \quad (60)$$

The  $(1/12)(1/\alpha - \alpha)$  is the conical-defect tip contribution, vanishing at smooth  $\alpha = 1$ .

**Combined heat trace.** Using G4-1's  $K_{S_{r_h}^2}(t) \sim 2r_h^2/t - r_h^2/3 + O(t)$  at small  $t$ , the conical-tip contribution to the cigar's heat trace is:

$$K_{\text{tip}}(t, \alpha) = \frac{1}{12} \left( \frac{1}{\alpha} - \alpha \right) \left[ \frac{2r_h^2}{t} - \frac{r_h^2}{3} + O(t) \right] \quad (61)$$

**Replica entropy derivation.** For Gaussian cutoff,  $S = K(1/\Lambda^2)$  gives the conical-tip spectral action contribution

$$I_E^{\text{conical}}(\alpha, \Lambda, r_h) = \frac{r_h^2 \Lambda^2}{6} \left( \frac{1}{\alpha} - \alpha \right) + O(1/\Lambda^2). \quad (62)$$

The  $\alpha$ -derivative at  $\alpha = 1$ :

$$\left. \frac{dI_E^{\text{conical}}}{d\alpha} \right|_{\alpha=1} = \frac{r_h^2 \Lambda^2}{6} \cdot (-2) = -\frac{r_h^2 \Lambda^2}{3}. \quad (63)$$

**Proposition 3** (Bekenstein–Hawking entropy from conical replica). *The spectral action's conical-defect contribution at the cigar tip generates the Bekenstein–Hawking entropy via the replica formula:*

$$S_{BH} = - \left. \frac{dI_E^{\text{conical}}}{d\alpha} \right|_{\alpha=1} = \frac{r_h^2 \Lambda^2}{3} = \frac{A \Lambda^2}{12\pi}, \quad (64)$$

where  $A = 4\pi r_h^2$ . Matching to  $S_{BH} = A/(4G_N)$  gives  $G_N = 3\pi/\Lambda^2$ .

**Factor-of-2 calibration with G7.** The Newton constant from §8.8 (G2's spectral action matching) is  $G_{\text{eff}} = 6\pi/\Lambda^2$ . The factor-of-2 difference from  $G_N = 3\pi/\Lambda^2$  is *not* a scalar-vs-Dirac conical-defect coefficient (the 2D Dirac and scalar cones share the coefficient  $(1/12)(1/\alpha - \alpha)$ ; Fursaev–Miele 1996, confirmed by the G4-4c discrete extraction): it is a bookkeeping artifact forced by Wald's theorem — two-term exactness implies pure Einstein gravity, so action- $G$  and entropy- $G$  are the *same* constant, and the 2 lives in a 4D-bulk vs factorized-2D  $\times$  2D normalization mismatch to be closed by convention audit (Paper 51, Wald-forces-entropy-relation remark). The structural emergence of  $A \cdot \Lambda^2$  from the conical tip is the load-bearing finding; the precise prefactor calibration is downstream.

**Discrete-substrate analog.** For G4-3 onwards (the extended program):

- Define  $\alpha$ -deformation of the discrete substrate at the tip
- Compute the conical contribution from the discrete Dirac
- Verify the discrete substrate reproduces the  $(1/\alpha - \alpha)$  form
- Apply replica method to extract  $S_{BH}$  from the discrete substrate spectral action

The conceptual framework is now in place; the full implementation program follows.

Sprint G4-2 driver: `debug/g4.2.conical.replica.py` and memo `debug/g4.2.conical.replica.memo.md`.



### 8.11 Cutoff-function dependence of the gravity predictions (Sprint G8)

Sprints G2 and G7 derived  $G_{\text{eff}} = 6\pi/\Lambda^2$  and  $\Lambda_{cc} = 6\Lambda^2$  assuming the Gaussian cutoff  $f(x) = e^{-x}$ . Sprint G8 generalizes these to arbitrary cutoff functions and identifies which framework predictions are cutoff-independent (structural) versus cutoff-dependent (calibration).

**Mellin transform of the cutoff.** For a general cutoff function  $f$ , the spectral content is captured by the Mellin transform:

$$\phi(s) := \int_0^\infty f(x) x^{s-1} dx. \quad (65)$$

The relevant moments for spectral action matching are  $\phi(1)$  (Einstein-Hilbert scale) and  $\phi(2)$  (cosmological-constant scale).

**General formulas.** From the G2 spectral action on  $S_R^3 \times S_\beta^1$ ,

$$S(R, \beta, \Lambda) = \phi(2) \cdot \frac{\beta R^3}{4} \Lambda^4 - \phi(1) \cdot \frac{\beta R}{8} \Lambda^2, \quad (66)$$

matching against Einstein-Hilbert plus cosmological constant gives:

$$G_{\text{eff}}(f, \Lambda) = \frac{6\pi}{\phi(1) \Lambda^2}, \quad \Lambda_{cc}(f, \Lambda) = \frac{6\phi(2)}{\phi(1)} \Lambda^2, \quad (67)$$

$$R_{\text{crit}} \Lambda = \sqrt{\frac{\phi(1)}{6\phi(2)}}. \quad (68)$$

Table 6: Gravity predictions for three cutoff functions.

| Cutoff $f(x)$         | $\phi(1)$      | $\phi(2)$ | $G_{\text{eff}} \Lambda^2$ | $\Lambda_{cc}/\Lambda^2$ |
|-----------------------|----------------|-----------|----------------------------|--------------------------|
| Gaussian $e^{-x}$     | 1              | 1         | $6\pi$                     | 6                        |
| Polynomial $e^{-x^2}$ | $\sqrt{\pi}/2$ | 1/2       | $12\sqrt{\pi}$             | $6/\sqrt{\pi}$           |
| Sharp $\Theta(1-x)$   | 1              | 1/2       | $6\pi$                     | 3                        |

#### Three cutoff examples.

**No cutoff-independent gravity prediction.** Direct algebra on (67)–(68) shows that every combination of  $G_{\text{eff}}$  and  $\Lambda_{cc}$  depends on  $\phi(1)$  and/or  $\phi(2)/\phi(1)$ :

- $G_{\text{eff}} \cdot \Lambda^2 = 6\pi/\phi(1)$  (depends on  $\phi(1)$ )
- $\Lambda_{cc}/\Lambda^2 = 6\phi(2)/\phi(1)$  (depends on ratio)
- $G_{\text{eff}} \cdot \Lambda_{cc} = 36\pi\phi(2)/(\phi(1)^2\Lambda^2)$  (depends on both)
- $R_{\text{crit}}\Lambda = \sqrt{\phi(1)/(6\phi(2))}$  (depends on ratio)

**Class 1 calibration data.** Per the framework’s external-input three-class partition (see CLAUDE.md §1.7 and project memory), the cutoff function  $f$  is *Class 1 calibration data* alongside Yukawas,  $Z_2/\delta m$  multi-loop renormalization data, and the Born rule. The framework determines:

- the *structural form* of  $S[D]$  (spectrum, Seeley-DeWitt coefficients),
- the *proportionality* of  $G_{\text{eff}}$  and  $\Lambda_{cc}$  to specific  $\phi$ -moments,
- the *functional form* of  $G_{\text{eff}}(f)$ ,  $\Lambda_{cc}(f)$ ,  $R_{\text{crit}}(f)$ .

The framework does *not* determine:

- the numerical value of  $G_{\text{eff}}$  or  $\Lambda_{cc}$  in absolute units,
- which cutoff function to use.

**Consistency with standard CC.** This is the standard situation in the Chamseddine-Connes spectral action [31, 32]: the cutoff function is external input, the framework determines the structural shape.

**Structural vs numerical scope.** Sprints G1, G2, G3, G4, G5, G6-Diag-Full produced *structural* findings (closed forms, theorems, mechanisms) that are cutoff-independent. Sprint G7 produced *numerical* predictions (specific values of  $G_{\text{eff}}$ ,  $\Lambda_{cc}$ ) that depend on the Gaussian cutoff choice. Sprint G8 clarifies that the framework’s gravity-arc predictions are best understood as:

- structural skeleton: cutoff-independent, determined by GeoVac’s spectral structure;
- numerical calibration: cutoff-dependent, proportional to  $\phi$ -moment combinations.

Sprint G8 driver: `debug/g8_cutoff_dependence.py`; memo: `debug/g8_cutoff_dependence_memo.md`.

## 8.12 Discrete warped-product substrate for the cigar (Sprint G4-3 opening)

Sprints G1 through G8 produced the continuum gravity arc: closed-form spectral action on  $S_R^3$  and  $S^3 \times S_\beta^1$ , two-term exactness theorems,  $S^2$  Dirac structure, Bekenstein-Hawking entropy via the conical-defect / replica method, and the cutoff-function dependence of the gravity predictions. Sprint G4-3 *opens* the discrete-substrate gravity arc: rebuild the gravity-arc results on GeoVac’s discrete substrate rather than at the continuum level.

This sprint is *scoping + first-pass*: define the discrete substrate, identify the load-bearing pieces, and specify the sub-sprint sequence G4-3a through G4-6.

**Discrete substrate.** For the cigar’s near-horizon limit  $ds^2 = d\rho^2 + \rho^2 d\phi^2 + r_h^2 d\Omega_2^2$  (constant warp  $r = r_h$ ):

$$\mathcal{G}_{\text{cigar}} = \mathbb{Z}_+(a)|_{N_\rho} \times \mathbb{Z}/N_\phi \times \text{Fock}(S^2, l_{\text{max}}) \quad (69)$$

where:

- $\mathbb{Z}_+(a)|_{N_\rho} = N_\rho$ -truncation of  $\mathbb{Z}_+$  with lattice spacing  $a$ ,  $\rho_k = ka$ ;
- $\mathbb{Z}/N_\phi =$  discrete  $\phi$ -circle,  $N_\phi$  sites;
- $\text{Fock}(S^2, l_{\text{max}}) =$  standard  $S^2$  Fock projection with  $l \leq l_{\text{max}}$  angular cutoff.

The conical-defect parameter is  $\alpha = N_\phi/N_0$  where  $N_0$  is the reference smooth-tip count.

**Constant-warp factorization.** At  $r(\rho) = r_h$  constant, the Laplacian factorizes:

$$\Delta_{\text{cigar}} = \Delta_{D^2}(\rho, \phi) + \frac{1}{r_h^2} \Delta_{S^2}.$$

The two factors act on independent sectors, so by construction:

$$K_{\text{cigar}}(t) = K_{D^2}(t) \cdot K_{S_{r_h}^2}(t).$$

This factorization holds at the operator level on the discrete substrate.

**Sub-sprint sequence.** G4-3 opens the extended discrete-substrate track:

- **G4-3a** (this): constant-warp substrate + scoping
- G4-3a-cleanup: Hermitian discrete polar Laplacian (engineering)
- G4-3b: variable warp  $r(\rho)$  for asymptotic Schwarzschild
- G4-3c: discrete conical-defect deformation ( $N_\phi$  sweeps)
- G4-3d: continuum-limit heat-kernel asymptotics verification
- G4-4: warped Dirac spectrum
- G4-5: discrete replica method
- G4-6: full discrete-substrate  $S_{BH}$  derivation

Estimated commitment: a substantial multi-stage program, well beyond sprint scale.

**Honest scope of G4-3a.** G4-3a is *scoping*: substrate definition, constant-warp factorization at operator level, and sub-sprint plan. The sprint-scale driver uses a naive non-Hermitian discrete polar Laplacian; numerical heat-trace values are quantitatively wrong but the structural framework (factorization, conical parameterization, continuum-limit targets) is intact. A Hermitian discrete radial Laplacian on  $L^2(\rho d\rho)$  via standard symmetrization is sprint-scale engineering for G4-3a-cleanup.

**Connection to G4-2 continuum derivation.** G4-2 derived  $S_{BH} = A\Lambda^2/(12\pi)$  at the continuum level via the conical-defect / replica method. G4-3 onwards rebuilds this on the discrete substrate. The target is to show:

1. Discrete substrate has analog conical-defect contribution  $\rightarrow (1/12)(1/\alpha - \alpha)$  in continuum limit
2. Discrete Dirac on warped substrate  $\rightarrow$  G4-1  $S^2$  Dirac at constant warp
3. Discrete replica method gives  $S_{BH} \rightarrow A\Lambda^2/(12\pi)$

**Connection to G8 cutoff dependence.** The  $\Lambda^2$  prefactor in  $S_{BH}$  depends on the cutoff function  $f$  per G8. The discrete substrate inherits this calibration: any discrete-substrate  $S_{BH}$  derivation will have prefactor  $\propto \phi(2)/\phi(1)$ . The cutoff dependence is structural, not a feature of G4-3 specifically.

**Gravity-arc status.** Continuum gravity arc: G1–G8 complete (sprint-scale closure). Discrete-substrate gravity arc: G4-3a opens; G4-3b through G4-6 run to the full  $S_{BH}$  derivation on the discrete substrate. This is the natural commitment for the next major project arc.

Sprint G4-3a driver: `debug/g4_3_warped_substrate.py`; memo: `debug/g4_3_warped_substrate_memo.md`.

## 9 Two-Loop Identified Constant: $S_{\min}$

**Correction note (round-2 dossier, 2026-06-10).** The earlier subsection heading “Extended PSLQ irreducibility” and the claim that  $S_{\min}$  is “confirmed irreducible” were a basis-coverage artifact: all 15 PSLQ attempts (47-element and 100-element bases at 150–200 dps) never contained the level-2 weight-5–8 monomials ( $\pi^4 \ln 2$ ,  $\pi^6 \ln 2$ ,  $\text{Li}_5(1/2)$ ,  $\text{Li}_4(1/2) \ln 2$ ,  $\zeta(3) \ln^2 2$ , ...) that the correct room requires. The present text replaces those claims with the identified decomposition; see driver `debug/smin_dossier_round1.py` and the round-1 dossier `debug/smin_dossier_round1_memo.md` for the complete reduction chain.

The CG-weighted two-loop sunset sum is:

$$S_{\min} = \sum_{k=1}^{\infty} T(k)^2, \quad T(k) = 2\zeta(2, k + \tfrac{3}{2}) - \tfrac{1}{2}\zeta(4, k + \tfrac{3}{2}). \quad (70)$$

This is a depth-2 multiple Hurwitz zeta value at half-integer shift  $a = 3/2$ , intrinsic to the Dirac spectrum on  $S^3$ .

### 9.1 200-digit computation

$S_{\min}$  is computed to 200 verified digits using `mpmath.nsum` with Levin  $u$ -transform acceleration. The first 80 digits agree with the Sprint 5 three-method verification (direct summation + explicit tail; Levin  $u$ -transform; Euler–Maclaurin 26-term asymptotic):

$$S_{\min} = 2.47993\,69380\,34222\,55441\,35795\,00082\,93821\,44688\dots \quad (71)$$

### 9.2 Motivic identification: Hoffman double $t$ -values

$S_{\min}$  decomposes *exactly* into Hoffman double  $t$ -values — classical level-2 alternating MZVs, periods of  $\mathcal{MT}(\mathbb{Z}[1/2])$ . Notation:  $\lambda(s) = (1 - 2^{-s})\zeta(s) = \sum_{o \geq 1, \text{odd}} o^{-s}$ ;  $t(b, c) = \sum_{o_2 > o_1 \geq 1, \text{odd}} o_2^{-b} o_1^{-c}$  (Hoffman [20] descending convention: first index  $b$  on the larger summation variable).

**8- $t$ -value decomposition** (verified to residual  $1.8 \times 10^{-129}$  at 130 dps, re-verified at 260 dps):

$$\begin{aligned} S_{\min} = & 64[t(2, 1) - t(4, 1) - 3t(2, 2) + 3t(4, 2) - t(2, 3) + t(4, 3) + 3t(2, 4) - 3t(4, 4)] \\ & + 32 \sum_{s=3}^8 d_s (\lambda(s) - 1 - 3^{-s}), \end{aligned} \quad (72)$$

where the diagonal coefficients are  $d_3 = 1$ ,  $d_4 = -3$ ,  $d_5 = -2$ ,  $d_6 = 6$ ,  $d_7 = 1$ ,  $d_8 = -3$ .

**Stuffle reductions** (exact algebraic identities from the ordering decomposition  $\lambda(b)\lambda(c) = t(b, c) + t(c, b) + \lambda(b + c)$ ):

$$t(2, 2) = \tfrac{1}{2}(\lambda(2)^2 - \lambda(4)), \quad (73)$$

$$t(4, 4) = \tfrac{1}{2}(\lambda(4)^2 - \lambda(8)), \quad (74)$$

$$t(2, 4) + t(4, 2) = \lambda(2)\lambda(4) - \lambda(6). \quad (75)$$

(The antisymmetric combination  $t(2, 4) - t(4, 2)$  enters  $S_{\min}$  with coefficient zero, so only the reducible symmetric combination appears.)

**Odd-weight depth-1 reductions** (verified at  $\geq 1 \times 10^{-130}$ ; both are verbatim rows of Hoffman [20] Appendix A):

$$t(2, 1) = -\frac{7}{16} \zeta(3) + \frac{1}{8} \pi^2 \ln 2, \quad (76)$$

$$t(2, 3) = -\frac{31}{64} \zeta(5) + \frac{1}{16} \pi^2 \zeta(3). \quad (77)$$

**Remaining odd-weight components — identified.** PSLQ at 220 dps against the full conjectural level-2 bases (dim 8 at weight 5; 19 product monomials at weight 7) identifies both remaining components (residuals  $2.8 \times 10^{-221}$  and  $3.6 \times 10^{-222}$ ; `debug/data/s3_pslq_stageA.json`):

$$t(4, 1) = \frac{1}{96} \pi^4 \ln 2 - \frac{1}{64} \pi^2 \zeta(3) - \frac{31}{64} \zeta(5), \quad (78)$$

$$t(4, 3) = \frac{1}{128} \pi^4 \zeta(3) - \frac{5}{128} \pi^2 \zeta(5) - \frac{127}{256} \zeta(7). \quad (79)$$

Both identified forms are likewise verbatim rows of Hoffman [20] Appendix A ( $t(4, 1)$  directly;  $t(4, 3)$  after converting his  $t$ -basis) — the PSLQ runs independently rediscovered the published table (grounding verified against the source PDF, 2026-06-11). Assembling the full decomposition yields the explicit closed form

$$S_{\min} = 8\pi^2 \ln 2 - \frac{2}{3} \pi^4 \ln 2 - 3\pi^2 \zeta(3) + \frac{1}{2} \pi^4 \zeta(3) - \frac{5}{2} \pi^2 \zeta(5) + \frac{\pi^6}{4} - \frac{3}{2} \pi^4 - \frac{\pi^8}{96}, \quad (80)$$

verified against the Hurwitz-tail anchor at 130 dps (residual  $1.7 \times 10^{-129}$ ; `debug/smin_final_assembly.py`). Two structural features are exact:  $\zeta(7)$  cancels against the  $\lambda(7)$  diagonal term, and all rational constants cancel to zero. Hence  $S_{\min}$  lies in  $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ : the two-loop constant carries *no irreducible depth-2 content at all* — every double  $t$ -value in its decomposition reduces to depth  $\leq 1$ .

**Basis-coverage audit of the prior “irreducible” claim.** The 47-element basis (all 15 PSLQ failures at 150 dps) and the 100-element basis (200 dps) both contained zero level-2 monomials in the weight-5–8 rooms. Specifically, none of  $\pi^4 \ln 2$ ,  $\pi^6 \ln 2$ ,  $\text{Li}_4(1/2) \ln 2$ ,  $\text{Li}_5(1/2)$ ,  $\text{Li}_6(1/2)$ ,  $\text{Li}_7(1/2)$ ,  $\zeta(3) \ln^2 2$ ,  $\dots$  were included. Of the conjectural weight-5 level-2 basis (dimension 8), those tests contained only 2 out of 8 elements ( $\zeta(5)$  and  $\pi^2 \zeta(3)$ ); at weight 7, approximately 4 out of 21. The “PSLQ-irreducible” label was a coverage artifact, not a statement about motivic depth.

### 9.3 Inner photon sum: closed-form channel count

The inner photon sum  $I(n_1, n_2) = \sum_{q \text{ allowed}} W(n_1, n_2, q) \cdot d_q^T / \mu_q^p$  (at  $p = 1$ :  $d_q^T / \mu_q = 1$ ) counts the total SO(4) CG-weighted vertex channels. Four structural properties:

- (i)  $I(0, n_2) = 0$  for all  $n_2$  (structural zero: vertex parity at  $n_{\text{ext}} = 0$  forces  $2n_2$  odd, impossible).
- (ii)  $I(n_1, n_2) = I(n_2, n_1)$  (symmetric).
- (iii)  $I(n, n) = 2n - 2 = 2(n - 1)$  for  $n \geq 1$ .
- (iv) **Closed form** (proved in dossier round-1c; verified against `geovac/qed_vertex.py` for all  $n_1, n_2 \leq 8$  in the frozen test `tests/test_smin_decomposition.py`; the archived dossier run swept to  $\leq 12$ ):

$$I(n_1, n_2) = 2 \min(n_1, n_2) - 1 - [n_1 = n_2], \quad n_1, n_2 \geq 1. \quad (81)$$

## 9.4 Exact sunset relation

The closed form (81) gives, by direct substitution  $I = 2 \min - 1 - \delta$  into the double sum:

$$S_{\text{sunset}} = 2 S_{\text{min}} - P^2 - Q, \quad (82)$$

where  $S_{\text{sunset}} = \sum_{n_1, n_2 \geq 1} I(n_1, n_2) \phi(n_1) \phi(n_2)$  is the CG-exact two-loop sunset,

$$P = D(4) - \frac{64}{81}, \quad D(4) = \pi^2 - \frac{\pi^4}{12}, \quad (83)$$

$$Q = 4 \zeta(4, \frac{5}{2}) - 2 \zeta(6, \frac{5}{2}) + \frac{1}{4} \zeta(8, \frac{5}{2}), \quad (84)$$

and the numerical value  $S_{\text{sunset}} = 3.89451620519784104455 \dots$  is verified at 60 dps in `tests/test_smin_decomposition`. The correction terms  $P^2$  and  $Q$  are depth-1 elements of  $\mathcal{MT}(\mathbb{Z})$  (pure-Tate,  $\pi$ -even);  $S_{\text{min}}$  itself reduces to depth  $\leq 1$  (Eq. (80)), so the two-loop sunset carries no irreducible depth-2 content.

## 10 Three-Loop Results

### 10.1 Factorized $O(N^3)$ algorithm

The three-loop iterated sunset has chain topology: three electron lines  $(n_1, n_2, n_3)$  connected by two photon lines  $(q_1, q_2)$ , with vertex selection at each junction. The chain topology permits factorization into partial sums:

$$S^{(3)} = \sum_{n_2=0}^N \frac{g_{n_2}}{|\lambda_{n_2}|^a} \cdot L(n_2) \cdot R(n_2), \quad (85)$$

where

$$L(n_2) = \sum_{\substack{n_1, q_1 \\ \text{allowed}}} \frac{W(n_1, n_2, q_1) g_{n_1} d_{q_1}^T}{|\lambda_{n_1}|^a \mu_{q_1}^p}, \quad R(n_2) = \sum_{\substack{n_3, q_2 \\ \text{allowed}}} \frac{W(n_2, n_3, q_2) g_{n_3} d_{q_2}^T}{|\lambda_{n_3}|^a \mu_{q_2}^p}. \quad (86)$$

$W(n_1, n_2, q)$  is NOT symmetric in  $(n_1, n_2)$ , so  $L$  and  $R$  are computed independently. This reduces the five-fold sum  $O(N^5)$  to  $O(N^3)$  (each of  $L, R$  is  $O(N^2)$  precomputed, then the outer sum is  $O(N)$ ).

**Verification:** The factorized algorithm matches the direct  $O(N^5)$  sum to  $< 10^{-78}$  at  $n_{\text{max}} = 10, 15$  across four parameter combinations (default  $a = 4, p = 1$ ; vertex-restricted; higher exponents  $a = 6, p = 2$ ; with and without CG weights). The factorized algorithm at  $n_{\text{max}} = 50$  completes in  $< 60$  seconds. Tests: 7 in `TestFactorizedMatchesDirect` and `TestFactorizedConvergence` (all passing; the frozen factorized-vs-direct gate is  $10^{-25}$  — the  $< 10^{-78}$  agreement quoted above is the measured value at the working precision, re-confirmed 2026-07-04).

### 10.2 Convergence analysis (corrected 2026-06-11)

At default exponents ( $a = 4, p = 1$ ) with CG weights, the raw partial sums converge far too slowly for direct evaluation. An empirical power-law fit over  $N \in [20, 120]$  gave

$$\Delta S^{(3)}(N) \approx 270 \cdot N^{-1.31}, \quad (87)$$

but the closure-sprint analysis showed the fitted exponent is an artifact: by the channel-count closed form (81) the inner partial sums grow as  $L(n_2) = 4 \ln n_2 + O(1)$ , so the increments follow

$(\ln N)^2/N^2$  — a log-modulated law whose local log-slope  $2 - 2/\ln N$  runs through 1.3–1.6 over the fit window and masquerades as a power law. The true remaining tail at  $N = 120$  is  $\approx 7$  units, i.e. *zero* reliable digits from raw partial sums (cf. Table 9: the  $n_{\max} = 50$  partial sum is 15.23 against a limit of 31.57).

The reliable direct evaluation is a rigorous bracket with no series acceleration anywhere: the factorized sum has positive terms, so partial sums with the closed-form full-range inner sum give a strict lower bound, and the explicit inequalities  $L(m) \leq 4(\ln m + 1)$ ,  $\phi(m) \leq 2/(m + 1)^2$  give an integral tail bound. At  $N = 4 \times 10^6$ ,

$$S^{(3)} \in [31.57063, 31.57300] \quad (88)$$

(driver `debug/s3_closure_anchor.py`). The corrected decomposition assembly (Sec. 10.4) gives the canonical value, inside the bracket:

$$S^{(3)} = 31.57256\,12075\,12022\,75476\,19295\,45069\dots \quad (89)$$

### 10.3 PSLQ identification attempts (historical)

At the fast-converging exponents  $a = 8$ ,  $p = 2$ , approximately 20 reliable digits are available at  $n_{\max} = 50$ . PSLQ with an 18-element depth-3 MZV basis (including  $S_{\min}$ ,  $\zeta(3)^2$ ,  $\pi^2 S_{\min}$ ,  $G \cdot S_{\min}$ , and all cross-products with  $\pi^2$ ,  $\zeta(3)$ ,  $G$ ,  $\beta(4)$ ) returns no relation. This route is superseded by the decomposition closure below.

The unrestricted three-loop sum factorizes exactly as  $D(a)^3$ : at  $a = 4$ ,  $D(4)^3 = (\pi^2 - \pi^4/12)^3 \approx 5.379$ , purely  $\pi^{\text{even}}$ .

### 10.4 Decomposition closure: the four trailing-1 identifications

Notation:  $t_3(b_1, b_2, b_3) = \sum_{o_1 > o_2 > o_3 \geq 1 \text{ odd}} o_1^{-b_1} o_2^{-b_2} o_3^{-b_3}$ , the Hoffman triple  $t$ -value in the same descending convention as  $t(b, c)$ . The channel-count closed form (81) expands the CG-weighted chain sum into 12 triple  $t$ -values, 26 double  $t$ -values, and  $\lambda$ /boundary terms (exact in Fraction arithmetic at every truncation; `tests/test_s3_decomposition.py`).

Two errors in the stage-1 numerics were diagnosed and closed on 2026-06-11 (memo `debug/sprint_s3_closure`) both with one root cause: Levin series acceleration silently mis-converges on summands with log-arithmic factors, with *precision-dependent* error. (i) The evaluator for trailing argument  $b_3 = 1$  fed the odd-harmonic factor  $\tilde{H}(j) \sim \frac{1}{2} \ln j$  into the accelerator;  $t_3(2, 1, 1)$  was off by  $-4.7 \times 10^{-4}$  at 130 dps and  $-2.9 \times 10^{-4}$  at 220 dps, poisoning the earlier evaluation  $S^{(3)} \approx 30.615$  through its table coefficient +2048 — the corrected shift +0.9572 is accounted for term-by-term to  $1.7 \times 10^{-15}$ . (ii) The same disease invalidated the factorized-anchor evaluation 30.2197 ( $L(m) \sim 4 \ln m$ ); the bracket (88) replaces it. The repaired evaluator uses Abel summation over the harmonic weight,  $t_3(b_1, b_2, 1) = \sum_{k \geq 1} (2k - 1)^{-1} A(k)$  with  $A(k) = \sum_{j \geq k} a_j$  — both sums log-free. With corrected values the stuffle witnesses pass at  $5 \times 10^{-216}$  (R2; stage-1 residual  $6.4 \times 10^{-12}$ ) and  $8 \times 10^{-217}$  (R6; was  $9.8 \times 10^{-20}$ ); the stage-1 residuals decompose exactly into the individual evaluator errors.

PSLQ at 220/340 dps against the complete conjectural level-2 weight-homogeneous bases identifies all four previously pending components (residuals  $2.6 \times 10^{-216}$  to  $7.3 \times 10^{-339}$  in the archived dossier run; the frozen regression `tests/test_s3_decomposition.py` re-gates the identifications at

$10^{-30}/45$  dps):

$$t_3(2, 1, 1) = \frac{1}{2}\text{Li}_4\left(\frac{1}{2}\right) + \frac{1}{48}\ln^4 2 + \frac{1}{24}\pi^2 \ln^2 2 - \frac{19}{5760}\pi^4, \quad (90)$$

$$t_3(4, 1, 1) = \frac{1}{192}\pi^4 \ln^2 2 - \frac{1}{64}\pi^2 \ln 2 \zeta(3) - \frac{11}{322560}\pi^6 - \frac{1}{2}t(5, 1) - \frac{7}{128}\zeta(3)^2, \quad (91)$$

$$t_3(3, 2, 1) = \frac{3}{64}\pi^2 \ln 2 \zeta(3) - \frac{1}{9216}\pi^6 - \frac{1}{2}t(5, 1) - \frac{49}{256}\zeta(3)^2, \quad (92)$$

$$t_3(3, 4, 1) = \frac{1}{768}\pi^4 \ln 2 \zeta(3) + \frac{5}{128}\pi^2 \ln 2 \zeta(5) + \frac{61}{2903040}\pi^8 - \frac{1}{256}\pi^2 \zeta(3)^2 \\ + \frac{1}{2}t(5, 3) - \frac{1}{2}t(7, 1) - \frac{217}{512}\zeta(3)\zeta(5). \quad (93)$$

Every component of the  $S^{(3)}$  decomposition except the weight-10 graded piece ( $W_{10} = -2048 t_3(4, 3, 3) - 1024 t_3(4, 4, 2) - 512 t_2(8, 2) - 1536 t_2(4, 6)$ , the full four-term graded piece; level-2 basis dimension 89, not executed; parity heuristic predicts depth  $\leq 2$ ) is now an identified element of the level-2 ring, with the genuine depth-2 generators  $t(5, 1)$  (weight 6) and  $t(5, 3)$ ,  $t(7, 1)$  (weight 8) entering at coefficients  $\pm \frac{1}{2}$ . **Realized motivic depth at three loops:  $\leq 2$  on every identified component** — one rung above  $S_{\min}$ 's depth  $\leq 1$ , all catalogued, nothing new minted. Falsifier: `tests/test_s3_decomposition.py` (including the rigorous-bracket regression guard for the Levin/log failure mode).

**Full assembly and the weight-10 block.** Assembling the complete decomposition with all identified components (symbolic assembly verified against the canonical value (89) at residual  $4.1 \times 10^{-197}$ ) yields a stronger statement than the entry-level reductions: *every depth-2 generator cancels in the total*.  $t(5, 1)$ ,  $t(5, 3)$ ,  $t(7, 1)$ , and  $\text{Li}_4(\frac{1}{2})$  all carry total coefficient zero, as does standalone  $\zeta(11)$  — the  $S^{(3)}$  echo of  $S_{\min}$ 's exact  $\zeta(7)$  cancellation. The identified part of  $S^{(3)}$  is a polynomial in  $\pi^2$ ,  $\ln 2$ , and odd zeta values alone. The remaining weight-10 block reduces, by exact Fraction Gaussian elimination on the complete weight-10 quasi-shuffle closure (rank 157 on 255 words; identity verified numerically at  $7 \times 10^{-59}$ ), to

$$W_{10}/256 = 4\lambda(10) - \lambda(2)\lambda(8) - 3\lambda(4)\lambda(6) - 4\lambda(3)a(4, 3) - 2\lambda(4)a(4, 2) \\ + a(8, 2) + 4a(7, 3) - 3a(6, 4) - 4t_3(2, 4, 4) - 8t_3(3, 3, 4), \quad (94)$$

where  $a(b, c) := t(b, c) - t(c, b)$  and  $\lambda(s) = (1 - 2^{-s})\zeta(s)$ ; the lower-weight  $a(4, 2)$ ,  $a(4, 3)$  are identified objects, leaving *five* surviving generators: the antisymmetric doubles  $a(8, 2)$ ,  $a(7, 3)$ ,  $a(6, 4)$  and the triples  $t_3(2, 4, 4)$ ,  $t_3(3, 3, 4)$ . A structural obstruction protects this block from stuffle methods: the depth- $\leq 3$  relation content of the quasi-shuffle closure saturates already at the  $\lambda \cdot t$  product stage, so no stuffle-product engineering at any depth can reduce it further.

**The weight-10 block closed (2026-06-12).** The missing input is the Charlton–Hoffman symmetry theorem [24] — level-2 regularized-double-shuffle content invisible to the stuffle-only system. Its generating-series form (their Thm. 2.21) at depth  $m = 3$ ,  $\phi = \underline{0}$ , extracted at the coefficient of  $y_1^{a-1}y_2^{b-1}y_3^{c-1}$  by exact-Fraction series algebra, gives the reversal-pair sums with every product term explicit (each instance used was verified numerically at  $10^{-78}$  or better; driver `debug/s3.w10_ident.py`, 24/24 gates):

$$t_3(4, 4, 2) + t_3(2, 4, 4) = \frac{5}{6144}\pi^4 \zeta(3)^2 - \frac{5}{256}\pi^2 \zeta(3)\zeta(5) - \frac{1}{256}\pi^2 \zeta(5, 3) + \frac{137}{99532800}\pi^{10}, \quad (95)$$

$$t_3(4, 3, 3) + t_3(3, 3, 4) = \frac{37}{12288}\pi^4 \zeta(3)^2 - \frac{25}{1024}\pi^2 \zeta(3)\zeta(5) - \frac{889}{2048}\zeta(3)\zeta(7) - \frac{1}{256}\pi^2 \zeta(5, 3) + \frac{251}{58060800}\pi^{10}, \quad (96)$$



where  $\zeta(5, 3) = \sum_{m>n\geq 1} m^{-5}n^{-3}$  is the classical weight-8 depth-2 MZV generator (conjecturally irreducible over products of single zetas), entering through the weight-8 layer of the theorem's  $\zeta$ -side product terms. The pair *differences* follow from four ordinary stuffles  $t(x)t_2(y, z)$ ,  $(x, y, z) \in \{(2, 4, 4), (4, 2, 4), (3, 3, 4), (3, 4, 3)\}$ , whose pairwise combinations eliminate the auxiliary triples  $t_3(4, 2, 4)$ ,  $t_3(3, 4, 3)$ . Solving the two  $2 \times 2$  systems identifies all four weight-10 triples individually:

$$t_3(2, 4, 4) = \frac{1}{4}a(8, 2) - \frac{1}{4}a(6, 4) - \frac{1}{6144}\pi^4\zeta(3)^2 - \frac{5}{512}\pi^2\zeta(3)\zeta(5) - \frac{1}{512}\pi^2\zeta(5, 3) + \frac{373}{174182400}\pi^{10}, \quad (97)$$

$$t_3(3, 3, 4) = \frac{1}{4}a(7, 3) - \frac{1}{4}a(6, 4) + \frac{1}{12288}\pi^4\zeta(3)^2 + \frac{5}{1024}\pi^2\zeta(3)\zeta(5) - \frac{889}{4096}\zeta(3)\zeta(7) - \frac{1}{512}\pi^2\zeta(5, 3) + \frac{251}{116121600}\pi^{10}, \quad (98)$$

each gated against its independently computed 200-digit value at residual  $\leq 6 \times 10^{-197}$  (likewise  $t_3(4, 4, 2)$ ,  $t_3(4, 3, 3)$ ).

**Collapse theorem and the full closed form.** Substituting Eqs. (97)–(98) into the full  $S^{(3)}$  assembly, the three antisymmetric doubles  $a(8, 2)$ ,  $a(7, 3)$ ,  $a(6, 4)$  *cancel identically*: the explicit  $a$ -content of the weight-10 corollary is exactly absorbed by the  $a$ -content of the two triples (symbolic verification in the frozen falsifier). The weight-10 block therefore contributes

$$S^{(3)} - \text{Ident} = -7\pi^4\zeta(3)^2 + 70\pi^2\zeta(3)\zeta(5) + 6\pi^2\zeta(5, 3) + \frac{179}{170100}\pi^{10}, \quad (99)$$

where Ident is the identified part of the symbolic assembly, and the complete closed form

$$S^{(3)} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5), \zeta(7), \zeta(9), \zeta(5, 3)] \quad (100)$$

is explicit, with  $\zeta(5, 3)$  the *only* constant beyond the depth- $\leq 1$  ring, at coefficient  $6\pi^2$ . The closed form reproduces the canonical value (89) at residual  $1.2 \times 10^{-198}$  (200 dps). Two structural readings: (i) the transient-generator cancellation pattern of the symbolic assembly ( $t(5, 1)$ ,  $t(5, 3)$ ,  $t(7, 1)$ ,  $\text{Li}_4(\frac{1}{2})$  all cancel) extends through weight 10 — the level-2 antisymmetric doubles  $a(8, 2)$ ,  $a(7, 3)$ ,  $a(6, 4)$  also carry total coefficient zero, and the *value* of  $S^{(3)}$  is independent of these still-unreduced generators; (ii) the genuinely new depth-2 content at three loops enters on the *level-1* side ( $\zeta(5, 3) \in \mathcal{MT}(\mathbb{Z})$ ), while the level-2-specific content (the  $\ln 2$  sector) stays at realized depth  $\leq 1$  — new depth arrives from level 1, new level stays at depth  $\leq 1$ . Falsifier: `tests/test_s3_w10_identification.py` (symmetry-pair sums, individual triples, symbolic  $a$ -collapse, full closed form at 50 and 200 dps with  $\zeta(5, 3)$  computed live).

**Literature grounding (verified against the sources).** All weight- $\leq 7$  closed forms in this section and Sec. 9.2 are verbatim rows of Hoffman's Appendix A [20] — including the triples  $t_3(2, 1, 1)$ ,  $t_3(4, 1, 1)$ ,  $t_3(3, 2, 1)$ , after eliminating  $\zeta(\bar{5}, 1)$  via his own  $t(5, 1)$  row. The PSLQ identifications above therefore independently rediscovered the published table — a two-route verification. The weight- $\geq 8$  reductions rest on the theorem-grade chain: Hoffman Cor. 4.1 (exact conversion to alternating MZVs) + parity/reduction theorems for alternating sums [21, 22] + the data mine's proven weight- $\leq 12$  reductions [23]. The explicit form Eq. (93) appears to be new (checked against the explicit families of [24], whose  $t$ -value convention reverses the argument order).

## 11 Depth- $k$ Tower

**Proposition 4** (Depth- $k$  tower, corrected). *At  $k$  loops on  $S^3$  with vertex restrictions and CG weights:*

- $k = 1$ :  $\pi^{\text{even}}$  only (Theorem 1, proven).
- $k = 2$ :  $S_{\min}$  is an identified element of  $\mathcal{MT}(\mathbb{Z}[1/2])$  at level 2. The depth is bounded by 2 (the nesting depth of the min-weight double sum) but the content decomposes into Hoffman double  $t$ -values, which are classical period-ring elements (Sec. 9.2). The prior label “irreducible” was a basis-coverage artifact.
- $k = 3$ :  $S^{(3)}$  is a fully identified element of  $\mathbb{Q}[\pi^2, \ln 2, \zeta(\text{odd}), \zeta(5, 3)]$  with realized depth  $\leq 2$  unconditionally (Eq. (100)): the sole depth-2 constant is the level-1 generator  $\zeta(5, 3)$ , at coefficient  $6\pi^2$ ; every level-2 depth-2 generator ( $t(5, 1)$ ,  $t(5, 3)$ ,  $t(7, 1)$ ,  $a(8, 2)$ ,  $a(7, 3)$ ,  $a(6, 4)$ ) appears transiently and cancels in the total (Sec. 10.4). The earlier “independence from  $S_{\min}$ ” statement (a 20-digit PSLQ at non-physical exponents) is superseded by the identification.
- $k = 4$ : the realized depth of  $S^{(4)}$  is  $\leq 3$ . Depth-4 content enters  $S^{(4)}$  only through the four-chain core  $C_4$  (the  $o$ -space relation gives  $C_4$  coefficient +1; the other cores are  $\leq$  depth-3), so the verdict reduces to whether  $C_4$ ’s depth-4 part reduces to depth  $\leq 3$ . At weights 9, 11, 13 the quasi-shuffle (stuffle) relation system spans the entire depth-4 word space (rank = number of depth-4 words: 35, 84, 165; exact rank over two independent primes), collapsing  $C_4$ ’s depth-4 content to depth  $\leq 3$  unconditionally. At weights 5, 7 the stuffle system is rank-deficient, but these weights lie below the threshold for any depth- $\geq 3$  irreducible: weight-5 values reduce to single polylog constants (depth  $\leq 1$ ) and weight-7 to depth  $\leq 2$  (the  $w \leq 7$  GeoVac multiple- $t$  values match Hoffman’s Appendix-A reductions, verified previously). The explicit closed form of  $S^{(4)}$  remains open; the depth verdict does not depend on it.

Loop order bounds the depth of the classical level-2 multiple- $t$ -value content; it does not mint new irreducible transcendentals automatically. Realized depth so far grows strictly more slowly than loop order ( $\leq k - 1$  at  $k = 2, 3, 4$ ).

The mechanism: each vertex coupling creates one level of Hurwitz nesting. One vertex produces a depth-2 double  $t$ -value sum ( $S_{\min}$ ). Two vertices produce a depth- $\leq 3$  triple  $t$ -value sum ( $S^{(3)}$ ). The prediction extends:  $k$  vertices produce a depth- $\leq (k + 1)$  multiple  $t$ -value in  $\mathcal{MT}(\mathbb{Z}[1/2])$ . Whether the surviving irreducible depth- $k$  component is new (not reducible by parity/stuffle within the level-2 algebra) is a question for each  $k$  separately — the structural nesting is genuine, but the motivic content is classical at  $k = 2$  and, on all identified components, at  $k = 3$  (realized depth  $\leq 2$ ). This is the  $S^3$  analog of the increasing motivic depth observed in flat-space multi-loop Feynman integrals [15, 19].

## 12 Fine-Structure Euler Sum Cancellation

The Dirac Dirichlet series  $D(s)$  studied in the preceding sections sums over the *free* Camporesi–Higuchi spectrum with uniform degeneracy weights  $g_n$ . A complementary probe arises from the *exact* Dirac hydrogen spectrum, where the degeneracy-weighted energy expansion in powers of  $(Z\alpha)^2$  generates Dirichlet sums with state-dependent coefficients that mix harmonic numbers of varying depth.

### 12.1 Setup

The exact Dirac energy for quantum numbers  $(n, j)$  is

$$E_{nj} = mc^2 \left( 1 + \frac{(Z\alpha)^2}{(n_r + \sqrt{\kappa^2 - (Z\alpha)^2})^2} \right)^{-1/2} - mc^2, \quad (101)$$

where  $\kappa = j + \frac{1}{2}$  and  $n_r = n - \kappa$ . Expanding in powers of  $x = (Z\alpha)^2$ ,

$$\sum_j d(n, j) \frac{E_{nj}}{mc^2} = \sum_{p=1}^{\infty} c_p(n) x^p, \quad (102)$$

where  $d(n, j) = 2(2j+1)$  for  $j < n - \frac{1}{2}$  and  $d(n, j) = 2j+1$  for  $j = n - \frac{1}{2}$ . The degeneracy-weighted Dirichlet sums are

$$D_p = \sum_{n=1}^{\infty} c_p(n). \quad (103)$$

## 12.2 Closed forms for $c_p(n)$

Symbolic expansion of the exact Dirac formula and summation over  $j$  gives closed forms involving generalized harmonic numbers  $H_{n-1}^{(r)} = \sum_{k=1}^{n-1} k^{-r}$ :

$$c_2(n) = -\frac{5n-4}{4n^3}, \quad (104)$$

$$c_3(n) = \frac{19n-20}{8n^5} - \frac{3}{2} \frac{H_{n-1}}{n^4} - \frac{1}{2} \frac{H_{n-1}^{(2)}}{n^3}. \quad (105)$$

At order  $p=2$  and  $p=3$ , the sums  $D_2$  and  $D_3$  collapse to pure single-zeta values:

$$D_2 = -\frac{5}{4} \zeta(2) + \zeta(3), \quad D_3 = \frac{19}{8} \zeta(4) - \frac{11}{4} \zeta(5). \quad (106)$$

## 12.3 $\zeta(3)$ complementarity

The  $p=2$  coefficient defines the *fine-structure shell sum*

$$\mathcal{S}(n) = \sum_j d(n, j) \left( \frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) = \frac{n(5n-4)}{2}, \quad (107)$$

so that  $c_2(n) = -\mathcal{S}(n)/(2n^4)$ . This decomposes into a *conformal* piece  $-3n^2/2$  (from the scalar curvature  $R/8 = 3/4$  on unit  $S^3$ ) and a *quantum defect* piece  $2n(2n-1)$ , the double hexagonal number.

In the Fock convention ( $n \geq 1$ ), the single-chirality Dirac mode degeneracy is  $g_n^{\text{Dirac}} = 2n(n+1)$ .

**Theorem 7** ( $\zeta(3)$  complementarity). *For all  $n \geq 1$ ,*

$$\mathcal{S}(n) + g_n^{\text{Dirac}} = \frac{9n^2}{2} = |\lambda_0|^2 \times n_{\text{Schr}}^2, \quad (108)$$

where  $|\lambda_0| = 3/2$  is the lowest Dirac eigenvalue on  $S^3$  and  $n_{\text{Schr}}^2 = 2n^2$  is the Schrödinger degeneracy (including spin). At the Dirichlet series level,

$$D_{\mathcal{S}}(s) + D_g(s) = \frac{9}{2} \zeta(s-2) \quad \text{for all } s, \quad (109)$$

where  $D_{\mathcal{S}}(s) = \sum_{n \geq 1} \mathcal{S}(n) n^{-s}$  and  $D_g(s) = \sum_{n \geq 1} g_n^{\text{Dirac}} n^{-s}$ . The  $\zeta(s-1)$  terms cancel identically.

*Proof.*  $\mathcal{S}(n) = 5n^2/2 - 2n$  and  $g_n^{\text{Dirac}} = 2n^2 + 2n$ ; their sum is  $9n^2/2$ . Therefore  $D_{\mathcal{S}}(s) = \frac{5}{2} \zeta(s-2) - 2 \zeta(s-1)$  and  $D_g(s) = 2 \zeta(s-2) + 2 \zeta(s-1)$ ; summing gives  $\frac{9}{2} \zeta(s-2)$ .  $\square$

The cancellation of  $\zeta(s-1)$  is the physical content: at odd  $s$ ,  $\zeta(s-1)$  is even-zeta ( $\pi^{\text{even}}$ ) and harmless, but at even  $s$ ,  $\zeta(s-1)$  carries odd-zeta content ( $\zeta(3), \zeta(5), \dots$ ). At  $s = d_{\text{max}} = 4$ :

$$D_S(4) = \frac{5}{2} \frac{\pi^2}{6} - 2\zeta(3) = \frac{5F}{2} - 2\zeta(3), \quad (110)$$

confirming that the quantum defect (the  $-2n$  piece of  $\mathcal{S}$ ) is the physical mechanism injecting  $\zeta(3)$  into the fine-structure spectrum—with sign opposite to the  $+2\zeta(3)$  that appears in  $D_g(4) = 2\zeta(2) + 2\zeta(3)$ . Their sum  $\frac{9}{2}\zeta(2) = 3\pi^2/4$  is purely  $\pi^{\text{even}}$ , returning to the second-order (scalar) sector.

The prefactor  $9/2 = |\lambda_0|^2 \times 2 = (\text{Dirac mass gap})^2 \times (\text{spinor rank})$  links through the Lichnerowicz formula  $D^2 = \Delta + R/4$  back to the  $S^3$  geometry: the complementarity sum recovers the scalar Laplacian sector exactly, closing the chain from the packing construction through  $S^3$  and the Dirac operator to the fine-structure Dirichlet series.

This is Theorem 1 expressed in physical terms: the  $D^2$  spectral zeta is even-zeta only *because* the fine-structure energy distribution and the Dirac mode counting are transcendently complementary. Paper 18's taxonomy predicted this separation: calibration ( $\pi^{\text{even}}$  from the second-order curvature  $R/8$ ) versus spinor-intrinsic (mixed zeta from the first-order quantum defect).

## 12.4 $D_4$ at weight 7: product zeta survives

At order  $p = 4$ ,

$$\begin{aligned} c_4(n) = & -\frac{5(41n-40)}{64n^7} + \frac{15}{4} \frac{H_{n-1}}{n^6} - \frac{1}{4} \frac{H_{n-1}^{(2)}}{n^5} \\ & - \frac{3}{4} \frac{H_{n-1}^{(3)}}{n^4} - \frac{1}{4} \frac{H_{n-1}^{(4)}}{n^3}. \end{aligned} \quad (111)$$

The harmonic-number sums decompose into weight-7 Euler sums  $S_{r,s} = \sum_{n \geq 1} H_n^{(r)}/n^s$  via the identity  $H_{n-1}^{(r)} = \zeta(r) - \zeta(r, n)$  and Euler's formula  $S_{1,6} = 4\zeta(7) - \zeta(2)\zeta(5) - \zeta(3)\zeta(4)$  [26].

**Observation 1** (Euler sum cancellation at weight 7). *At 200 dps, PSLQ identifies*

$$D_4 = -\frac{205}{64} \zeta(6) + \frac{71}{8} \zeta(7) - \frac{9}{2} \zeta(3)\zeta(4), \quad (112)$$

*with residual  $< 10^{-195}$  (archived PSLQ run; the value is pinned live in `tests/test_paper28_gap_closure_pins.py` and the  $D_4/D_5$  decimals were independently cross-verified by direct partial summation, 2026-07-04). The product  $\zeta(2)\zeta(5)$  cancels algebraically: its coefficient vanishes exactly (verified in exact rational arithmetic from the Euler sum decomposition), while  $\zeta(3)\zeta(4)$  survives with coefficient  $-9/2$ .*

This breaks the pure-single-zeta pattern:  $D_2$  and  $D_3$  involve only  $\zeta(s)$  values, whereas  $D_4$  introduces the first zeta product. The four weight-7 Euler sums, independently identified by PSLQ, are

| Sum       | $\zeta(7)$ | $\zeta(2)\zeta(5)$ | $\zeta(3)\zeta(4)$ |
|-----------|------------|--------------------|--------------------|
| $S_{1,6}$ | 4          | -1                 | -1                 |
| $S_{2,5}$ | -10        | 5                  | 2                  |
| $S_{3,4}$ | 18         | -10                | 0                  |
| $S_{4,3}$ | -17        | 10                 | 1                  |
| $S_{5,2}$ | 11         | -4                 | -2                 |

The stuffle relations  $S_{r,s} + S_{s,r} = \zeta(r)\zeta(s) + \zeta(r+s)$  are satisfied to  $< 10^{-200}$ .

## 12.5 $D_5$ at weight 9: product survival rule

At order  $p = 5$ ,

$$c_5(n) = \frac{7(71n-72)}{128n^9} - \frac{105}{16} \frac{H_{n-1}}{n^8} + \frac{45}{16} \frac{H_{n-1}^{(2)}}{n^7} + \frac{5}{4} \frac{H_{n-1}^{(3)}}{n^6} - \frac{3}{8} \frac{H_{n-1}^{(4)}}{n^5} - \frac{15}{32} \frac{H_{n-1}^{(5)}}{n^4} - \frac{5}{32} \frac{H_{n-1}^{(6)}}{n^3}, \quad (113)$$

verified as exact rational equality against direct sympy expansion for  $n = 1, \dots, 15$ . The Dirichlet sum decomposes via weight-9 Euler sums  $S_{r,s}$  for  $r + s = 9$  (computed via Hurwitz zeta to 250 dps, stuffle relations verified to  $< 10^{-200}$ ):

| Sum       | $\zeta(9)$ | $\zeta(2)\zeta(7)$ | $\zeta(3)\zeta(6)$ | $\zeta(4)\zeta(5)$ |
|-----------|------------|--------------------|--------------------|--------------------|
| $S_{2,7}$ | $-35/2$    | 7                  | 2                  | 4                  |
| $S_{3,6}$ | $85/2$     | -21                | 0                  | -6                 |
| $S_{4,5}$ | $-125/2$   | 35                 | 0                  | 5                  |
| $S_{5,4}$ | $127/2$    | -35                | 0                  | -4                 |
| $S_{6,3}$ | $-83/2$    | 21                 | 1                  | 6                  |
| $S_{7,2}$ | $37/2$     | -6                 | -2                 | -4                 |

PSLQ at 200 dps identifies

$$D_5 = \frac{497}{128} \zeta(8) - \frac{467}{16} \zeta(9) + \frac{385}{32} \zeta(3)\zeta(6) + \frac{75}{8} \zeta(4)\zeta(5), \quad (114)$$

with residual  $< 10^{-190}$ . The product  $\zeta(2)\zeta(7)$  cancels algebraically (verified from the Euler sum collection), while both  $\zeta(3)\zeta(6)$  and  $\zeta(4)\zeta(5)$  survive. This extends the pattern from  $D_4$ : product terms grow from 0 (at  $D_2, D_3$ ) to 1 (at  $D_4$ ) to 2 (at  $D_5$ ).

## 12.6 $D_6$ at weight 11: product survival rule extended

At order  $p = 6$  the weight-11 Euler sums  $S_{r,s}$  for  $r + s = 11$  are computed via Hurwitz zeta to 400 dps, with stuffle relations verified to  $< 10^{-300}$ . Seven nontrivial sums ( $r = 1, \dots, 8, r \neq 3$ ) contribute through the  $\beta_r$  coefficients of the  $c_6(n)$  expansion. Individual decompositions are found by PSLQ against the weight-11 depth-1 MZV basis  $\{\zeta(11), \zeta(2)\zeta(9), \zeta(3)\zeta(8), \zeta(4)\zeta(7), \zeta(5)\zeta(6)\}$ :

| Sum        | $\zeta(11)$ | $\zeta(2)\zeta(9)$ | $\zeta(3)\zeta(8)$ | $\zeta(4)\zeta(7)$ | $\zeta(5)\zeta(6)$ |
|------------|-------------|--------------------|--------------------|--------------------|--------------------|
| $S_{1,10}$ | 6           | -1                 | -1                 | -1                 | -1                 |
| $S_{2,9}$  | -27         | 9                  | 2                  | 6                  | 4                  |
| $S_{4,7}$  | $-329/2$    | 84                 | 0                  | 21                 | 4                  |
| $S_{5,6}$  | $463/2$     | -126               | 0                  | -21                | 0                  |
| $S_{6,5}$  | $-461/2$    | 126                | 0                  | 21                 | 1                  |
| $S_{7,4}$  | $331/2$     | -84                | 0                  | -20                | -4                 |
| $S_{8,3}$  | -82         | 36                 | 1                  | 15                 | 6                  |

The  $\zeta(3)\zeta(8)$  column has a sparse support pattern: nonzero only for  $S_{1,10}$ ,  $S_{2,9}$ , and  $S_{8,3}$ .

Collecting through the  $\beta_r$  coefficients and the rational  $\zeta(10)$ ,  $\zeta(11)$  parts:

$$D_6 = -\frac{2289}{512} \zeta(10) + \frac{1589}{16} \zeta(11) - \frac{1617}{64} \zeta(3)\zeta(8) - \frac{1785}{64} \zeta(4)\zeta(7) - \frac{2065}{64} \zeta(5)\zeta(6), \quad (115)$$

verified to 62 digits against direct Euler-sum evaluation at 400 dps. The product  $\zeta(2)\zeta(9)$  cancels algebraically, and all three surviving products  $\zeta(3)\zeta(8)$ ,  $\zeta(4)\zeta(7)$ ,  $\zeta(5)\zeta(6)$  are present, matching the prediction  $\max(0, \lfloor (2 \cdot 6 - 5)/2 \rfloor) = 3$ .

**Observation 2** (Product survival rule). *At each Sommerfeld order  $p \geq 2$ , the Dirichlet sum  $D_p$  decomposes into weight- $(2p-1)$  zeta values. The product  $\zeta(2)\zeta(2p-3)$  cancels algebraically at every order. Exactly  $\max(0, \lfloor (2p-5)/2 \rfloor)$  product terms survive, all of the form  $\zeta(a)\zeta(b)$  with  $a+b = 2p-1$ ,  $a$  odd  $\geq 3$ ,  $b$  even  $\geq 4$ .*

Verified through  $D_6$  ( $p = 6$ , weight 11). The systematic expulsion of  $\zeta(2)$  products is driven by the  $n^2$  Fock degeneracy that weights the Sommerfeld sums. This is the same  $n^2$  that produces  $F = \zeta(2) = D_{n^2}(4)$  in the Phase 4F identification (Paper 2 [9]): the Fock degeneracy protects its own Dirichlet evaluation from cross-contamination with odd-zeta products.

## 12.7 Transcendental structure

Substituting  $\zeta(6) = \pi^6/945$  and  $\zeta(4) = \pi^4/90$ , equation (112) becomes

$$D_4 = -\frac{41\pi^6}{12096} + \frac{71}{8}\zeta(7) - \frac{\pi^4}{20}\zeta(3). \quad (116)$$

This is a three-tier decomposition:  $\pi^6$  (weight-6 even-zeta),  $\zeta(7)$  (weight-7 odd-zeta), and the mixed product  $\pi^4 \cdot \zeta(3)$  (weight-4 even  $\times$  weight-3 odd). In Paper 18's operator-order taxonomy,  $\pi^{\text{even}}$  comes from 2nd-order operators and  $\zeta(\text{odd})$  from 1st-order operators. The mixed product  $\pi^4 \cdot \zeta(3)$  is the first instance in the framework where these two tiers appear multiplicatively in a single observable.

The  $\zeta(2)\zeta(5)$  cancellation is non-trivial: four of the five contributing Euler sums carry nonzero  $\zeta(2)\zeta(5)$  coefficients ( $S_{1,6}$ ,  $S_{2,5}$ ,  $S_{4,3}$ ,  $S_{5,2}$ ), and the cancellation emerges only in the specific linear combination that the Dirac degeneracy weights and harmonic-number structure select. In contrast,  $S_{3,4}$  alone is free of  $\zeta(3)\zeta(4)$ , but the remaining four sums contribute  $\zeta(3)\zeta(4)$  terms that do not cancel.

## 12.8 K–Sommerfeld structural separation

The product survival rule (Observation 2) raises a natural question: does the all-orders Sommerfeld cancellation pattern explain why Paper 2's  $K = \pi(B + F - \Delta)$  has pure  $\pi$ -content?

PSLQ at 200 dps tests whether  $K/\pi$  is a rational linear combination of  $\{D_2, D_3, D_4, D_5, D_6, 1\}$ . The result is *null*:  $K/\pi$  is not in the  $\mathbb{Q}$ -span of the Sommerfeld sums through sixth order.  $K$  inhabits a different algebraic space from the  $D_p$ .

The structural reason:  $K$  is composed of spectral-action ingredients —  $B$  (truncated Casimir trace),  $F = \zeta(2)$  (Fock Dirichlet at  $s = 4$ ),  $\Delta^{-1} = g_3^{\text{Dirac}}$  (Dirac mode count) — all of which live in the spectral-zeta world where Theorem 1 guarantees  $\pi$ -polynomial content. The Sommerfeld  $D_p$ , by contrast, are sums of *physical energy coefficients* dressed with harmonic numbers  $H_{n-1}^{(r)}$ , which access the full Euler–Zagier MZV ring including irreducible odd-zeta at every order.

The  $\zeta(2)$  cancellation mechanism is nonetheless a genuine bridge: the  $n^2$  Fock degeneracy that generates  $F = \zeta(2)$  simultaneously drives the systematic expulsion of  $\zeta(2) \times \zeta(\text{odd})$  products from the Sommerfeld sums. The Fock degeneracy organizes both  $K$  (through  $F$ ) and the  $D_p$  (through the cancellation pattern), but the operator/energy distinction prevents  $K$  from being a Sommerfeld functional.

## 13 Flat-Space Consistency

On  $S^3$  of radius  $R$ , the Dirac eigenvalues scale as  $|\lambda_n(R)| = (n + 3/2)/R$  and the Dirichlet series becomes  $D(s, R) = R^s \cdot D(s, 1)$ . The ratio  $D(s, R)/R^s$  is  $R$ -independent *exactly*, confirming that

the even/odd  $s$ -parity discriminant is structural, not a finite- $R$  artifact.

Weyl’s law is verified: the cumulative Dirac mode count  $N(n_{\max}) = 2(n_{\max}+1)(n_{\max}+2)(n_{\max}+3)/3$  matches the Weyl prediction  $(2/3)R^3 p_{\max}^3$  with  $O(1/n_{\max})$  convergence.

The *same* transcendental constants— $\zeta(3)$ , Catalan  $G$ , Dirichlet  $\beta(4)$ —appear in both the  $S^3$  spectral sums and flat-space two-loop QED [13, 14]. Coefficient matching for the full Petermann–Sommerfield two-loop anomalous magnetic moment  $a_2 = 197/144 + \pi^2/12 + 3\zeta(3)/4 - \pi^2 \ln 2/2$  [16, 17] requires matching specific diagram topologies and renormalization subtractions, which is beyond the scope of spectral sums. At one loop, the  $S^3$  computation (Sec. 7) gives  $F_2/[\alpha/(2\pi)] = 1.084$  at  $n_{\text{ext}} = 1$  on unit  $S^3$ , matching the Parker–Toms leading curvature correction  $1 + R/(12|\lambda|^2) = 1.080$  within 0.4%. The 5-point mode-dependent curve ( $n_{\text{ext}} = 1, \dots, 5$ ) follows a power law  $\approx 5.91 |\lambda|^{-1.84}$  (Table 5), consistent with the flat-space limit being  $R \rightarrow \infty$  at fixed physical mass, not  $|\lambda| \rightarrow \infty$  at fixed  $R$ . In the proper flat-space limit, the curvature excess vanishes, recovering the Schwinger value [10].

The sunset sum scales exactly as  $R^{2s_1+2s_2+2p}$  (power = 10 at default parameters), with the ratio  $S(R)/R^{\text{power}}$   $R$ -independent to machine precision. The  $D(5) = 14\zeta(3) - 31\zeta(5)/2$  identification is  $R$ -independent, confirming that the odd-zeta content is structural.

## 14 Comprehensive Transcendental Tables

Table 7: Even/odd decomposition of  $D(s)$ : PSLQ at 80+ dps.

| $s$ | $D_{\text{even}}(s)$                                  | $D_{\text{odd}}(s)$                                   | $D_{\text{even}} - D_{\text{odd}}$ |
|-----|-------------------------------------------------------|-------------------------------------------------------|------------------------------------|
| 4   | $\frac{\pi^2}{2} - \frac{\pi^4}{24} - 4G + 4\beta(4)$ | $\frac{\pi^2}{2} - \frac{\pi^4}{24} + 4G - 4\beta(4)$ | $8(\beta(4) - G)$                  |
| 5   | odd-zeta + Dirichlet                                  | odd-zeta – Dirichlet                                  | $16(\beta(5) - \beta(3))$          |
| 6   | $\pi^{\text{even}} + \text{Dirichlet}$                | $\pi^{\text{even}} - \text{Dirichlet}$                | $32(\beta(6) - \beta(4))$          |

Table 8: Verification of Theorem 3:  $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$ .

| $s$ | $2^{s-1}$ | Residual  LHS – RHS |
|-----|-----------|---------------------|
| 2   | 2         | 0 (sympy)           |
| 3   | 4         | 0 (sympy)           |
| 4   | 8         | $< 10^{-100}$       |
| 5   | 16        | $< 10^{-100}$       |
| 6   | 32        | $< 10^{-100}$       |
| 7   | 64        | $< 10^{-100}$       |
| 8   | 128       | $< 10^{-100}$       |
| 9   | 256       | $< 10^{-100}$       |
| 10  | 512       | $< 10^{-100}$       |
| 15  | 16384     | $< 10^{-100}$       |

Table 9: Three-loop factorized convergence at  $a = 4$ ,  $p = 1$ , CG-weighted. The limit is  $S^{(3)} = 31.5726\dots$  (Eq. (89)): raw partial sums carry zero reliable digits at these truncations — the  $(\ln N)^2/N^2$  increment law, not the naively fitted power law, governs the tail (Sec. 10). Rows recomputed 2026-07-04 from `geovac/qed_three_loop.py` and pinned in `tests/test_paper28_gap_closure_pins.py`; an earlier printing carried stale values (1.399/3.741/5.836/7.729/15.23) from a pre-refactor run.

| $n_{\max}$ | $S^{(3)}(n_{\max})$ | $\Delta$ from previous | Time (s)              |
|------------|---------------------|------------------------|-----------------------|
| 5          | 1.062               | —                      | < 0.1                 |
| 10         | 3.572               | 2.510                  | < 0.1                 |
| 15         | 5.908               | 2.336                  | 0.2                   |
| 20         | 7.872               | 1.964                  | 0.5                   |
| 50         | 14.798              | —                      | < 60                  |
| $\infty$   | 31.5726...          | —                      | (decomposition route) |

## 15 Graph-Native QED

The preceding sections computed QED on  $S^3$  via infinite spectral sums over the Camporesi–Higuchi Dirac modes. This section reports the complementary computation: QED formulated entirely on the *finite* GeoVac graph, using no infinite sums, no regularization, and no renormalization. All quantities are finite matrix traces over the Dirac-labeled graph nodes and Fock-labeled graph edges.

### 15.1 Construction

The graph-native QED infrastructure consists of four layers:

1. **Fock graph** (nodes = Fock states  $(n, l, m)$  at  $n \leq n_{\max}$ , edges = nearest-neighbor Fock ladders) with node Laplacian  $L_0 = BB^T$ , edge Laplacian  $L_1 = B^T B$ , and exact Hodge decomposition (Paper 25 [5] vocabulary).
2. **Electron propagator**  $G_e = D^{-1}$ , where  $D$  is the Dirac graph operator in the  $(n, \kappa, m_j)$  basis with hopping parameter  $t$ . At  $t = 0$ ,  $G_e = \text{diag}(2/(2n + 3))$ .
3. **Photon propagator**  $G_\gamma = L_1^+$ , the Moore–Penrose pseudoinverse of the edge Laplacian. This is the graph analog of the Feynman-gauge photon propagator.
4. **Vertex coupling tensor**  $V[a, b, e]$ , built from the Clebsch–Gordan projection matrix  $P$  that maps Dirac labels  $(n, \kappa, m_j)$  onto Fock nodes  $(n, l, m)$ .

At  $n_{\max} = 2$ : 10 Dirac states, 5 Fock nodes, 3 Fock edges. At  $n_{\max} = 3$ : 28 Dirac states, 14 Fock nodes, 13 Fock edges.

### 15.2 Vacuum polarization

The graph vacuum polarization tensor is

$$\Pi_{\text{graph}}[e_1, e_2] = \sum_{a, b} V[a, b, e_1] G_e[a, b] V[b, a, e_2],$$

the complete one-loop photon self-energy on the finite graph. At  $n_{\max} = 2$ ,  $\text{Tr}(\Pi) = 224/75$  (pure rational). At  $n_{\max} = 3$ ,  $\text{Tr}(\Pi) = 3872/735$ ; the  $13 \times 13$  matrix decomposes into  $l$ -sector blocks



( $l = 0$ :  $2 \times 2$ ,  $l = 1$ :  $7 \times 7$ ,  $l = 2$ :  $4 \times 4$ ) with zero cross-block coupling. Four entries in the  $l = 2$  sector contain  $\sqrt{6}$  from CG coefficients; all others are rational.

### 15.3 Self-energy and vertex correction

The one-loop self-energy is

$$\Sigma[a, b] = \sum_{e, e'} G_\gamma[e, e'] V_e \cdot V_{e'}^T,$$

where  $V_e[a, c] = V[a, c, e]$ . At  $n_{\max} = 2$ ,  $t = 0$ :  $\text{Tr}(\Sigma) = 44/3$ ; eigenvalues  $\{0^{(\times 5)}, 4/3, 2^{(\times 2)}, 4, 16/3\}$ ; positive semidefinite; Hermitian.

The one-loop vertex correction is

$$\Lambda[a, b] = \sum_{e', e''} G_\gamma[e', e''] V_{e'} \cdot G_e \cdot V_{e''}^T.$$

At  $n_{\max} = 2$ ,  $t = 0$ :  $\text{Tr}(\Lambda) = 32/9$ ; eigenvalues  $\{0^{(\times 5)}, 8/45, (28 \pm 4\sqrt{145})/45, 4/5, 4/3\}$ .

The graph anomalous magnetic moment is extracted as  $F_2 = \text{Tr}(\Lambda)/\text{Tr}(V_{\text{bare}} \cdot G_e) = 5\sqrt{2}/3 \approx 2.357$  at  $t = 0$ .

At the physical coupling  $t = \kappa = -1/16$ , the self-energy is *identical* (the formula does not involve  $G_e$ ), but the vertex correction changes:  $\text{Tr}(\Lambda) = 16\sqrt{3}/8985 + 19255/5391 \approx 3.575$ .  $\Lambda$  acquires one negative eigenvalue ( $-0.447$ ), becoming indefinite. The anomalous moment shifts to

$$F_2(\kappa) = \frac{48\sqrt{3} + 96275}{3(15 + 16\sqrt{6} + 9616\sqrt{2})} \approx 2.3525,$$

a 0.2% shift from the  $t = 0$  value.  $F_2(\kappa)$  remains algebraic irrational in  $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$ , the CG ring.

### 15.4 Neumann series structure of the electron propagator

The coupling-dependence of  $F_2(t)$  admits a transparent algebraic explanation. Write the Dirac graph operator as  $D = \Lambda + tA$ , where  $\Lambda = \text{diag}(|\lambda_n|)$  is the free (diagonal) Dirac spectrum and  $A$  is the adjacency-like hopping matrix. The electron propagator admits a Born/Neumann expansion

$$G_e(t) = (\Lambda + tA)^{-1} = \Lambda^{-1} \sum_{k=0}^{\infty} (-t \Lambda^{-1} A)^k,$$

which converges when  $\rho(t \Lambda^{-1} A) < 1$ . The free propagator  $\Lambda^{-1} = \text{diag}(2/(2n+3))$  is trivially rational. Each subsequent order multiplies by the matrix  $-t \Lambda^{-1} A$ , whose entries lie in  $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$  (the CG ring). The expansion therefore preserves the algebraic ring at every order: no new transcendentals enter from the coupling.

By the Cayley–Hamilton theorem, the series terminates exactly at order  $N - 1$ , where  $N = \dim(\text{Dirac space})$ . Consequently,  $G_e(t)$  — and hence  $F_2(t)$  — is a *rational function* of  $t$  with coefficients in the CG ring. At the physical coupling  $t = \kappa = -1/16$ , the spectral radius  $\rho(\kappa \Lambda^{-1} A) \approx 0.2$  at  $n_{\max} = 2$ , so the first few Born orders dominate. The 0.2% shift  $F_2(\kappa)/F_2(0) \approx 0.998$  is consistent with first-order perturbation in  $\kappa$ .

This structure connects to Paper 18’s exchange-constant taxonomy [3]: the hopping parameter  $t$  enters  $F_2$  as a polynomial perturbation of the free value, preserving the INTRINSIC algebraic ring at every order. The Neumann expansion makes explicit that the graph-native anomalous moment

is a *finite rational expression* in  $\kappa$  and CG coefficients, with no transcendental content at any truncation  $n_{\max}$ . For larger  $n_{\max}$  (where direct symbolic inversion of  $D$  becomes expensive), the rapid convergence of the Born series at  $|\kappa| = 1/16$  provides an efficient perturbative alternative.

#### 15.4.1 $F_2(t)$ is an even rational function

The Neumann expansion reveals that  $F_2(t)$  is an *even* function of  $t$ : all odd Taylor coefficients vanish identically. The mechanism is the decomposition  $D(t) = \Lambda + tA$ , where  $\Lambda$  (diagonal eigenvalues  $\pm|\lambda_n|$ ) is even in  $t$  and  $A$  (the  $T_{\pm}$  ladder operators connecting adjacent shells) is odd. The vertex correction involves the bilinear  $V \cdot G_e \cdot V^T$ , which is invariant under  $t \rightarrow -t$  because the adjacency contribution enters quadratically via  $G_e(t) = G_e(-t)$  in the trace.

At  $n_{\max} = 2$ , the Taylor expansion is

$$F_2(t) = \frac{5\sqrt{2}}{3} + c_2 t^2 + c_4 t^4 + O(t^6), \quad (117)$$

with  $c_0 = 5\sqrt{2}/3 \approx 2.357$  (algebraic),  $c_1 = c_3 = c_5 = 0$  (exact),  $c_2 \approx -1.163$  (algebraic in  $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$ ),  $c_4 \approx 1.685$  (algebraic). The even-ness means the anomalous moment depends only on  $t^2 = \kappa^2$ , not on the sign of the hopping coupling.

#### 15.4.2 $F_2(\kappa)$ convergence

| $n_{\max}$ | $N_{\text{Dirac}}$ | $E_{\text{Fock}}$ | $F_2(\kappa)$ | $F_2(0)$ | $F_2 \cdot n_{\max}$ |
|------------|--------------------|-------------------|---------------|----------|----------------------|
| 2          | 10                 | 3                 | 2.3525        | 2.3570   | 4.705                |
| 3          | 28                 | 13                | 1.8731        | 1.8756   | 5.619                |
| 4          | 60                 | 34                | 1.5892        | 1.5898   | 6.357                |
| 5          | 110                | 70                | 1.3958        | 1.3954   | 6.979                |
| 6          | 182                | 125               | 1.2532        | 1.2523   | 7.519                |

Table 10: Graph-native anomalous moment  $F_2$  at the physical coupling  $\kappa = -1/16$ .  $F_2$  decreases monotonically; a power-law fit gives  $F_2 \sim 3.507 n_{\max}^{-0.573}$  ( $R^2 = 0.99990$ ).

The graph-native  $F_2(\kappa)$  decreases monotonically with  $n_{\max}$  but remains  $O(1)$ , orders of magnitude above the Schwinger value  $\alpha/(2\pi) \approx 1.16 \times 10^{-3}$ . The suppression from graph to continuum requires both the coupling constant  $\alpha$  (one power per loop) and the vector photon structure (SO(4) selection rules that eliminate most graph-allowed channels).

**Power-law convergence (caveat: 5-point fit, asymptote inferred not derived).** A least-squares fit across  $n_{\max} = 2, \dots, 6$  gives  $F_2(\kappa) \sim 3.507 n_{\max}^{-0.573}$  with  $R^2 = 0.99990$ . Local log-log exponents increase monotonically in magnitude:  $-0.562$  ( $n = 2-3$ ),  $-0.571$  ( $3-4$ ),  $-0.582$  ( $4-5$ ),  $-0.591$  ( $5-6$ ), consistent with a slowly steepening power law. Successive ratios  $F_2(n+1)/F_2(n) = 0.796, 0.849, 0.878, 0.898$  approach 1 from below, confirming slow (sub-1/ $n$ ) decay. Richardson extrapolation with adaptive exponent estimation ( $p \approx 0.51$ ) gives the unphysical  $F_{\infty} \approx -0.22$ , which we interpret as consistent with  $F_2 \rightarrow 0$  within the zero-offset power-law family.

This conclusion is fit-based, not derived: a 5-point power-law fit with high  $R^2$  does not by itself rule out a small constant offset  $F_{\infty}^{\text{true}} \neq 0$  that the truncated data cannot distinguish from the leading power law. Larger  $n_{\max}$  data, or a derivation of the leading large- $n$  behavior of  $F_2$  from the spinor heat-kernel curvature expansion, would be needed to fix the asymptote on structural rather than fit grounds. The interpretation “ $F_2 \rightarrow 0$  on a fixed-radius sphere” (consistent with form-factor

suppression at  $|\lambda| \rightarrow \infty$  on  $R = 1$ , see §15.4.2 discussion above) is the most natural reading of the data, but should be treated as suggestive rather than proven.

The sign of  $F_2(\kappa) - F_2(0)$  flips at  $n_{\max} = 5$ :  $\delta = -0.0045, -0.0025, -0.0006, +0.0004, +0.0009$  for  $n_{\max} = 2, \dots, 6$ . This zero-crossing indicates a subtle interplay between the  $t$ -dependence and the truncation level; both sequences track within  $< 0.1\%$  and converge to the same graph limit.

## 15.5 Broken structural zero and the pendant-edge theorem

Theorem 4 states that  $\Sigma(n_{\text{ext}} = 0) = 0$  in the continuum, by vertex parity impossibility. On the finite graph, this structural zero is **broken**:

$$\Sigma_{\text{graph}}^{\text{GS}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \neq 0.$$

The CG projection maps Dirac  $(n, \kappa, m_j)$  states onto Fock  $(n, l, m)$  nodes, opening vertex couplings invisible to the  $\text{SO}(4)$  channel count that enforces the continuum selection rule.

A three-layer investigation reveals the complete mechanism.

### 15.5.1 Layer 1: Anatomy of the ground-state self-energy

At  $n_{\max} = 2$ , the entire ground-state self-energy comes from a single edge pair. Let  $v_0 = |1, 0, 0\rangle$  (the ground-state Fock node) and  $v_1 = |2, 0, 0\rangle$  (the first radial excitation); these are connected by the edge  $e_0$ . All Clebsch–Gordan coefficients equal 1 because  $l = 0$  forces trivial angular projection. Two intermediate s-wave Dirac states ( $m_j = \pm 1/2$ ) contribute equally. The formula reduces to

$$\Sigma[a, b] = 2 \cdot G_\gamma[e_0, e_0] \cdot 1,$$

where  $G_\gamma = L_1^+$  is the photon propagator. At  $n_{\max} = 2$ ,  $L_1$  is a  $3 \times 3$  matrix and  $G_\gamma[e_0, e_0] = 1/2$ , giving  $\Sigma[a, b] = 1$  for all entries of the  $2 \times 2$  ground-state block.

### 15.5.2 Layer 2: The pendant-edge theorem

The anatomy at  $n_{\max} = 2$  generalizes exactly to all  $n_{\max}$ .

The ground-state node  $v_0 = |1, 0, 0\rangle$  is a *leaf* (degree 1) of the Fock graph at every  $n_{\max}$ . The single edge  $e_0$  incident on  $v_0$  is a *pendant edge*. The s-wave ( $l = 0$ ) radial edges form a tridiagonal sub-block of  $L_1 = B^T B$  — the standard discrete Laplacian on a path graph of length  $N = n_{\max} - 1$ . This path-graph Laplacian  $T_N$  has an explicitly known inverse: its  $(0, 0)$  entry is

$$T_N^{-1}[0, 0] = \frac{N}{N+1} = \frac{n_{\max} - 1}{n_{\max}}.$$

Because the pendant edge  $e_0$  decouples from all non-radial edges in  $L_1$  (s-wave edges share no endpoint with edges at  $l \geq 1$ ),  $G_\gamma[e_0, e_0] = T_N^{-1}[0, 0]$  exactly.

**Proposition 5** (Pendant-edge theorem for the graph self-energy). *On the GeoVac Fock graph at any finite  $n_{\max} \geq 2$ , the graph-native ground-state self-energy satisfies*

$$\Sigma_{\text{graph}}(\text{GS}) = \frac{2(n_{\max} - 1)}{n_{\max}}, \quad (118)$$

where the value is exact (rational) and degenerate across both  $m_j = \pm 1/2$  ground-state components. In particular,

$$\lim_{n_{\max} \rightarrow \infty} \Sigma_{\text{graph}}(\text{GS}) = 2 \neq 0. \quad (119)$$

The structural zero of Theorem 4 is never recovered in the graph limit.

*Proof.* The ground-state Fock node  $v_0 = |1, 0, 0\rangle$  has degree 1 at every  $n_{\max}$  (its only neighbor is  $|2, 0, 0\rangle$ ). The self-energy formula  $\Sigma[a, b] = \sum_{e, e'} G_\gamma[e, e'] V_e \cdot V_{e'}^T$  at the ground state receives contributions only from edges incident on  $v_0$ , of which there is exactly one ( $e_0$ ). All CG coefficients at  $l = 0$  equal 1, and two Dirac intermediate states ( $m_j = \pm 1/2$ ) contribute, giving  $\Sigma[a, b] = 2 G_\gamma[e_0, e_0]$ . The s-wave radial edges form a path graph of  $N = n_{\max} - 1$  edges in  $L_1$ , decoupled from all  $l \geq 1$  edges. The Moore–Penrose pseudoinverse of the path-graph Laplacian  $T_N$  satisfies  $T_N^{-1}[0, 0] = N/(N + 1)$ . Therefore  $\Sigma_{\text{graph}}(\text{GS}) = 2N/(N + 1) = 2(n_{\max} - 1)/n_{\max}$ .  $\square$

**Verification:** Eq. (118) is continuously verified by `tests/test_graph_qed_self_energy.py`: at  $n_{\max} = 2$  in exact `sympy` rational arithmetic (GS block  $[[1, 1], [1, 1]]$ ), and at  $n_{\max} = 3, 4$  numerically to  $10^{-10}$  (`TestPendantEdgeSweep`, added 2026-07-03 after the `graph_qed_photon`  $n_{\max} \geq 4$  eigenvalue-sort crash fix; the exact-`sympy` photon pseudoinverse at  $n_{\max} \geq 3$  exceeds any reasonable test budget, so those points are numeric). The formula itself is proven in the theorem above for all  $n_{\max}$ .

### 15.5.3 Layer 3: Selection-rule gap

The continuum possesses *two* independent protection mechanisms that enforce  $\Sigma(n_{\text{ext}} = 0) = 0$ :

1. **Vertex parity.** The selection rule requires  $n_1 + n_2 + q$  odd (Eq. (6)). At  $n_{\text{ext}} = 0$ , the triangle inequality forces  $q = n_{\text{int}}$ , so the sum is  $2n_{\text{int}}$  (always even). No allowed triple exists.
2. **SO(4) channel count.** The CG multiplicity  $W(0, n_{\text{int}}, q) = 0$  for all  $q$ . The ground state has  $j_R = 0$  in the  $\text{SU}(2)_L \times \text{SU}(2)_R$  decomposition; the right-factor triangle degenerates (one component fails the  $\text{SU}(2)_R$  triangle; the other passes all triangles but is killed by parity).

The finite graph has *neither* mechanism. The graph photon lives on edges—it is a 1-cochain, not a vector field. The CG projection is scalar  $\text{SU}(2)$ , not the  $\text{SU}(2)_L \times \text{SU}(2)_R$  double triangle of  $\text{SO}(4)$  vector harmonics. Both protection mechanisms arise from the *spin-1 vector nature* of the continuum photon, which is absent on the graph.

### 15.5.4 Interpretation: scalar vs. vector QED

The graph-native QED of this section is a *scalar* QED: the photon carries no spin-1 structure on the graph (it is a 1-cochain in the Hodge complex, not a vector field on the manifold). The continuum QED adds vector structure via  $\text{SO}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$ , and Theorem 4 is a vector-QED selection rule that scalar QED does not possess.

In Paper 18’s exchange-constant taxonomy [3], the photon’s spin-1 vector nature is a CALIBRATION exchange constant: it is injected when projecting from the algebraic graph to the continuous manifold. The pendant-edge theorem (Proposition 5) quantifies the cost of this omission: a ground-state self-energy of 2 (in graph units) that the continuum’s vector selection rules eliminate exactly. The structural zero is not a fine cancellation that the finite graph misses due to truncation—it is a *qualitative* selection rule tied to the photon’s vector character, which the graph does not encode.

**Independent reading: Connes–van Suijlekom operator-system gap.** The same broken structural zero admits a parallel structural interpretation through the spectral-truncation framework of Connes and van Suijlekom [30]: the truncated algebra-image  $\mathcal{O}_{n_{\max}} = P_{n_{\max}} C^\infty(S^3) P_{n_{\max}}$  is  $*$ -closed but *not* multiplicatively closed—taking its  $C^*$ -envelope  $C^*(\mathcal{O}_{n_{\max}})$  collapses to a “non-informative full matrix algebra” that loses precisely the operator-system data carrying the off-diagonal Gaunt structure. The CG projection that “opens vertex couplings invisible to the  $\text{SO}(4)$

channel count” (Sec. 15.5) is exactly the failure-of-closure that distinguishes  $\mathcal{O}_{n_{\max}}$  from its envelope. Read this way, the pendant-edge value  $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max}$  is not an artifact of finite truncation but the operator-system content the envelope would discard.

This reading is sharpened by an independent quantitative match: the GeoVac operator system  $\mathcal{O}_{n_{\max}}$  has propagation number  $\text{prop}(\mathcal{O}_{n_{\max}}) = 2$  at  $n_{\max} \in \{2, 3, 4\}$ , matching the value Connes & van Suijlekom prove for the Toeplitz operator system  $C(S^1)^{(n)}$  on the circle (their Proposition 4.2:  $\text{prop} = 2$  independently of  $n$ ); see Paper 32 [7] (propagation-number proposition). The witness pair underlying the propagation-number computation is the same boundary mechanism as the pendant-edge theorem here: a “raise-then-lower at the top shell escapes the truncation” deficit, viewed once from the QED self-energy side and once from the operator-system side. The fact that two structurally distinct constructions on the same Fock-projected  $S^3$  graph both quantitatively realize the same boundary correction is internal evidence that the operator-system reading of GeoVac is not a formal relabelling but the structural source of both phenomena. See `debug/wh1_connes_vs_mapping_memo.md` (round-1 mapping) and `debug/wh1_round2_propagation_memo.md` (round-2 propagation computation) for the full record.

## 15.6 Selection rule survival census

A systematic test of 8 continuum QED selection rules on the finite Fock graph at  $n_{\max} = 2$  reveals that only *one* survives:

| Rule                             | Verdict         | Tier       | Key metric                       |
|----------------------------------|-----------------|------------|----------------------------------|
| Gaunt/CG sparsity                | <b>survives</b> | intrinsic  | density 16%→2.6%→0.6%            |
| Angular momentum $\Delta m_j$    | broken          | intrinsic  | 20 violations                    |
| Spatial parity (E1)              | broken          | intrinsic  | 100% parity violations           |
| Charge conjugation (C)           | broken          | intrinsic  | 8 C-violations in $\Sigma$       |
| Furry’s theorem                  | broken          | intrinsic  | tadpole = $16\sqrt{2}/15 \neq 0$ |
| Vertex parity ( $n_1+n_2+q$ odd) | broken          | structural | 0% forbidden                     |
| SO(4) channel count ( $W > 0$ )  | broken          | structural | 100% have $W = 0$                |
| Ward identity                    | broken          | structural | residual = 5.08                  |

Table 11: Selection rule survival census at  $n_{\max} = 2$ . STRUCTURAL rules require vector photon structure absent from the graph; INTRINSIC rules are broken by graph kinematics.

The 7 broken rules partition cleanly into two classes:

- **3 structural** (vertex parity, SO(4) channel count, Ward identity): require the photon to be a vector harmonic on  $S^3$ . The graph’s scalar 1-cochain photon has no mechanism to carry parity-flip, angular momentum transfer, or gauge derivative structure. These enter at the CALIBRATION tier when projecting to vector QED.
- **4 intrinsic** ( $\Delta m_j$ , spatial parity, charge conjugation, Furry): broken by graph kinematics. The CG projection allows couplings forbidden in vector QED because the scalar photon carries different angular momentum content.

Notable graph-native invariants:  $\text{Tr}(\Sigma)/\text{Tr}(\Lambda) = 33/8$ ; tadpole =  $16\sqrt{2}/15$ ; bubble =  $32/9$ ; triangle =  $3584\sqrt{2}/3375$ .

## 15.7 Native Dirac graph QED

The scalar Fock graph recovers only 1 of 8 continuum QED selection rules (Table 11). A natural question is whether building QED directly on the *Dirac* graph—with nodes labeled by  $(n_{\text{Fock}}, \kappa, m_j)$ ,

the native quantum numbers of the Dirac operator on  $S^3$ —recovers additional rules by encoding spinor structure at the graph level.

### 15.7.1 Two adjacency rules

Two natural edge rules are tested:

- **Rule A ( $\kappa$ -preserving):** lifts the scalar Fock ladders  $\Delta N = \pm 1$  within each  $\kappa$  sector. The graph decomposes into independent  $\kappa$ -blocks, analogous to the  $l$ -shell splitting of the scalar graph. At  $n_{\max} = 2$ :  $V = 10$ ,  $E = 8$ ,  $\beta_1 = 1$ .
- **Rule B (E1 dipole):** enforces  $\Delta l = \pm 1$  (parity flip),  $|\Delta j| \leq 1$ ,  $|\Delta m_j| \leq 1$ . Mixes  $\kappa$  values; the graph is a single connected component. At  $n_{\max} = 2$ :  $V = 10$ ,  $E = 20$ ,  $\beta_1 = 11$ .

### 15.7.2 Selection rule recovery

| Rule                      | Scalar Fock | Dirac A    | Dirac B    | Tier   |
|---------------------------|-------------|------------|------------|--------|
| Gaunt/CG sparsity         | ✓           | ✓          | ✓          | always |
| $\Delta m_j$ conservation | ×           | ✓          | ✓          | spinor |
| Spatial parity (E1)       | ×           | ×          | ✓          | spinor |
| Furry’s theorem           | ×           | ✓          | ✓          | spinor |
| Vertex parity             | ×           | ×          | ×          | vector |
| SO(4) channel count       | ×           | ×          | ×          | vector |
| Ward identity             | ×           | ×          | ×          | vector |
| Charge conjugation        | ×           | ×          | ×          | vector |
| <b>Total surviving</b>    | <b>1/8</b>  | <b>3/8</b> | <b>4/8</b> |        |

Table 12: Selection rule survival on the scalar Fock graph versus the two Dirac graph rules. The 8 rules partition into three tiers: 1 always survives, 3 are spinor-recoverable, and 4 require vector photon structure.

The selection rules partition cleanly into three tiers:

1. **Always survives (1):** Gaunt/CG angular sparsity, which is a property of the coupling coefficients regardless of graph topology.
2. **Spinor-recoverable (3):**  $\Delta m_j$  conservation (both rules—Dirac nodes carry explicit  $m_j$ ), spatial parity (Rule B only—E1 edges enforce  $\Delta l = \pm 1$  by construction), and Furry’s theorem (both rules—the identity vertex on the Dirac graph is off-diagonal, making odd-loop traces vanish at  $t = 0$ ). These are INTRINSIC to the spinor quantum numbers.
3. **Vector-photon-required (4):** vertex parity, SO(4) channel count, Ward identity, and charge conjugation. These require promoting the photon from a scalar 1-cochain to a vector harmonic with  $SU(2)_L \times SU(2)_R$  structure. They are CALIBRATION content from the continuum embedding.

### 15.7.3 Self-energy structure

The ground state is **not** a pendant vertex on either Dirac graph: degree 2 (Rule A) and degree 5 (Rule B), both constant across  $n_{\max} = 2, 3, 4$ . Consequently, the pendant-edge theorem (Proposition 5) does not apply, and the structural zero  $\Sigma(\text{GS}) = 0$  is not recovered on either Dirac graph.

Exact algebraic results at  $n_{\max} = 2$ :

- $\text{Tr}(\Sigma_A) = 17/2$ ,  $\text{Tr}(\Sigma_B) = 103/20$ .
- Rule A GS block:  $\text{diag}(5/8, 5/8)$  (diagonal, no  $m_j$  mixing).
- Rule B GS block:  $\begin{pmatrix} 103/160 & 21/160 \\ 21/160 & 103/160 \end{pmatrix}$  (off-diagonal  $m_j$  mixing from dipole edges).
- $\Sigma$  is strictly positive definite for both rules (no zero eigenvalues), in contrast to the scalar Fock graph's 5 zero eigenvalues from the photon kernel.
- Rule A eigenvalues:  $\{1/2 (\times 4), 3/4 (\times 2), 5/4 \pm \sqrt{2}/2 (\times 2)\}$ .
- Rule B eigenvalues:  $\{1/5 (\times 2), 1/4 (\times 3), 49/100, 33/40, 27/20, 267/400 \pm 3\sqrt{881}/400\}$ .

#### 15.7.4 $\pi$ -free algebraic content

All Dirac graph QED quantities are  $\pi$ -free. Rule A lives in  $\mathbb{Q}[\sqrt{2}]$ —*simpler* than the scalar Fock graph's  $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}]$ —because the  $\kappa$ -preserving adjacency avoids the CG irrationals from cross- $l$  couplings. Rule B lives in  $\mathbb{Q}[\sqrt{2}, \sqrt{17}, \sqrt{41}, \sqrt{881}]$ , the irrationals arising from the edge Laplacian's characteristic polynomial rather than from CG coefficients.

#### 15.7.5 Interpretation: spinor barrier and vector barrier

The Dirac graph is an *intermediate* construction between the scalar Fock graph (1/8 rules) and continuum vector QED (8/8 rules). The spinor-recoverable rules (3 rules) are **INTRINSIC** to the electron's spinor quantum numbers—they are restored simply by using Dirac  $(n, \kappa, m_j)$  nodes instead of scalar  $(n, l, m)$  Fock nodes. The vector-photon-required rules (4 rules) are **CALIBRATION** content that enters when the 1-cochain photon is promoted to a vector harmonic on  $S^3$ .

In Paper 18's exchange-constant taxonomy [3], the path from graph to continuum QED therefore has two distinct barriers: (i) the *spinor barrier*, crossed by adopting Dirac node labels (graph-intrinsic, zero cost); (ii) the *vector barrier*, requiring the photon's spin-1 structure (calibration tier, continuum embedding). The three-tier partition of Table 12 makes this classification precise.

### 15.8 Vector QED sprint: five negative results

A systematic five-track investigation (VQ-1 through VQ-5) tested whether the 4 vector-photon-required selection rules can be recovered from the graph's intrinsic geometry without explicitly promoting the photon to a vector harmonic. All five tracks are negative for selection rule recovery.

#### 15.8.1 Approaches tested

1. **VQ-1 ( $\sigma \cdot L$  decomposition):** the spin-orbit coupling operator  $\sigma \cdot L$  is zero on all graph edges because it is purely intra-shell ( $L|l=0\rangle = 0$ ), producing no inter-shell coupling.
2. **VQ-2 (pure  $\sigma_\mu$  vertex):** Pauli matrices as the vertex coupling. The graph is disconnected (no inter-shell edges); 2/8 rules survive but this is *worse* than the Dirac Rule B baseline (4/8).
3. **VQ-3 ( $\sigma_\mu \otimes V_{\text{scalar}}$  vertex):** inserting  $\sigma_\mu$  at the CG vertex while keeping the same photon propagator. The Pauli trace identity  $\sum_\mu \sigma_\mu^2 = 3I$  forces  $\text{Tr}(\Sigma_{\text{vec}}) = 3 \times \text{Tr}(\Sigma_{\text{scalar}})$  and  $F_{2,\text{vec}} = 3 \times F_{2,\text{scalar}}$ —a trivial rescaling. Selection rules: 1/8 (same as scalar;  $\Delta m_j$  conservation and Furry's theorem are *broken* by the sigma dressing).

4. **VQ-4 (direction-resolved Hodge decomposition):** classifying Fock graph edges into T-channel (radial,  $\Delta n = \pm 1$ ) and L-channel (angular,  $\Delta m = \pm 1$ ), with per-channel sub-Laplacians and photon propagators. At  $n_{\max} = 2$  the channels are spectrally anisotropic (T eigenvalues  $\{2\}$  vs L eigenvalues  $\{1, 3\}$ ) and the full propagator is block-diagonal in T/L. At  $n_{\max} = 3$ , cross-channel coupling emerges ( $\|G_{TL}\|/\|G_{\text{full}}\| = 5.8\%$ ), mixing radial and angular photon modes. This gives the scalar graph photon effective “polarization” from pure topology—a positive *structural* finding, though insufficient for selection rule recovery.
5. **VQ-5 (full vector QED):** combining direction-labeled photon channels (VQ-4) with channel-specific  $\sigma_\mu$  vertex coupling (VQ-3). All 15  $\sigma$ -to-channel configurations tested (5 named mappings + exhaustive  $3 \times 3$  scan). Three distinct  $F_2$  values found ( $1.0\times$ ,  $1.4\times$ ,  $1.6\times$  scalar), but all configurations produce  $1/8$  surviving selection rules.

### 15.8.2 Structural conclusion

The VQ sprint confirms the three-tier partition of Table 12 from a complementary direction: even when the graph’s T/L edge anisotropy provides effective polarization structure and the  $\sigma_\mu$  matrices provide spin coupling, the 4 vector-required rules (vertex parity,  $\text{SO}(4)$  channel count, Ward identity, charge conjugation) remain broken across all tested configurations. These rules require the *photon* to carry angular momentum quantum numbers  $(L, M_L)$ —promoting it from a scalar 1-cochain to a vector harmonic—which is calibration exchange constant content in Paper 18’s taxonomy [3].

The positive structural finding from VQ-4 (spectral anisotropy of T/L channels, cross-channel coupling at  $n_{\max} = 3$ ) shows that the Fock graph topology does encode directional information, but this information is insufficient to generate selection rules that depend on the photon’s *spin-1* quantum numbers rather than its *propagation direction*.

## 15.9 Transverse photon sector from plaquettes

The preceding sections use the edge Laplacian  $L_1 = B^\top B$  as the photon propagator. On a graph (simplicial 1-complex without higher-dimensional faces), this captures *only* the exact (longitudinal/gauge) sector of the Hodge decomposition. The co-exact (transverse/physical) sector requires 2-cells—faces bounded by closed edge paths.

Paper 30’s  $\text{SU}(2)$  Wilson gauge construction [6] provides precisely these faces as plaquettes (minimal non-backtracking closed walks of length  $\geq 6$  on the Fock graph). With boundary operator  $d_1$  mapping edges to faces, the co-exact Laplacian is  $L_1^{\text{co}} = d_1 d_1^\top$  and the transverse photon propagator is its pseudoinverse:

$$G_T = (d_1 d_1^\top)^+. \quad (120)$$

### 15.9.1 Ground-state protection recovered

Computing the self-energy  $\Sigma_T[a, b] = \sum_{e, e'} V_e G_T[e, e'] V_{e'}^\top$  with the transverse propagator:

| $n_{\max}$ | $\Sigma_{\text{scalar}}(\text{GS})$ | $\Sigma_T(\text{GS})$ | Plaquettes | Co-exact rank |
|------------|-------------------------------------|-----------------------|------------|---------------|
| 3          | $4/3$                               | <b>0</b> (exact)      | 4          | 2             |
| 4          | $3/2$                               | <b>0</b> (exact)      | 33         | 8             |

The transverse propagator recovers the structural zero of Theorem 4 at machine precision ( $< 10^{-16}$ ) while the scalar (longitudinal) propagator breaks it via the pendant-edge theorem (Proposition 5).



### 15.9.2 Mechanism: pendant-edge exclusion from plaquettes

The ground state maps to the Fock node  $n = 1$ , a pendant vertex (degree 1) connected to the graph by a single edge  $e_0$ . A pendant edge cannot participate in any closed walk: leaving a degree-1 vertex and returning requires immediate backtracking, which is excluded from plaquette enumeration. Therefore  $d_1[e_0, f] = 0$  for all faces  $f$ , giving  $(d_1 d_1^\top)[e_0, :] = 0$  and hence  $G_T[e_0, :] = 0$ . Since the ground-state vertex tensor  $V[a, :, e]$  couples only through  $e_0$ , the self-energy vanishes:

$$\Sigma_T(\text{GS}) = 0 \quad (\text{topological, exact at all } n_{\max}). \quad (121)$$

This is the graph analog of the continuum vertex-parity argument: vertex parity ( $n_1 + n_2 + q$  odd) with  $n_2 = 0$  forces  $2n_1$  odd (impossible); plaquette exclusion of pendant edges with pendant GS forces  $G_T[e_0, :] = 0$  (topological). Both mechanisms protect the ground state from self-energy corrections, but via structurally different routes.

### 15.9.3 Refined selection rule partition

The three-tier partition of Table 12 is refined to four tiers:

1. **Always survives** (1 rule): Gaunt/CG angular sparsity.
2. **Spinor-recoverable** (3 rules):  $\Delta m_j$  conservation, spatial parity, Furry’s theorem—from Dirac graph node labels.
3. **Plaquette-recoverable** (1 rule): ground-state structural zero—from co-exact Hodge topology of the face structure.
4. **Vector-quantum-number-required** (3 rules):  $\text{SO}(4)$  channel count, Ward identity, charge conjugation—requiring explicit  $(L, M_L)$  labeling of co-exact modes.

The boundary between “graph-intrinsic” and “calibration” has shifted: ground-state protection is recoverable from the graph’s own Hodge structure (plaquettes from Paper 30), not requiring any photon angular momentum quantum numbers.

### 15.9.4 Mode-resolved analysis: why the final 3 rules require calibration

The co-exact eigenvalues  $\mu_k$  of  $d_1 d_1^\top$  approach the continuum photon eigenvalues  $q(q+2)$  for low-lying modes (e.g.  $\mu = 15.0 = 3 \times 5$  at  $n_{\max} = 3$  matches  $q = 3$  exactly;  $\mu = 48.0 = 6 \times 8$  at  $n_{\max} = 4$  matches  $q = 6$ ). However, the *eigenvectors* have the wrong support. Testing the  $\text{SO}(4)$  triangle inequality  $|n_1 - n_2| \leq q \leq n_1 + n_2 - 2$  on the mode-resolved self-energy  $\Sigma_k[a, b] = \sum_{e, e'} V_e v_k v_k^\top G_T[e, e'] V_{e'}^\top$ :

| $n_{\max}$ | Mode | $q_{\text{assign}}$ | Triangle inequality |
|------------|------|---------------------|---------------------|
| 4          | 0–3  | $q = 3, 4$          | PASS (0% violation) |
| 4          | 4–5  | $q = 5$             | FAIL (66%)          |
| 4          | 6    | $q = 5$             | FAIL (63%)          |
| 4          | 7    | $q = 6$             | FAIL (86%)          |

The structural obstruction is topological: the Fock graph has only nearest-neighbor edges ( $|\Delta n| \leq 1$ ), so every co-exact eigenvector is a linear combination of  $\Delta n = 0, \pm 1$  edges. A continuum photon of angular momentum  $q$  couples shells separated by up to  $q$  levels, but the

graph’s plaquette eigenvectors cannot span that range. The Ward identity diagnostic confirms:  $\|[\Sigma_T, H_0]\|/\|\Sigma_T\| \approx 0.58$  at both  $n_{\max} = 3, 4$  (zero would indicate energy conservation structure).

This identifies the Tier 1/Tier 2 boundary precisely: the graph gives the correct *count* and approximately correct *energies* of transverse modes, but not the angular momentum structure within each mode. Promoting graph 1-cochains to sections of a vector bundle indexed by  $(q, m_q)$  is the minimal calibration required for the final 3 selection rules.

## 15.10 Continuum bridge and projection exchange constant

The graph VP (trace) and the continuum spectral-sum VP (Sec. 3) are related at finite truncation by a *rational* projection constant  $C = \Pi_{\text{cont}}/\text{Tr}(\Pi_{\text{graph}})$ . At  $n_{\max} = 3$ :

$$C = \frac{50471424}{1779441125} \approx 0.0284.$$

**$C$  is rational. No  $\pi$  enters at finite truncation.**

Transcendental numbers ( $\pi^{\text{even}}$  from Hurwitz zeta,  $\text{Vol}(S^3) = 2\pi^2$ ) enter exclusively via two mechanisms: (i) infinite-sum regularization ( $n_{\max} \rightarrow \infty$ ) producing  $\pi^{\text{even}}$  from Bernoulli numbers; (ii) volume normalization converting to dimensionful quantities. Both are Paper 18’s CALIBRATION tier [3].

The graph is strictly richer than the continuum at finite truncation: at  $n_{\max} = 3$ , the graph captures  $\sim 35\times$  more VP content than the  $W$ -weighted continuum formula.

### 15.10.1 Topology-dependent projection

The VP projection constant  $C$  is specific to the vacuum polarization diagram. A natural hypothesis is that  $C \times F_2 \rightarrow \alpha/(2\pi)$  as  $n_{\max} \rightarrow \infty$ , providing a single-constant projection from graph-native scalar QED to physical vector QED. This hypothesis is **falsified**:

| $n_{\max}$ | $C$                   | $F_2(\kappa)$ | $C \times F_2$        | $\alpha/(2\pi)$       | ratio |
|------------|-----------------------|---------------|-----------------------|-----------------------|-------|
| 2          | 0                     | 2.3525        | —                     | $1.16 \times 10^{-3}$ | —     |
| 3          | $2.84 \times 10^{-2}$ | 1.8731        | $5.31 \times 10^{-2}$ | $1.16 \times 10^{-3}$ | 45.7  |
| 4          | $4.74 \times 10^{-2}$ | 1.5892        | $7.54 \times 10^{-2}$ | $1.16 \times 10^{-3}$ | 64.9  |
| 5          | —                     | 1.3958        | —                     | $1.16 \times 10^{-3}$ | —     |
| 6          | —                     | 1.2532        | —                     | $1.16 \times 10^{-3}$ | —     |

Table 13:  $C \times F_2$  grows with  $n_{\max}$ , diverging from  $\alpha/(2\pi)$ . The VP projection constant is not the vertex correction projection.  $C$  not computed at  $n_{\max} = 5, 6$ ; the established divergence at  $n_{\max} = 3, 4$  precludes convergence.

The product  $C \times F_2$  *grows* with  $n_{\max}$  because  $C$  increases faster than  $F_2$  decreases. The correct graph-to-continuum projection for different diagram topologies involves different spectral-density factors: the VP projection (sensitive to photon self-energy) and the vertex projection (sensitive to electron self-energy and vertex structure) are structurally distinct. In Paper 18’s taxonomy, these are independent CALIBRATION exchange constants.

### 15.10.2 Three-diagram projection constants

The topology-dependent projection is characterized by computing *independent* projection constants for each diagram type:  $C_{\text{VP}} = \Pi_{\text{cont}}/\Pi_{\text{graph}}$ ,  $C_{\text{SE}} = \text{Tr}(\Sigma_{\text{cont}})/\text{Tr}(\Sigma_{\text{graph}})$ , and  $C_{F_2} = [\alpha/(2\pi)]/F_{2,\text{graph}}$ .

| $n_{\max}$ | $C_{\text{VP}}$ | $C_{\text{SE}}$ | $C_{F_2}$             | $C_{\text{VP}}/C_{\text{SE}}$ |
|------------|-----------------|-----------------|-----------------------|-------------------------------|
| 2          | 0               | 0               | $4.94 \times 10^{-4}$ | —                             |
| 3          | 0.0284          | 0.275           | $6.20 \times 10^{-4}$ | 0.1030                        |
| 4          | 0.0474          | 0.453           | $7.31 \times 10^{-4}$ | 0.1047                        |
| 5          | 0.0599          | 0.582           | $8.32 \times 10^{-4}$ | 0.1029                        |

Table 14: Cross-diagram projection constants at  $n_{\max} = 2$ –5.  $C_{\text{VP}}$  uses the photon-propagator trace  $\text{Tr}(\Pi)$ . The three constants have structurally independent power-law scaling:  $C_{\text{VP}} \sim n^{1.48}$ ,  $C_{\text{SE}} \sim n^{1.48}$ ,  $C_{F_2} \sim n^{0.57}$ .

Power-law fits give  $C_{\text{VP}} \sim 0.0070 \times n^{0.93}$  ( $R^2 = 0.95$ ),  $C_{\text{SE}} \sim 0.056 \times n^{1.48}$  ( $R^2 = 0.99$ ), and  $C_{F_2} \sim 3.3 \times 10^{-4} \times n^{0.57}$  ( $R^2 = 1.00$ ). The  $C_{F_2}$  exponent (+0.57) is the negative of the  $F_2$  convergence exponent (−0.57, Table 10), so  $C_{F_2} \times F_2 \rightarrow \text{const} = \alpha/(2\pi)$  by construction. However,  $C_{\text{VP}}$  and  $C_{\text{SE}}$  grow with *different* exponents, confirming that no single focal-point constant projects all diagram types simultaneously.

**VP/SE constant ratio.** The ratio  $C_{\text{VP}}/C_{\text{SE}} = 0.1035 \pm 0.0009$  (CV = 0.83%) is constant to sub-percent precision across  $n_{\max} = 3, 4, 5$ . The closest simple rational is  $3/29 = 0.10345$ . VP and SE share the same power-law exponent ( $\sim n^{1.48}$ ), differing only by a constant prefactor  $\sim 1/10$ . The vertex correction  $C_{F_2}$  has fundamentally different scaling ( $n^{0.57}$ ) because it is defined relative to the asymptotic Schwinger value. This reveals a **two-tier calibration structure**: one shared VP/SE tier (scaling as  $\sim n^{1.48}$ , with constant internal ratio), and one vertex-specific tier (scaling as  $\sim n^{0.57}$ , set by the  $F_2$  convergence rate).

The per-mode analysis (Sec. 15.10.1) shows that the per-mode projection ratio  $\rho(n_{\text{ext}}, n_{\text{int}})$  varies from 0 to 1.66 with no clean pattern — the graph-to-continuum projection is *topological*, not multiplicative. The root cause is that the Fock graph has nearest-neighbor connectivity ( $\Delta n = \pm 1$  only), while the continuum  $\text{SO}(4)$  channel count  $W$  obeys a triangle inequality allowing non-nearest-neighbor contributions. Some modes have graph = 0, continuum  $\neq 0$  (and vice versa).

The  $\chi_{-4}$  Dirichlet character mechanism that produces Catalan’s  $G$  and Dirichlet  $\beta(4)$  in the continuum (Theorem 3) is **not accessible** from finite-graph parity decomposition: 24 PSLQ targets against  $\{G, \beta(4), \pi^2, \zeta(3)\}$  all returned null. Graph  $F_2$  even/odd ratios alternate with  $n_{\max}$  (driven by degeneracy counting, not physics), while continuum even/odd ratios converge to  $\sim 1$ . The Catalan/ $\beta$  content lives in the continuum’s Hurwitz zeta at quarter-integer shifts, which are analytic objects with no finite-graph analog.

### 15.11 Transcendental certificate

All graph-native QED quantities are  $\pi$ -free:

- VP trace: rational at every  $n_{\max}$ .
- Self-energy  $\Sigma$ : diagonal entries rational; off-diagonals contain  $\sqrt{2}$ ,  $\sqrt{3}$ ,  $\sqrt{6}$  from CG coefficients.
- Vertex correction  $\Lambda$ : eigenvalues involve  $\sqrt{145}$ .
- Anomalous moment  $F_2 = 5\sqrt{2}/3$  ( $t = 0$ );  $(48\sqrt{3} + 96275)/[3(15 + 16\sqrt{6} + 9616\sqrt{2})]$  ( $t = \kappa$ ): algebraic irrational at both couplings.
- Projection exchange constant: rational.

The algebraic irrationals enter through Clebsch–Gordan coefficients in the Dirac-to-Fock projection. In Paper 18’s taxonomy, all graph-native QED lives in the INTRINSIC tier; transcendentals enter only when projecting onto the continuum (CALIBRATION tier).

**Rational function structure.**  $F_2(t)$  is an even rational function of  $t$  (all odd Taylor coefficients exactly zero, from  $t \rightarrow -t$  symmetry of the vertex bilinear). The degree and number field grow rapidly with  $n_{\max}$ : at  $n_{\max} = 2$ ,  $F_2(t)$  has degree 2/2 over  $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$ ; at  $n_{\max} = 3$ , degree 16/16 over  $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{10}, \sqrt{15})$ , the new irrationals arising from CG coefficients involving  $l = 2$  states. The rapid degree growth ( $2 \rightarrow 16$ ) reflects the combinatorial explosion of CG coupling channels. At  $n_{\max} = 3$ , the Taylor coefficients  $c_k$  for  $k \geq 2$  are *all negative* ( $c_2 \approx -0.650$ ,  $c_4 \approx -0.608$ ,  $c_6 \approx -0.431$ ,  $c_8 \approx -0.385$ ), in contrast to  $n_{\max} = 2$  where the signs alternate. The uniform sign indicates that the poles of the  $n_{\max} = 3$  rational function are farther from the origin, yielding better convergence of the Taylor series at  $t = \kappa$ .

## 15.12 Vector-photon QED: 7/8 scalar, 8/8 Dirac selection rules recovered

The preceding graph-native QED sections treat the photon as a scalar 1-cochain—a number on each edge. This produces  $\pi$ -free results in  $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \dots]$  but recovers only 1 of 8 continuum QED selection rules (Gaunt/CG sparsity on the scalar Fock graph) or 4 of 8 on the native Dirac graph (Rule B). The transverse photon from Wilson plaquettes (Sec. 15.9) recovers the ground-state structural zero, bringing the count to 5 of 8.

### 15.12.1 Pre-investigation: $L_1$ block-diagonalization

A natural question is whether the edge Laplacian  $L_1 = B^\top B$  already contains photon angular momentum structure that can be extracted by reorganization—i.e., does  $L_1$  block-diagonalize by  $(q, m_q)$ ?

The answer is **no**. Photon quantum numbers describe the *transition* between endpoint labels (the angular momentum *difference*), not a conserved quantum number at each vertex.  $L_1$  couples edges that share a vertex, and at that vertex, edges with different photon quantum numbers meet. The cross-sector Frobenius fraction is 20% at  $n_{\max} = 3$  on the scalar Fock graph and 57% on the Dirac Rule B graph. The *only* decomposition that works is by connected components:  $l$ -sector (Fock) or  $\kappa$ -sector (Dirac Rule A).

Vector photon structure must therefore be added from outside. It is genuinely new structure, not a reorganization of what the graph contains.

### 15.12.2 Construction

The electron register uses Fock graph nodes  $(n, l, m)$  or Dirac labels  $(n, \kappa, m_j)$ ; both configurations are tested. The photon register adds modes labeled by  $(q, m_q)$  with  $q = 1, \dots, q_{\max}$  and  $m_q = -q, \dots, q$ . The photon propagator is  $G_\gamma(q) = 1/[q(q+2)]$ , the vector Laplacian eigenvalues on  $S^3$ .

The vertex coupling is

$$V(a, b, q, m_q) = \begin{pmatrix} l_a & q & l_b \\ -m_a & m_q & m_b \end{pmatrix} \times \delta_{l_a + l_b + q \text{ odd}}, \quad (122)$$

where the Wigner  $3j$  symbol enforces the triangle inequality and  $m$ -conservation, and the parity constraint  $l_a + l_b + q$  odd enforces E1 vector coupling. Radial matrix elements are set to unity ( $R = 1$ ), isolating angular structure.

The self-energy is

$$\Sigma(a, c) = \sum_{b, q, m_q} V(a, b, q, m_q) G_e(b) G_\gamma(q) V(c, b, q, m_q), \quad (123)$$

where  $G_e(b)$  is the electron propagator.

### 15.12.3 Selection rule census

Table 15: Selection rule recovery with vector photon quantum numbers. Scalar electrons recover 7 of 8 rules; Dirac electrons recover **8 of 8** at  $n_{\max} = 2$ .

| # | Rule                      | Scalar + vec      | Dirac + vec       | Mechanism             |
|---|---------------------------|-------------------|-------------------|-----------------------|
| 1 | Gaunt/CG sparsity         | PASS              | PASS              | $3j$ symbol           |
| 2 | GS structural zero        | PASS              | PASS              | $2l_{\text{int}}$ odd |
| 3 | SO(4) channel count       | PASS              | PASS              | triangle ineq.        |
| 4 | $\Delta m_j$ conservation | PASS              | PASS              | $3j$ $m$ -rule        |
| 5 | Spatial parity            | PASS              | PASS              | E1 parity             |
| 6 | Furry's theorem           | FAIL              | PASS              | Dirac spinor phase    |
| 7 | Ward identity             | PASS <sup>a</sup> | PASS <sup>a</sup> | $\Sigma$ block-diag.  |
| 8 | Charge conjugation        | PASS              | PASS              | $\Sigma$ Hermiticity  |

passes at  $n_{\max} = 2$ ; truncation artifacts at  $n_{\max} \geq 3$  (ratio 0.61 Dirac, 3.1 scalar).

### 15.12.4 Ground-state structural zero: angular momentum mechanism

$\Sigma(\text{GS}, \text{GS}) = 0$  exactly (machine precision) at all tested  $n_{\max}$ . The mechanism: for the ground state with  $l = 0$ , the triangle inequality forces  $q = l_{\text{int}}$ , and the parity rule requires  $l_{\text{int}} + l_{\text{int}} + q = 2l_{\text{int}} + q$  odd. With  $q = l_{\text{int}}$ , this gives  $3l_{\text{int}}$  odd — but more precisely, with  $l_{\text{ext}} = 0$ :  $l_{\text{int}} + q$  odd, and the triangle with  $l_{\text{ext}} = 0$  gives  $q = l_{\text{int}}$ , so  $2l_{\text{int}}$  must be odd. Impossible for any integer.

This is the rule broken by the pendant-edge theorem on the scalar graph ( $\Sigma_{\text{scalar}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max} \rightarrow 2$ ). The vector photon restores it through angular momentum conservation alone.

### 15.12.5 Calibration exchange constant: $1/(4\pi)$

The  $l = 1$  self-energy at  $n_{\max} = 2$  is **exactly**  $1/(4\pi)$  to machine precision. This is the Weyl exchange constant for  $S^2$  (the density-of-states constant for the 2-sphere), entering through the normalization of the vector spherical harmonics:  $\sqrt{(2l+1)(2q+1)(2l'+1)/(4\pi)}$  in the vertex coupling.

The entire scalar graph QED is  $\pi$ -free ( $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \dots]$ ). The vector photon introduces **exactly one transcendental per loop**:  $\pi$ , through the  $S^2$  solid angle normalization. In Paper 18's taxonomy [3], this is precisely where calibration content enters: the  $1 \rightarrow 7$  transition from graph topology to angular momentum conservation costs exactly one transcendental ( $\pi$ , via  $1/(4\pi)$  per loop).

### 15.12.6 Furry's theorem: scalar obstruction, Dirac resolution

For **scalar electrons**, Furry's theorem fails: the diagonal vertex coupling  $V(a, a, q, m_q)$  is nonzero when  $q$  is odd (parity  $2l + q$  odd is satisfied), and the  $\begin{pmatrix} j & q & j \\ -m & 0 & m \end{pmatrix}$  does not vanish for generic  $m$ .

For **Dirac electrons**, the diagonal coupling vanishes identically by the *Dirac spinor phase constraint*. The physical mechanism: the Dirac spinor  $\psi = (f \cdot \Omega_\kappa, ig \cdot \Omega_{-\kappa})$  has an  $i$ -factor on the small component, and the Dirac  $\alpha$ -matrix  $\alpha = \begin{pmatrix} 0 & \sigma \\ \sigma & 0 \end{pmatrix}$  is off-diagonal in large/small component space. For  $a = b$ , the vertex decomposes as a Hermitian pair whose small-component  $i$  phase reduces it to an imaginary part,  $V_{LS} + V_{SL} \propto fg \operatorname{Im} \langle \Omega_\kappa | \sigma \cdot \varepsilon | \Omega_{-\kappa} \rangle$ , and  $m_l$  conservation forces that imaginary part to vanish — the mechanism is  $m_l$  conservation, not Hermiticity alone (Paper 33’s mechanism remark; sympy-exact in `tests/test_paper33_furry_derivation.py`). The cross terms cancel exactly, making  $V(a, a, q, m_q) = 0$  for all  $q$  and  $m_q$ .

This is *not* Furry’s theorem in the field-theoretic sense. In continuum QED, Furry works because for every electron loop there is a positron loop that exactly cancels — charge conjugation symmetry of the *field*, not the *state labels*. The Dirac spinor phase constraint is a kinematic identity of the single-particle vertex: it kills the tadpole at the vertex level, before any loop sum is performed. The two mechanisms are independent, and the Dirac constraint is the stronger one.

### 15.12.7 The 1–7–8 partition

The 8 continuum QED selection rules partition cleanly into three tiers. The partition depends on whether the electron basis is scalar or Dirac:

| Tier             | Rules | Source                              | Scalar / Dirac |
|------------------|-------|-------------------------------------|----------------|
| Graph topology   | 1     | Gaunt/CG sparsity (always survives) | both           |
| Angular momentum | 6     | CG + E1 parity + triangle ineq.     | both           |
| 8th rule         | 1     | Furry / Dirac spinor phase          | FAIL / PASS    |
| Total:           |       |                                     | 7/8 / 8/8      |

The  $1 \rightarrow 7$  transition costs exactly one transcendental:  $\pi$ , entering as  $1/(4\pi)$  per loop from the  $S^2$  solid angle normalization. For **scalar electrons**, the  $7 \rightarrow 8$  transition requires the full field-theoretic structure of charge conjugation — a categorically different ingredient that is absent from the graph/vector construction. For **Dirac electrons**, the  $7 \rightarrow 8$  transition is closed by the Dirac spinor phase constraint at zero additional cost: the off-diagonal  $\alpha$ -matrix combined with the  $i$ -factor on the small component reduces  $V(a, a, q, m_q)$  to an imaginary part that  $m_l$  conservation forces to zero identically, killing the tadpole at the vertex level. No second-quantized field theory is needed.

## 16 Discussion

The five theorems, two propositions, and two observations of this paper, together with the anomalous magnetic moment computation (Sec. 7), the graph-native QED infrastructure (Sec. 15), and the vector-photon QED construction (Sec. 15.12), complete the first comprehensive treatment of perturbative QED on  $S^3$  in both the continuum spectral-sum and finite-graph formulations. The vector-photon construction recovers 7 of 8 continuum selection rules for scalar electrons and **8 of 8 for Dirac electrons**, with a single transcendental cost ( $1/(4\pi)$  per loop). For Dirac electrons, the 1–7–8 partition closes entirely at the kinematic level: graph topology  $\rightarrow$  angular momentum conservation  $\rightarrow$  Dirac spinor phase constraint. The three-axis taxonomy extends Paper 18’s exchange-constant classification [3] from free propagators to interaction vertices: *diagram topology* is an independent source of transcendental content beyond operator order and bundle type.

The self-energy structural zero (Theorem 4) and the anomalous magnetic moment extraction are the first physical-process QED results in the GeoVac framework, complementing the vacuum

diagrams of the prior QED modules. The self-energy demonstrates that the vertex parity selection rule has direct consequences for scattering processes on  $S^3$ . The pendant-edge theorem (Proposition 5) sharpens the broken-structural-zero finding: the graph ground-state self-energy  $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\text{max}} - 1)/n_{\text{max}}$  is an exact rational expression that converges to 2, not 0, as  $n_{\text{max}} \rightarrow \infty$ . This is not a finite-size artifact that heals in the continuum limit; it reflects a qualitative difference between graph QED (scalar, 1-cochain photon) and continuum QED (vector,  $\text{SO}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$  photon). The photon’s spin-1 vector nature is thereby classified as a CALIBRATION exchange constant in Paper 18’s taxonomy [3]. The selection rule survival census (Table 11) makes this classification precise: of 8 continuum QED selection rules, only Gaunt/CG sparsity survives on the scalar Fock graph; the 7 broken rules partition cleanly into 3 STRUCTURAL (requiring vector photon structure) and 4 INTRINSIC (graph kinematics). The native Dirac graph QED (Sec. 15.7) refines this classification: building QED on the  $(n, \kappa, m_j)$  Dirac graph with E1 dipole adjacency (Rule B) recovers 4 of 8 rules, identifying a three-tier partition—1 always survives (Gaunt/CG), 3 are spinor-recoverable ( $\Delta m_j$ , spatial parity, Furry), and 4 require vector photon structure (vertex parity,  $\text{SO}(4)$ , Ward, charge conjugation). The vector-photon QED construction (Sec. 15.12) then recovers 7 of 8 for scalar electrons and **8 of 8 for Dirac electrons** by adding explicit photon angular momentum labels  $(q, m_q)$  with Clebsch–Gordan vertex coupling and E1 parity enforcement. The path from graph to continuum QED has three distinct barriers: the spinor barrier (crossed by adopting Dirac labels, zero cost), the angular momentum barrier (crossed by promoting the photon from scalar 1-cochain to vector harmonic, calibration cost  $1/(4\pi)$  per loop), and — for scalar electrons only — the field-theory barrier (requiring second-quantized charge conjugation for Furry’s theorem). For Dirac electrons, no field-theory barrier exists: the Dirac spinor phase constraint ( $V(a, a, q, m_q) = 0$  identically from the off-diagonal  $\alpha$ -matrix, the  $i$ -factor on the small component, and  $m_l$  conservation) closes the  $7 \rightarrow 8$  gap at the single-particle vertex level, without invoking particle–antiparticle cancellation. The vector QED sprint (Sec. 15.8) provides five independent negative results confirming the three-tier selection rule partition from a complementary direction: neither Pauli spin matrices at the vertex (VQ-1–3), nor direction-resolved edge decomposition (VQ-4), nor their combination (VQ-5) recovers any of the 4 vector-required rules. The positive structural finding—spectral anisotropy of the T/L edge channels and cross-channel coupling at  $n_{\text{max}} = 3$ —shows that the graph topology encodes directional information, but directional propagation is not equivalent to spin-1 quantum numbers.

The even-ness of  $F_2(t)$  (Sec. 15.4.1) — all odd Taylor coefficients vanish — reflects the  $t \rightarrow -t$  symmetry of the vertex bilinear and confirms that the anomalous moment depends only on  $\kappa^2$ , not on the sign of the hopping coupling.

The topology-dependent projection result (Tables 13 and 14) falsifies the hypothesis that a single projection constant  $C$  converts graph-native QED to physical QED:  $C \times F_2$  grows with  $n_{\text{max}}$  ( $45.7\times$  and  $64.9\times$  above  $\alpha/(2\pi)$  at  $n_{\text{max}} = 3, 4$ ), not converges. The three-diagram analysis (Table 14) sharpens this:  $C_{\text{VP}}$ ,  $C_{\text{SE}}$ , and  $C_{F_2}$  have structurally independent power-law scalings ( $n^{1.48}$ ,  $n^{1.48}$ ,  $n^{0.57}$  respectively), but VP and SE share the *same* exponent with a constant prefactor ratio  $C_{\text{VP}}/C_{\text{SE}} = 0.1035 \pm 0.0009$  ( $\text{CV} = 0.83\%$ , closest rational  $3/29$ ). This reveals a two-tier calibration structure: one shared tier for VP and SE (vector photon structure and non-nearest-neighbor connectivity calibrate both identically; the  $\sim 10\times$  prefactor difference is a pure topology-of-diagram effect), and one vertex-specific tier for  $F_2$  (requiring the full vertex correction projection). The extended  $F_2$  convergence data (Table 10) confirms this picture:  $F_2(\kappa)$  decreases as a slow power law  $\sim n_{\text{max}}^{-0.57}$ , with Richardson extrapolation giving an unphysical negative limit, confirming  $F_2 \rightarrow 0$  rather than converging to  $\alpha/(2\pi)$ . The graph-native anomalous moment vanishes in the infinite-graph limit; the physical Schwinger value requires both the coupling constant  $\alpha$  and the vector photon selection rules, neither of which is present on the finite graph.

The anomalous magnetic moment ( $F_2/[\alpha/(2\pi)] = 1.084$  at  $n_{\text{ext}} = 1$ ) provides the first quantitative curvature correction to a measurable QED observable in the framework, matching the Parker–Toms prediction within 0.4%.

The curvature coefficient identification  $c_2 = (2 - B\Delta - F\Delta - F/B)/5$  (Sec. 7.7) is the first concrete numerical bridge between Papers 2 and 28 (a numerical coincidence, not a derivation — §7.7): the three spectral objects  $B = 42$ ,  $F = \pi^2/6$ ,  $\Delta = 1/40$  that compose the  $\alpha$  construction  $K = \pi(B + F - \Delta)$  also appear as the geometric invariants controlling the second curvature correction to the anomalous magnetic moment on  $S^3$ . Since  $\alpha$  cancels in the ratio  $F_2/[\alpha/(2\pi)]$ , this is a statement about the spectral geometry of the Dirac operator on  $S^3$ , not about QED per se. The high-precision  $n_{\text{int}} = 50$  extraction (Sec. 7.7) gives  $c_3 = -5.946(3) \times 10^{-7}$  at  $200.9\sigma$ , confirming the expansion does not terminate at second order; the ratio  $|c_3 x^3|/|c_2 x^2| \approx 5.4 \times 10^{-6}$  indicates rapid but non-terminating convergence. The T9 theorem constrains  $c_3$  to lie in the  $\pi^{\text{even}}$ -polynomial ring; its closed-form identification remains open.

The 5-point mode-dependent curve (Table 5) reveals that the anomalous moment on a compact space is strongly mode-dependent: enhanced for the lowest mode and progressively suppressed for higher modes, with a power-law  $|\lambda|^{-1.84 \pm 0.06}$ . The dominant internal mode ( $n_{\text{int}} = 1$ ) approaches the Parker–Toms  $\lambda^{-2}$  scaling asymptotically, but higher internal modes contribute increasingly at larger  $n_{\text{ext}}$ , pulling the effective exponent above  $-2$ . This non-perturbative mode dependence is a structural consequence of the discrete spectral geometry on  $S^3$ .

The  $\chi_{-4}$  identity (Theorem 3) is a concrete spectral-zeta to Dirichlet- $L$  bridge. Combined with the RH-sprint finding that the spectral zeros of  $D(s)$ ,  $D_{\text{even}}(s)$ , and  $D_{\text{odd}}(s)$  exhibit GUE-like spacing statistics (CV  $\approx 0.35$ – $0.40$ , KS rejects Poisson; Paper 29 [4]), the Dirac spectral geometry on  $S^3$  realizes both axes of the Weil dictionary — though GUE spacing without a single critical line means the analogy to the Riemann hypothesis remains structural, not literal.

The depth- $k$  tower has a natural interpretation in the framework of Zagier’s motivic weight conjecture [18]: each loop order increases the motivic weight by the vertex coupling, and the algebra of multiple zeta values provides the natural receptacle for the resulting transcendentals. The half-integer shift  $a = 3/2$  from the Dirac spectrum on  $S^3$  produces Hurwitz (rather than Riemann) multiple zeta values, and the failure of 100-element PSLQ bases suggests that the  $S^3$  spectral geometry accesses a sector of the MZV algebra that is not reached by flat-space Feynman integrals at the same loop order.

The fine-structure Euler sum cancellation (Observations 1 and 2) connects the free-propagator Dirichlet series studied in Secs. 3–5 to the *interacting* Dirac hydrogen spectrum. The product survival rule —  $\zeta(2) \times \zeta(\text{odd})$  cancelled at every order, with exactly  $\max(0, \lfloor (2p-5)/2 \rfloor)$  products surviving — is verified through  $D_6$  and driven by the same  $n^2$  Fock degeneracy that produces  $F = \zeta(2)$  in Paper 2 [9]. Despite this structural link, PSLQ confirms that  $K/\pi$  does not lie in the  $\mathbb{Q}$ -span of  $\{D_2, \dots, D_6\}$ : the spectral-action ingredients of  $K$  inhabit a  $\pi$ -polynomial ring (Theorem 1), while the Sommerfeld sums carry irreducible odd-zeta content from the Euler–Zagier MZV algebra at every order. The operator/energy distinction — spectral zeta versus physical energy coefficients — is the structural separation between  $K$  and the  $D_p$ .

The three-loop factorized algorithm (Sec. 10) reduces the computational complexity from  $O(N^5)$  to  $O(N^3)$ , but raw partial sums at physical exponents carry zero reliable digits at any practical  $n_{\text{max}}$  (the  $(\ln N)^2/N^2$  increment law; the fitted  $N^{-1.31}$  was its local masquerade). The decomposition route supersedes direct summation entirely: the closure sprint (2026-06-11) gives  $S^{(3)} = 31.5726\dots$  inside the rigorous bracket  $[31.57063, 31.57300]$ , and the W10-identification sprint (2026-06-12) closes the full form:  $S^{(3)} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(\text{odd}), \zeta(5, 3)]$ , realized depth  $\leq 2$ , with  $\zeta(5, 3)$  the only depth-2 constant (Eq. (100)). The earlier “ $S^{(3)}$  independent of  $S_{\text{min}}$ ” 20-digit test at non-physical exponents is superseded.



## 16.1 Open questions

1. **Resolved (2026-06-11):**  $S_{\min}$  is identified in closed form, Eq. (80). The 100-element PSLQ null was a basis-coverage artifact, not irreducibility.
2. Does the depth- $k$  tower stabilize (i.e., do all constants beyond some  $k_0$  reduce to products of lower-depth constants)? **Updated (2026-06-12, W10-identification sprint):** at  $k = 2$  everything reduces to depth  $\leq 1$ ; at  $k = 3$  the full closed form is explicit with realized depth  $\leq 2$ , the sole depth-2 constant being the level-1 generator  $\zeta(5, 3)$  (Eq. (100)); all level-2 depth-2 generators cancel in the total. Realized depth  $\leq k - 1$  at  $k = 2, 3$ . **Updated (2026-06-13,  $S^{(4)}$  stage-2):** the depth bound extends to  $k = 4$  — realized depth  $\leq 3$  for  $S^{(4)}$  (Prop. 4,  $k = 4$  bullet), proven numerics-free at weights 9, 11, 13 (stuffle spans the full depth-4 space, two-prime exact rank) and settled at weights 5, 7 by the low-weight depth filtration. So realized depth  $\leq k - 1$  holds at  $k = 2, 3, 4$ . The explicit closed form of  $S^{(4)}$  (the depth- $\leq 3$  survivors) and the  $k \geq 5$  continuation remain open.
3. Is there a Picard–Fuchs interpretation of  $S_{\min}$ , analogous to Apéry’s recurrence for  $\zeta(3)$  [27]?
4. Can the self-energy structural zero at  $n_{\text{ext}} = 0$  be lifted by going to higher loop order, or does vertex parity enforce  $\Sigma^{(k)}(0) = 0$  at every order? **Partial resolution:** The structural zero is broken on the finite graph (Sec. 15.5): the pendant-edge theorem (Proposition 5) proves  $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\text{max}} - 1)/n_{\text{max}} \rightarrow 2$  exactly, never healing. The root cause is that the graph photon is a scalar 1-cochain lacking the spin-1 vector selection rules (vertex parity +  $\text{SO}(4)$  channel count) that enforce the continuum zero. **Further resolution:** Adding explicit vector photon quantum numbers  $(q, m_q)$  (Sec. 15.12) recovers the GS structural zero exactly through angular momentum conservation alone ( $2l_{\text{int}}$  odd is impossible), confirming that the broken zero was due to the scalar photon, not a fundamental graph obstruction. The continuum question at higher loop order remains open.
5. The three-loop CG-weighted sum at physical exponents has convergence rate  $\alpha \approx -1.31$ . Is there a resummation technique (e.g., Padé, conformal mapping) that can extract more digits from the partial sums?
6. The 5-point mode-dependent curve  $F_2/[\alpha/(2\pi)] \approx 5.91 |\lambda|^{-1.84 \pm 0.06}$  has an effective exponent  $2.8\sigma$  from  $-2$ . The dominant internal mode’s local slope converges toward  $-2$ , but higher internal modes ( $n_{\text{int}} \geq 2$ ) contribute increasingly at larger  $n_{\text{ext}}$  and pull the effective exponent above  $-2$ . Is the true asymptotic exponent exactly  $-2$  (Parker–Toms), or a distinct value? This could be resolved by computing  $n_{\text{ext}} = 6$ –10 or by analytical large- $\lambda$  asymptotics of the CG vertex sums.
7. The tree-level magnetic amplitude satisfies  $V_{\text{mag}} \rightarrow (4/3)/|\lambda|$  asymptotically (Richardson-extrapolated). Can this  $4/3$  prefactor be derived from the CG coefficient structure in closed form?
8. The exact algebraic expressions for  $V_{\text{mag}}$  and  $B(n_{\text{int}} = 1)$  at each  $n_{\text{ext}}$  are sums of square roots with rational coefficients, exhibiting a telescoping recurrence: the negative sqrt argument at step  $n$  equals the positive sqrt argument at step  $n - 1$ . Can this recurrence be proven from the CG coupling rules?
9. Can the flat-space limit  $R \rightarrow \infty$  be verified explicitly by computing  $F_2$  on spheres of increasing radius with  $n$  scaling as  $n \sim mR$  to keep the physical mass fixed?

10. The curvature coefficient identification  $c_2 = (2 - B\Delta - F\Delta - F/B)/5$  matches the spectral sum to 8 digits (Sec. 7.7). **RESOLVED (partially):** Extending to  $n_{\text{int}} = 50$  confirms  $c_2$  and gives  $c_3 = -5.946(3) \times 10^{-7}$  at  $200.9\sigma$ . Deriving  $c_2$  analytically from the Schwinger–DeWitt heat kernel remains open.
11. **RESOLVED.**  $c_3 = -5.946(3) \times 10^{-7}$  from  $n_{\text{int}} = 50$ , nonzero at  $200.9\sigma$ . The curvature expansion does not terminate at second order; the ratio  $|c_3 x^3|/|c_2 x^2| \approx 5.4 \times 10^{-6}$  shows rapid but non-terminating convergence. The closed-form identification of  $c_3$  in the T9  $\pi^{\text{even}}$ -polynomial ring remains open.
12. The multi- $n_{\text{ext}}$  extraction of curvature coefficients requires Wigner–Eckart decomposition into multipole form factors for higher- $j$  sectors. At  $n_{\text{ext}} \geq 2$ , the  $j = 1/2$  component is a  $2/[(n+1)(n+2)]$  fraction of the total degeneracy. Can the  $j$ -dependent form factors be separated cleanly to give independent  $c_2$  extractions at each  $n_{\text{ext}}$ ?
13. The VP/SE projection ratio  $C_{\text{VP}}/C_{\text{SE}} = 0.1035 \pm 0.0009$  (Table 14) is constant to sub-percent across  $n_{\text{max}} = 3, 4, 5$ , with closest simple rational  $3/29 = 0.10345$ . Is this a rational number? If so, what structural invariant fixes it? The constancy implies that VP and SE probe the same calibration content with a universal diagram-topology prefactor.
14. **RESOLVED.** The Euler sum cancellation extends to  $D_5$  (weight 9):  $\zeta(2)\zeta(7)$  cancels while  $\zeta(3)\zeta(6)$  and  $\zeta(4)\zeta(5)$  survive (Observation 2). The product survival rule  $\max(0, \lfloor (2p-5)/2 \rfloor)$  is verified through  $p = 5$ . The  $S_N$ -symmetry origin remains open, but the mechanism is identified: the  $n^2$  Fock degeneracy weight systematically expels  $\zeta(2)$  products via Euler sum reflection identities.

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